Hirzebruch-Riemann-Roch Theorem

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Abstract

The Riemann-Roch problem asks how many holomorphic, meromorphic, or algebraic sections a space admits (equivalently, what can be said about $H^0(M, \mathcal{F})$, and what the higher sheaf cohomology says when that group is smaller than one would expect). This paper is an exposition of the *Hirzebruch-Riemann-Roch Theorem*, which answers this for a holomorphic vector bundle E over a projective variety M:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \smile \mathrm{td}(M).$$

The proof is organized around two generating functions: the χ_y-characteristic, which packages the Euler characteristics, and the T-characteristic T_y, which packages the Chern character and Todd class data. The two agree at $y = 0$ with $\chi(M, E)$, and, for trivial E, at $y = 1$ with the signature $\tau(M)$; that second agreement is the bridge which forces the theorem, and cobordism then reduces the statement to the case $M = \mathbb{C}\mathrm{P}^n$.

The exposition deliberately reverses the order of Hirzebruch's original treatment, introducing each tool at the point where the problem calls for it rather than in advance: Chern classes and the Chern character, via the splitting principle and K-theory; Dolbeault cohomology and Kähler geometry, to relate the analytic and algebraic Euler characteristics; the Todd class, as a multiplicative sequence; and the oriented cobordism ring together with the Hirzebruch signature theorem. The final chapter closes with a sketch of Grothendieck-Riemann-Roch and a survey of the current generalizations in the literature.

Contents

This paper is an exposition on the *Hirzebruch-Riemann-Roch Theorem*, a generalization of the classical Riemann-Roch theorem for curves and Riemann surfaces. The Classical Riemann-Roch Theorem asserts that if D is a (Weil) divisor of a curve/Riemann surface C, the dimension of the vector space of meromorphic functions on C with poles bounded by D is controlled by $\deg(D)$ and the genus g. The goal of this paper is to show that the result generalizes to a projective variety M and show that the appropriate generalization of the divisor D is to consider vector bundles E on M.

In particular, this paper shall show that if M is a projective variety, E a holomorphic vector bundle (equivalently, a locally free coherent sheaf), $\mathrm{ch}(E)$ the *Chern character* of E, and $\mathrm{td}(M)$ the *Todd class* of M, then:

$$\chi(M,E) = \int_M \mathrm{ch}(E) \smile \mathrm{td}(M) \tag{1}$$

where $\chi(M,E)$ is the Euler characteristic for vector bundles and $\int_M$ is the evaluation of $\mathrm{ch}(E) \smile \mathrm{td}(M)$ at the fundamental class $[M]$. The high-level idea of the proof is to define two generating functions: one for the Euler characteristic for (complex) vector bundles and the other for something that shall be called the *T-Characteristic* with respect to M and E which shall hold the information of $\mathrm{ch}(-)$ and $\mathrm{td}(-)$. Let $\chi_y(M,E)$ and $T_y(M,E)$ represent these two generating functions with dummy variable y. Then, it shall be shown that when $y = 0$:

$$T_0(M,E) = \chi_0(M,E) = \chi(M,E)$$

and when $y = 1$:

$$T_1(M) = \chi_1(M) = \tau(M)$$

where the bundle E was purposefully omitted, and τ is the *signature* of M. The key connection is the signature. Showing that both χ_y and T_y agree with τ when $y = 1$ shall be a driving force in this paper, namely it shall be the "bridge" which gives the equality in equation (1). Furthermore, by the use of cobordism, it shall be enough to show the result in the case where $M = \mathbb{CP}^n$ for all n.

The proof of HRR requires many results from fields such as Hodge Theory, cobordism theory, and algebraic topology, as well as GAGA and some other miscellaneous results. This paper will include the "core" results that do not require much build-up given the assumed necessary background (see box 0.0.1). The paper shall focus on the necessary background to define the aforementioned concepts, leaving some proofs as references to other textbooks (or to my notes whenever I have proven it elsewhere to save space in this paper).

The first chapter is to motivate the generalization. The goal is for someone with the necessary background who is comfortable with the Riemann-Roch Theorem to be able to follow the generalizations and understand the choice of topics covered in this paper.

The second chapter reviews the necessary characteristic class theory[1], giving a proof of the splitting principle and overviewing the construction of the Chern and Pontryagin classes. The last section of the chapter goes over the definition of the Chern character and also sets up some motivation for the generalization of HRR to the Grothendieck-Riemann-Roch Theorem.

The third chapter goes into the cohomology of projective varieties. Wherever possible, the results are stated to work on almost-complex manifolds, but these can all be ignored without concern for future results in the paper. An important unique element that follows Hirzebruch's idea is the definition

[1] Throughout, this document prefers modern names even where they differ from the usage in Hirzebruch's [16]: *characteristic class theory* rather than the older abbreviation "characteristic theory", and *signature* rather than "index" (see the terminology remark in section 5.2).

of the *virtual Euler characteristic* which is the Euler characteristic for a filtration of codimension 1 sub-varieties at each inclusion:

$$V_r \subseteq V_{r-1} \subseteq V_{r-2} \subseteq \cdots \subseteq V_1 \subseteq M.$$

The key take-away will be to relate the Euler characteristic to the virtual Euler characteristic (theorem 3.4.6). This is for one of the reductions that Hirzebruch takes, namely proving HRR for line bundles first. As a codimension 1 sub-variety is associated to a line bundle, this shall allow for some important interplay between the space V_r and the line bundle $\mathcal{L}(V_r)$. The chapter also includes a bonus section for *vanishing theorems* which go over some special cases where the Euler characteristic $\chi(X, E)$ reduces to $\dim H^0(X, E)$.

The fourth chapter introduces the Todd class and T-characteristic. The Todd class can be thought of as holding the appropriate geometric information and generalizes the genus g. The Chern character shall hold the algebraic information, and the T-characteristic shall combine these two. Then just like in the previous chapter, the virtual T-characteristic is defined and related to the T-characteristic.

The fifth chapter introduces the necessary cobordism theory to define the signature and show it is a cobordism invariant, allowing for the reduction of the proof to $M = \mathbb{CP}^n$ for $n \in \mathbb{N}$. Due to the length of the paper, the proof that cobordism classes are determined by their Pontryagin numbers was omitted, but it is not a big leap with the background presented in this paper; a proof can be found in [16].

The sixth and final chapter shall prove the titular theorem. The first section goes over the classical Riemann-Roch and also gives a simple generalization to vector bundles over an algebraic curve, mainly to motivate some results in the theory of vector bundles in general (namely the splitting-off of trivial summands when the rank exceeds the dimension of the base space). The second section combines the results of all previous chapters and proves the HRR, first over line bundles (the hard part) and then over vector bundles.

There are two more sections in the last chapter covering generalizations of HRR. The third section covers Grothendieck-Riemann-Roch and follows Vakil's notes on intersection theory [19]. As this is a paper on HRR, all the intersection theory build-up is skipped and it is simply presented for exposure (or for the reader who already knows intersection theory, it is an overview of the proof). The final section covers the various directions in which the HRR has been generalized, pushing the envelope as late as 2017.

Notation:

1. As much as possible $\mathbb{F}$ will represent a field (of characteristic 0). The notation k, K is already used when appropriate for curvature, canonical divisor, K-genus, K-theory, and so forth.
2. $\mathbf{F}(q)$ will represent the flag manifold of $\mathbb{C}^q$.
3. If $H^i \times H^{n-i}$ is a perfect pairing, as the manifolds/varieties we are working with are nice enough we shall abuse notation due to de Rham's Theorem and write:

   $$\int_Y c$$

   for the pairing of the cohomology class c with Y. If we want to make it more explicit, we shall use the $\kappa_n[-]$ notation.
4. $A \sqcup B$ represents disjoint union

5. TM represents tangent bundles. $N_{N/M}$ represents the normal bundle of N with respect to M.

0.0.1 Background

1. Basics of Algebraic Geometry: Riemann-Roch, sheaf theory (in particular sheaves on projective space), divisor theory. Knowledge of sheaf cohomology is very useful but not strictly necessary.
2. GAGA will be assumed (there is no space to prove it). In chapter 1, this shall not be assumed and the line-bundle/divisor correspondence shall be sketched out.
3. Complex analysis up to Ahlfors (knowledge of Riemann-Roch from a complex perspective is good too, see for ex. [29]).
4. Basic vector bundle theory and characteristic class theory are necessary if the reader does not want to take the results in chapter 2 at face value (I cover the details in [12]).
5. Riemann Geometry up to John M. Lee [18].
6. Algebraic Topology up to Hatcher [9], as well as the Leray-Hirsch theorem (see [13])

Hartshorne will sometimes be quoted, but the reader could get by using any other textbook in algebraic geometry. I will introduce the necessary Hodge Theory and cobordism theory, and only the characteristic class theory that is necessary.

It should be noted that for the reader who is following this paper as a supplement to Hirzebruch's original book ([16]), my paper almost completely "reverses" the order of that book. Hirzebruch took a very "Bourbaki" approach where he does not motivate the proof of the necessary results and combines them only when it becomes strictly necessary, while I hope to motivate why the concepts are being covered as they are presented.

1

Motivation: Riemann-Roch Problem

A natural question to ask in complex and algebraic geometry is

"how many holomorphic/meromorphic/algebraic/analytic/etc. functions or sections are there on a (suitable) space X"

By letting $\mathcal{F}(X)$ represent our choice of functions/sections (i.e. the global sections of a [quasi-coherent] sheaf $\mathcal{F}$), we shall phrase this question as finding information about the 0th sheaf cohomology group

$$H^0(X, \mathcal{F})$$

which by [3] is equal to $\mathcal{F}(X)$. We denote $\mathcal{F}(X)$ in terms of sheaf cohomology because this group shall sometimes be trivial, and the reason for its triviality can be described by higher order sheaf cohomology groups:

$$H^1(X, \mathcal{F}), H^2(X, \mathcal{F}), \dots$$

Exactly how to extract this information is known as the *Riemann-Roch Problem.* To motivate the direction the problem took, let us answer this question in some simple cases, build up to how the Riemann-Roch question is answered for algebraic curves, and then motivate how the (classical) Riemann-Roch Theorem would be generalized to a result about sections on vector bundles.

1.1 Riemann-Roch Problem

The simplest case would be to deal with polynomial functions over $X = \mathbb{C}$ which are determined by a finite set of roots (counting multiplicity) and a constant:

$$p(z) = c(z - a_1) \cdots (z - a_n)$$

and similarly for rational functions. If we ask for holomorphic functions over $\mathbb{C}$, $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{H}(\mathbb{C}))$, then the Weierstrass Factorization Theorem (see [33, section 4.2]) tells us that $f \in \mathcal{H}(\mathbb{C})$ factors as an infinite product times a unit $e^{g(z)}$ along with correction factors for the linear terms:

$$z^m e^{g(z)} \prod_{k=1}^{\infty} \left[\left(1 - \frac{z}{a_k}\right) e^{p_k(z)} \right]$$

where

$$p_k(z) = \frac{z}{a_k} + \frac{1}{2}\left(\frac{z}{a_k}\right)^2 + \cdots + \frac{1}{m_k}\left(\frac{z}{a_k}\right)^{m_k}$$

is the correction factor and the a_k are roots that form a non-accumulating sequence. In particular, unlike the polynomial case, genuinely analytic data (the sequence of roots a_k and the entire function g) is required. A meromorphic function f on $\mathbb{C}$ can be determined by two entire functions $f = g/h$ (again see [33, section 4.2]); this is possible as for poles of f we can define a function h with the appropriate roots and multiplicities. In other words, determining meromorphic functions becomes a classification of non-accumulating sequences in $\mathbb{C}$. The situation requires even more analytic information in hyperbolic space D^2 (think of the existence of $\sum_n z^{n!}$ and $1/(1-z)$). To understand the algebraic behaviour, limiting the scope to *compact spaces* such as Riemann surfaces or projective varieties gets much better as any holomorphic/meromorphic function must have finitely many roots/poles by the Identity Theorem (that if two functions agree on a set with an accumulation point, they are equal), and so we can focus our efforts within the algebraic setting. This setting is plenty rich enough for study and often offers a good starting point to generalize to the non-compact case[1]. In fact, by the efforts of Serre, Chow, and Grothendieck, the category of coherent analytic sheaves on (suitable) compact spaces is equivalent to the category of coherent algebraic sheaves on these spaces ([34], [23]), meaning many compact spaces are "intrinsically" algebraic.

Let us try finding holomorphic/meromorphic functions on $X = \mathbb{C}\mathrm{P}^1$ and see if they are determined by the points of $\mathbb{C}\mathrm{P}^1$ just like in the case of $\mathbb{C}$. Immediately, $\mathbb{C}\mathrm{P}^1$ has only constant holomorphic functions by an easy generalization of Liouville or the maximum modulus principle on manifolds. By the Residue Theorem, the number of zeros minus the number of poles of a meromorphic function (counted with multiplicity) must be 0.

Let us represent this formally. Let

$$D = \sum_i n_i p_i \qquad n_i \in \mathbb{Z},\ p_i \in \mathbb{C}\mathrm{P}^1$$

be a finite formal sum; it is an element of the free (abelian) group $F^{\mathbb{Z}}(\mathbb{C}\mathrm{P}^1)$. Call D a *divisor*, and let $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$ represent this group. A divisor can be thought of as an object that remembers which roots/poles with which multiplicity we are interested in. Define:

$$\mathrm{div}(f) = \sum_{p \in \mathbb{C}\mathrm{P}^1} \nu_p(f) p$$

where $\nu_p(f)$ is the order of the zero/pole, and is negative if it is a pole. This forms a subgroup of $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$; represent it as $\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$ ("P" for principal), and elements of this subgroup are called *principal divisors*. This subgroup represents the divisors that are *representable* as a meromorphic function. Let us find which divisors can be represented. Let:

$$\deg D = \sum_i n_i$$

[1]These cases are still being studied, see for example [1]

Notice that $\deg(-)$ is certainly a group homomorphism. By our earlier remark:

$$\deg(\operatorname{div}(f)) = 0$$

which is a restriction applied to any meromorphic function on $\mathbb{C}\mathrm{P}^1$. It is easy to see that this restriction would work on any compact Riemann surface (see [33, chapter 6]). If $\deg(D) = 0$, there exists an f such that $\operatorname{div}(f) = D$: take $f = \prod_i (z - p_i)^{n_i}$, which is meromorphic at ∞ precisely because $\sum_i n_i = 0$. This shows this condition characterizes meromorphic functions $f \in \mathcal{M}(\mathbb{C}\mathrm{P}^1)$ (but it shall not on other Riemann surfaces, as demonstrated below).

Let us see to what extent the set of all divisors fails to represent a meromorphic function. Consider the (*Weil*) *divisor class group*:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) = \frac{\operatorname{Div}(\mathbb{C}\mathrm{P}^1)}{\operatorname{PDiv}(\mathbb{C}\mathrm{P}^1)}$$

for any $[D] \in \mathrm{Cl}(\mathbb{C}\mathrm{P}^1)$, $\deg([D])$ is well-defined as $\deg(D) = \deg(D + \operatorname{div}(f))$ for any $\operatorname{div}(f) \in \operatorname{PDiv}(\mathbb{C}\mathrm{P}^1)$. If $\deg(D) = \deg(D')$, then $\deg(D) - \deg(D') = \deg(D - D') = 0$, and hence $D - D' = \operatorname{div}(f)$. Therefore, we have that:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong \mathbb{Z}$$

which can be thought of as measuring the obstruction to a meromorphic function being defined by its points/roots.

Let us translate this intuition into sheaf cohomology, that is, let us translate this into a measure of the obstruction to gluing local information into global information. The natural guess would be that:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1})$$

However, $H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1})$ is trivial. This is because there is no obstruction to extending a function *additively*, in particular if $\mathbb{C}\mathrm{P}^1 = U_0 \cup U_1$ is the usual covering and if $f_{01} \in \Gamma(U_0 \cap U_1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1})$ then $f_{01} = g_0 - g_1$ (decompose f_{01} into a Laurent series, split the Laurent series into the principal and non-principal part, change coordinates in the principal part so that it becomes holomorphic on U_1). Instead, the obstruction is captured in the multiplicative structure[2]:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1})$$

The higher order sheaf cohomology groups of $\mathcal{O}_{\mathbb{C}\mathrm{P}^1}$ shall be trivial as the dimension of $\mathbb{C}\mathrm{P}^1$ is 1 (over $\mathbb{C}$); for $\mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}$, the vanishing follows from the exponential sequence together with the vanishing of $H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1})$, and hence we'll have:

$$\begin{aligned} H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) &\cong \mathbb{C} \\ H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}) &\cong \mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \\ H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) = H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}) &\cong \{0\} \\ &\vdots \end{aligned}$$

Doing a similar translation for the previously mentioned case of $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{O}_{\mathbb{C}})$ where $\mathcal{O}_{\mathbb{C}}$ is the sheaf of algebraic functions on $\mathbb{C} = \mathbb{A}^1$, it is not hard to see that:

$$\mathrm{Cl}(\mathbb{C}) = \frac{\operatorname{Div}(\mathbb{C})}{\operatorname{PDiv}(\mathbb{C})} \cong 0$$

[2]These two are linked via the natural exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}^*_X \to 0$, the details are left for [3]

which shall give us:

$$\begin{aligned} H^0(\mathbb{C}, \mathcal{O}_{\mathbb{C}}) &\cong \mathbb{C}[x] \\ H^1(\mathbb{C}, \mathcal{O}^*_{\mathbb{C}}) &\cong \{0\} \\ H^2(\mathbb{C}, \mathcal{O}_{\mathbb{C}}) &\cong \{0\} \\ &\vdots \end{aligned}$$

as every divisor can be represented by a principal divisor. We can interpret this as there being no impediments to using local information to construct meromorphic functions. Notice that the ring $\mathbb{C}[x]$ is a UFD; this is the algebraic characterization of a trivial class group (details in [3]), which intuitively can be thought of as there being a unique way of constructing a function from local information.

This answers the Riemann-Roch problem for holomorphic/meromorphic functions for $\mathbb{C}$ and $\mathbb{CP}^1$. Let us look at the next simplest nontrivial example. Let $X = \mathbb{C}/\Lambda$ where $\Lambda \subseteq \mathbb{C}$ is a lattice spanned by the $\mathbb{R}$-linearly independent elements 1 and τ and let $\mathcal{F} = \mathcal{M}_X$ (all meromorphic functions on X); X is called an *elliptic curve*, and so we shall label it as E. As E is compact, if $f \in \mathcal{M}_E$ then $\deg(\mathrm{div}(f)) = 0$. However, if $\deg(D) = 0$, this does not imply there exists an $f \in \mathcal{M}_E$ such that $\mathrm{div}(f) = D$. To see this, let's analyze $\mathcal{M}_E$ a little more closely. As $\mathbb{C}/\Lambda$ is the quotient by Λ, any meromorphic function must respect the Λ structure:

$$f(z + n + m\tau) = f(z) \qquad \forall n + m\tau \in \Lambda$$

Such a function is called *doubly periodic* or an *elliptic function*. Using this periodicity condition, it is simple to see that $\deg(\mathrm{div}(f)) = 0$ by looking at:

$$\frac{1}{2\pi i}\int_{\partial\mathcal{D}} \frac{f'(z)}{f(z)} dz = 0$$

where $\mathcal{D}$ is the fundamental domain[3], namely the opposing edges cancel out. We can modify this integral to find other properties, for example:

$$\frac{1}{2\pi i}\int_{\partial\mathcal{D}} z\frac{f'(z)}{f(z)} dz = n + m\tau \in \Lambda$$

which need not be zero, and so we see that if $D = \mathrm{div}(f)$:

$$\sum_i n_i p_i \equiv 0 \bmod \Lambda$$

where here $\sum_i n_i p_i \notin F^{\mathbb{Z}}(E)$ but $\sum_i n_i p_i \in E$, namely we are treating p_i as a complex number multiplied by n_i and summed. Certainly there is no such restriction on a general divisor $D \in \mathrm{Div}(E)$ of degree 0. We thus have shown that there is a surjective map:

$$\mathrm{Cl}^0(E) = \frac{\mathrm{Div}^0(E)}{\mathrm{PDiv}(E)} \twoheadrightarrow \mathbb{C}/\Lambda \tag{1.1}$$

where $\mathrm{Div}^0(E)$ are all divisors of degree 0. It is in fact an isomorphism, which comes from constructing an elliptic function using the Weierstrass $\wp$-function, the details are in [33, chapter 5]. Thus, we get the short exact sequence:

$$0 \to \mathbb{C}/\Lambda \to \mathrm{Cl}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

[3]If there are zeros/poles on the path, we simply shift around appropriately

often called the *Abel-Jacobi* exact sequence as $\mathbb{C}/\Lambda$ is a Jacobian variety, and a short exact sequence:

$$0 \to \mathrm{Cl}^0(E) \to \mathrm{Cl}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

Altogether, we get a short exact sequence and a commutative diagram:

$$0 \to \mathrm{PDiv}(E) \to \mathrm{Div}^0(E) \to E \to 0$$

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Div}^0(E) \arrow[r, hook] \arrow[d, two heads, "\pi"] & \mathrm{Div}(E) \arrow[r, "\deg"] \arrow[d, two heads, "\pi"] & \mathbb{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \mathrm{Cl}^0(E) \arrow[r, hook] & \mathrm{Cl}(E) \arrow[r, "\deg"] & \mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

In terms of cohomology, as the surjective map in equation (1.1) is an isomorphism:

$$\begin{aligned}
H^0(E, \mathcal{O}_E) &\cong \mathbb{C} \\
H^1(E, \mathcal{O}_E^*) &\cong E \times \mathbb{Z} \\
H^2(E, \mathcal{O}_E) = H^2(E, \mathcal{O}_E^*) &\cong \{0\} \\
&\vdots
\end{aligned}$$

Continuing on to other Riemann surfaces, the map in equation (1.1) generalizes to the *Abel-Jacobi map*, which is still an isomorphism onto a complex torus, the *Jacobian variety*; what we lose is that this torus is no longer the curve itself. This is the beginning of the study of Jacobian varieties (see [33, section 6.4]). The situation becomes even more complicated going to higher dimensions. The above strategy using divisors to find cohomology information becomes the beginning of the study of Weil and Cartier divisors[4]; the study of Cartier/Weil divisors can be thought of as the study of the obstruction to an object being constructed from codimension 1 objects, namely the study of $H^1(X, \mathcal{O}_X^*)$. This however does not comprehensively answer the problem of finding holomorphic/meromorphic functions on X. Overall, we get some useful information about the obstruction to finding global sections, but not necessarily concrete ways of finding $H^0(X, \mathcal{O}_X)$.

To answer the Riemann-Roch problem, a more granular approach to find which functions (of various kinds) exist on a space X must be taken. As divisors keep track of root/pole information in the 1-dimensional case, they are a great tool in classifying functions that satisfy the conditions given by a divisor. To make this precise, define a partial ordering on divisors $\mathrm{Div}(X)$ where (for now) X is either a Riemann surface or a smooth projective curve: let $D_2 \geq D_1$ if each coefficient of $D_2 - D_1$ is non-negative. Then we can consider:

$$\mathcal{O}_X(D)(X) = \{f \in \mathcal{M}_X(X)^* \mid \mathrm{div}(f) + D \geq 0\} \cup \{0\}$$

In other words, if $D = \sum_i n_i p_i$, then $\mathcal{O}_X(D)(X)$ is all meromorphic functions f where it has a pole at p_i of order at most n_i if $n_i > 0$ and roots at p_i of order at least $|n_i|$ if $n_i < 0$ (and $\mathcal{O}_X(D)$ is a [coherent] sheaf). From this, we can ask the Riemann-Roch problem for $(X, \mathcal{O}_X(D))$. This is answered by the famous *Riemann-Roch Theorem* where Riemann proved the Riemann inequality,

[4]and more generally Chow rings (see [3]). In section 6.3, we briefly cover how this characterizes rational cohomology under suitable algebraic assumptions

and Roch provided the necessary work to make it an equality, see [29]. Due to its generalizations, it is now known as *Riemann-Roch for curves* and is the equality:

$$\dim H^0(X, \mathcal{O}_X(D)) - \dim H^1(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g \tag{1.2}$$

where g is the genus of the Riemann surface or smooth projective curve (see [35] or [3] for defining the genus of a curve via scheme theory, or for a direct computational proof using normalization see [33]). In the original phrasing of the Riemann-Roch theorem, the group $H^1(X, \mathcal{O}_X(D))$ is replaced with $H^0(X, \mathcal{O}_X(K_X - D))^\vee$ where K_X is the canonical divisor of X; these groups are isomorphic by Serre Duality (see [3, section 6.3]). Then this space is identified with the space of meromorphic 1-forms ω with $\mathrm{div}(\omega) \geq D$ (either through some simple techniques from complex analysis, see Brill-Noether Reciprocity in either [29] or [33], or by combining Poincaré duality with de Rham's Theorem, see either [18] or [31]). Switching to the traditional notation $l(D) = \dim H^0(X, \mathcal{O}_X(D))$ and $i(D) = \dim H^0(X, \mathcal{O}_X(K_X - D))^*$ we get:

$$l(D) - l(K_X - D) = \deg(D) + 1 - g \qquad \text{(equiv.)} \qquad l(D) - i(D) = \deg(D) + 1 - g$$

This result will be proven in theorem 6.1.1. As we phrased the Riemann-Roch problem in the beginning, this formula would then be asking for the value of $l(D)$. There are cases where $l(K_X - D)$ disappears (for example whenever $\deg(D) > 2g - 2$, as then $\deg(K_X - D) < 0$), and there are many cases we can compute it easily enough. It is simple to compute the right hand side but in general it is harder to compute the left hand side. Thus, this formula becomes a new constraint that helps us find $l(D)$, namely it provides a lot of information towards answering the Riemann-Roch question for $(X, \mathcal{O}_X(D))$. There are a set of theorems known as *vanishing theorems* that specify under which conditions $l(K_X - D)$ (or more generally, higher cohomology groups) vanish (covered in section 3.5, but for all the details see [27]); under these conditions we recover $l(D)$ without a "fudge factor"[5].

This formula works for Riemann surfaces/projective curves. Let us work towards the intuition behind the generalization. The notation on the left hand side in equation (1.2) can be compactified by using the Euler-Poincaré characteristic (or simply the Euler characteristic): $\chi(X, \mathcal{O}_X(D)) = \sum_{i=0}^{\infty} (-1)^i \dim H^i(X, \mathcal{O}_X(D))$ so that:

$$\chi(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g$$

As X is a curve, $H^i(X, \mathcal{O}_X(D)) = 0$ for $i > 1$[6] giving the equality. The right hand side shall be drastically different, split into special invariants of vector bundles known as the *Chern character* and *Todd class* where the Chern character can be thought of as capturing some "algebraic" or "local" information given by D (this will generalize the $\deg(D)$) while the Todd class shall capture the "geometric" or "global" information (this will generalize the $1 - g$). In fact, the Euler Characteristic can easily be generalized to be an invariant on vector bundles by using sheaf cohomology instead of singular cohomology, and hence the Riemann-Roch theorem generalizes to a question about the existence of sections on vector bundles over compact complex manifolds, namely if $(X, \mathcal{F})$ is a compact complex manifold along with a vector bundle $\mathcal{F}$, then we want to compute:

$$H^0(X, \mathcal{F})$$

[5]The term "fudge factor" is used here informally for the correction terms that were endemic to the intersection theory of the Italian school. There is a connection in its use here and intersection theory due to the work of Fulton ([20]), but there is no space to cover the details in this paper

[6]This is a very common fact and can be proven in multiple different ways. Algebraic proof: [3, section 28.3], topological proof: [13, chapter 3], geometric proof: [12, chapter 3]

Note that there is a bijective correspondence between vector bundles and locally free sheaves (see [3, chapter 4] for the details), and hence we may treat $\mathcal{F}$ as either a vector bundle or a sheaf; the details for line bundles are given in section 1.2.1. The result of this will be:

$$\chi(X, \mathcal{F}) = \int_X \mathrm{ch}(\mathcal{F}) \smile \mathrm{td}(X) = \int_X \mathrm{ch}(\mathcal{F})\mathrm{td}(X)$$

where the cup product $\smile$ is often omitted. This equation is precisely the result of the *Hirzebruch-Riemann-Roch Theorem.*

1.2 Motivating the Generalization

Let us now motivate each component of the Hirzebruch-Riemann-Roch Theorem in more detail. The first section motivates how the divisor D can naturally be thought of as a line bundle, leading to the generalization to vector bundles. Then, the motivation for the Euler characteristic is covered to show how it inherently captures both geometric and arithmetic information. In the final section, we shall motivate how the use of characteristic theory will naturally give the generalization of $\deg(D)$ and the genus.

1.2.1 How do Vector Bundles enter the Picture

To motivate how vector bundles will enter the picture, it is worthwhile seeing how $\mathcal{O}_X(D)$ can be interpreted as a line bundle. To that end, recall that the projective curve $X = \mathbb{P}^1$ as defined in classical algebraic geometry only has constant global sections. Namely, $U_0, U_1 \cong \mathbb{A}^1$ are the usual cover with global sections (i.e. functions) $\mathcal{O}_X(U_0) = \mathbb{F}[x]$ and $\mathcal{O}_X(U_1) = \mathbb{F}[y]$, and the compatibility condition on $U_0 \cap U_1$ is $f(x) = g(1/x)$, namely:

$$\begin{tikzcd} U_0 & U_1 \\ U_0 \cap U_1 \arrow[u, hook] \arrow[r, dash, "\cong"] & U_0 \cap U_1 \arrow[u, hook] \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} \mathbb{A}^1 & \mathbb{A}^1 \\ \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \arrow[r, dash, "\cong"] & \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \end{tikzcd}$$

gives:

$$\begin{tikzcd} \mathcal{O}_X(U_0) \arrow[d] & \mathcal{O}_X(U_1) \arrow[d] \\ \mathcal{O}_X(U_0 \cap U_1) \arrow[r, dash, "\cong"] & \mathcal{O}_X(U_0 \cap U_1) \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} \mathbb{F}[x] \arrow[d] & \mathbb{F}[y] \arrow[d] \\ \mathbb{F}[x, 1/x] \arrow[r, dash, "\cong"] & \mathbb{F}[y, 1/y] \end{tikzcd}$$

where $x \mapsto 1/y$ and $1/x \mapsto y$. Then if $F([x' : y']) \in \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$ is a global function where $x = x'/y'$ and $y = y'/x'$ (when $y', x' \neq 0$ respectively), then $F([x : 1]) \to f(x) \in \mathcal{O}_X(U_0)$ and $F([1 : y]) \to g(y) \in \mathcal{O}_X(U_1)$ are the restrictions of the function to U_0 and U_1 respectively, and by the above diagram it must be that these functions map to the same element in the restriction, namely:

$$f(x) \to f(x) = g(1/x) = g(y) \leftarrow g(y)$$

but it is impossible that $f(x) = g(1/x)$ in $\mathbb{F}[x]$ unless f, g are the same constant. Thus, it must be that the only global functions on the projective line are constants:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = \mathbb{F}$$

The fundamental limitation here is that the restriction $f(x) = g(1/x)$ is very strong. What if we can loosen this restriction? For example what if

$$f(x) = xg\left(\frac{1}{x}\right)$$

Then we can in fact form linear homogeneous functions! For example consider

$$F([x' : y']) = ax' + by'$$

Then F maps to $f(x) = ax + b$ and $g(y) = a + by$. Then:

$$f(x) = ax + b = x\left(a + b\frac{1}{x}\right) = xg\left(\frac{1}{x}\right)$$

which works! More generally, the condition $f(x) = x^n g(1/x)$ allows for degree n homogeneous polynomials, and the homogeneity can happen for any number of variables. Denote the degree n-homogeneous polynomials for $\mathbb{P}^k$ by $\mathcal{O}_{\mathbb{P}^k}(n)(\mathbb{P}^k)$.

Now, recall that an n-dimensional vector bundle $\pi : E \to X$ over a field $\mathbb{F}$ can be defined by taking open sets $U_i \subseteq X$ such that $\pi^{-1}(U_i) \cong U_i \times \mathbb{F}^n$ and ensuring that for any intersection $U_i \cap U_j$ we have a gluing map:

$$g_{ij} : U_i \cap U_j \to \mathrm{GL}_n(\mathbb{F})$$

such that $g_{ij}g_{jk} = g_{ik}$ (the cocycle condition). When limiting to line bundles, the gluing map becomes:

$$g_{ij} : U_i \cap U_j \to \mathbb{F}^*$$

Intuitively, this gives a point-wise gluing condition on the bundle: if $v \in \pi^{-1}(U_i)$ and $w \in \pi^{-1}(U_j)$ represent the same point of the bundle over $U_i \cap U_j$, then:

$$v = g_{ij}w$$

Now, let us correspond $\mathcal{O}_{\mathbb{P}^1}(n)$ to a line bundle $\mathcal{L}(n)$. Let $\mathcal{L}(n)$ be locally trivial on U_0, U_1: $\Gamma(U_i, \mathcal{L}(n)) = \mathcal{O}_{\mathbb{P}^1}(n)(U_i)$. On their intersection define the gluing map:

$$g_{ij} \in \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j) = \left\{cx^k \;\middle|\; c \in \mathbb{F}^*,\ k \in \mathbb{Z}\right\}$$

where we take $\mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$ in place of $\mathbb{F}^*$ as these are the units (i.e. isomorphisms) on this intersection. For a given f, g to restrict to the same functions on the intersection, we now "twist" g to be:

$$f(x) = x^n g(1/x) \qquad x^n \in \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$$

The word "twist" comes from both complex analysis where z^n twists $\mathbb{C}$ onto itself around the branch point, and from algebraic geometry where the non-triviality of bundles is partly captured by characteristic classes that represent the obstruction to a global trivialization via twisting (we shall cover this intuition in chapter 2).

This shows how we can construct a line bundle $\mathcal{L}(n)$ for sheaves $\mathcal{O}_{\mathbb{P}^1}(n)$. The converse works as well: the details of the correspondence are given in [3, section 4.1] (especially how to transition from points of a bundle to sections of a sheaf). Overall, we see that $\mathcal{O}_{\mathbb{P}^1}(n)(\mathbb{P}^1)$ is the space of global sections of a line bundle; more generally $\mathcal{O}_{\mathbb{P}^1}(n)$ remembers all the sections over each open set U, as $\mathcal{O}_{\mathbb{P}^1}(n)$ is a sheaf.

With this intuition, we can see how to interpret $\mathcal{O}_X(D)$ as a line bundle. Choosing an affine cover U_i for X, we can define:

$$D|_{U_i} = \sum_{p \in U_i} n_p p$$

which is the representation of the divisor D in U_i. Then for each affine U_i, choose $s_i \in \mathcal{M}_X(U_i)$ so that $\langle s_i \rangle_{\mathcal{O}_X(U_i)} = \mathcal{O}_X(D)(U_i)$, i.e. s_i generates the local sections as an $\mathcal{O}_X(U_i)$-module, and:

$$\operatorname{div}(s_i) = -D|_{U_i}$$

so that s_i has roots/poles of the appropriate order. Such a section exists, for D is locally principal: refining the cover if necessary, we can certainly find an s_i with exactly zeros and poles at $-D|_{U_i}$ by taking a product of powers of local uniformizers at the finitely many points of $D|_{U_i}$, and it is an exercise in ring theory to show this is a generator (think of choosing a uniformizing parameter for a local ring).

To find an $f \in \mathcal{O}_X(D)(X)$ (i.e. a global section), we need to ensure we can glue together nicely on intersections. Consider (nonempty) $U_i \cap U_j$ and take the generators $s_i \in \mathcal{O}_X(D)(U_i)$ and $s_j \in \mathcal{O}_X(D)(U_j)$. It is not hard to see that $\operatorname{div}(s_i)|_{U_i \cap U_j} = \operatorname{div}(s_j)|_{U_i \cap U_j}$. Then certainly on $U_i \cap U_j$:

$$\operatorname{div}(s_i) = \operatorname{div}(s_j) = -D|_{U_i \cap U_j}$$

and so, as we are considering meromorphic functions, we can consider:

$$\operatorname{div}\left(\frac{s_j}{s_i}\right) = \operatorname{div}(s_j) - \operatorname{div}(s_i) = 0$$

As s_j/s_i has no zeros/poles on $U_i \cap U_j$, this means that it is holomorphic and nowhere vanishing on $U_i \cap U_j$. But then:

$$g_{ij} = \frac{s_j}{s_i} \in \mathcal{O}_X^*(U_i \cap U_j)$$

Showing that we have the desired transition functions to define a line bundle! Thus the g_{ij} glue the local trivializations into a line bundle whose sheaf of sections is $\mathcal{O}_X(D)$. In most cases (and in most spaces that will be considered in this paper), every line bundle $\mathcal{L}$ is induced by a divisor D; section 3.2 demonstrates this correspondence for Kähler manifolds, and for the correspondence more generally on (the appropriate type of) scheme see [3, section 4.3].

1.2.2 Motivating Euler Characteristic

The origins of the Euler characteristic stem from an observation by Leonhard Euler about the relation between the vertices, edges, and faces of a convex polyhedron. If V, E, F represents the respective quantities, then:

$$V - E + F = 2$$

For example, a cube has 8 vertices, 12 edges, and 6 faces, so $8 - 12 + 6 = 2$. Similarly, a connected planar graph holds the same property (counting the outside of the planar graph as a face). This sum is called the *Euler Characteristic* of the polyhedron. If the polyhedron was not convex, this relation need not hold; however, it is possible to have a non-convex polyhedron whose Euler characteristic is 2. This shows that the Euler characteristic captures some rough information about the geometry of the shape.

Generalizing this, we can form a CW-complex structure C on a topological space X and define the Euler characteristic as:

$$\chi(X, C) = \sum_{i=0}^{\infty} (-1)^i k_i$$

where k_i is the number of i-cells. One of the crowning results of algebraic topology in the 20th century is the invariance of this number under the choice of CW-complex. This led to the development of *singular homology* with the homology groups $H_i(X, \mathbb{Z})$ and the *Betti numbers* $b_i = \mathrm{rk}(H_i(X, \mathbb{Z}))$ and:

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i b_i = \sum_{i=0}^{\infty} (-1)^i \mathrm{rk}(H_i(X, \mathbb{Z}))$$

As this is a homological constant, it is sometimes called the *Euler-Poincaré* characteristic to distinguish it from the Euler characteristic for polyhedra, but this shall rarely be done in this paper. As homology groups are homotopy invariants, the Euler characteristic is invariant under homotopy. It is often computationally easier to work with the cohomology groups. By the universal coefficient theorem (see [13]), $\mathrm{rk}H^i(X, \mathbb{Z}) = \mathrm{rk}H_i(X, \mathbb{Z})$, giving the equivalent sum:

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i \mathrm{rk}(H^i(X, \mathbb{Z}))$$

How should we interpret the alternating nature of the sum when summing over homology/cohomology? The key idea is to notice that the alternating sum mimics the pattern when invoking the inclusion-exclusion principle. Let us go back to Euler's original $V - E + F$ formula and take for example a cube. Then we can think of each face as including the vertices and edges, and the edges as including the vertices. By counting all the faces, we have in fact double counted the edges. We thus subtract the sum of the edges. But this shall at the same time remove all the vertices from the sum, and hence we add them back in. This is exactly the inclusion-exclusion principle.

When converting to a cohomological perspective, we recover the inclusion-exclusion principle from the perspective of *Čech cohomology*. In particular, the intuition is to cover your space X in open sets, define the desired type of functions on each open set, and measure the obstruction to gluing these functions on intersections, intersection of intersections, and so on and so forth:

$$\begin{tikzcd}
\check{C}^0(\mathcal{U}, \mathcal{F}) \arrow[r, "d^0"] \arrow[d, equal] & \check{C}^1(\mathcal{U}, \mathcal{F}) \arrow[r, "d^1"] \arrow[d, equal] & \check{C}^2(\mathcal{U}, \mathcal{F}) \arrow[r, "d^2"] \arrow[d, equal] & \cdots \\
\prod_i \mathcal{F}(U_i) \arrow[r, "\delta^0"] & \prod_{i<j} \mathcal{F}(U_i \cap U_j) \arrow[r, "\delta^1"] & \prod_{i<j<k} \mathcal{F}(U_i \cap U_j \cap U_k) \arrow[r, "\delta^2"] & \cdots
\end{tikzcd}$$

Representing this visually:

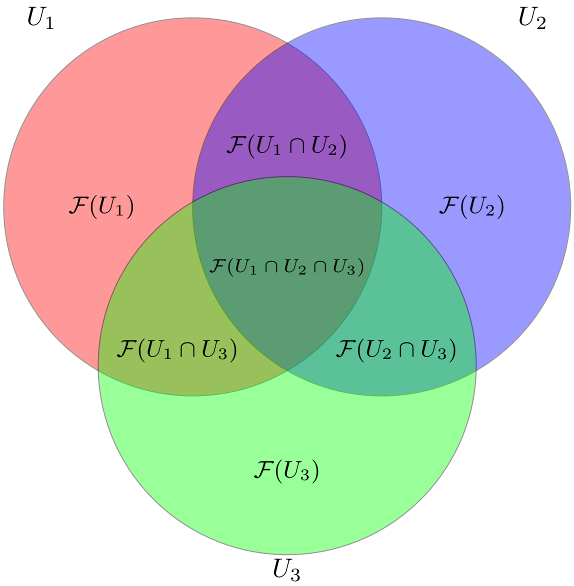

Figure 1.1: Venn diagram for an open cover with three sets U_1, U_2, U_3 and their sections.

How should we interpret what invariance is captured by the Euler characteristic? The case of the convex and non-convex polyhedron hints at a notion of curvature: indeed, by the Gauss-Bonnet theorem, if M is a compact Riemann surface (without boundary) with a Riemannian metric ω which induces a Gaussian curvature K, then the total curvature of M equals 2π times the Euler characteristic:

$$\chi(M) = \frac{1}{2\pi} \int_M K dA$$

This further generalizes to higher-dimensional Riemannian manifolds (as shall be mentioned shortly). Another interpretation from the theory of vector fields is that the number of zeros/poles as well as points with non-trivial curl are given by the topology of the surface, and hence they are represented by χ (see [13]).

These are very useful invariants, especially in 2 dimensions where the classification of Riemann surfaces up to homotopy is first broken down into the cases where the Euler characteristic is positive, zero, or negative (representing spherical, flat, and hyperbolic curvature respectively), and these values are strongly tied to the properties of vector fields on the surface. However, the Euler characteristic can be defined on any space admitting a cohomology theory, suggesting that a purely cohomological (characteristic-class) interpretation is the more suitable generalization. The following section covers these ideas.

1.2.3 Motivating Characteristic Class Theory

By the discussion in section 1.2.1, $\deg(D)$ and $1 - g$ should be interpreted as properties of a bundle. Let us return to the Gauss-Bonnet Theorem to find the motivation of this generalization. Recall that if M is a compact two-dimensional Riemannian manifold with empty boundary and K is the

Gaussian curvature of M, then:

$$\chi(M) = \frac{1}{2\pi}\int_M K dA$$

It was proven by Chern (the *Chern Theorem* or *Chern-Gauss-Bonnet Theorem*) that this generalizes to an even dimensional closed Riemannian manifold:

$$\chi(M) = \int_M e(\Omega)$$

where e is the *Euler class* (which is a characteristic class), and Ω is the curvature form (an $\mathrm{End}(TM)$-valued 2-form); $e(\Omega)$, given by the Pfaffian $\mathrm{Pf}(\Omega/2\pi)$, is then a $2n$-form, i.e. a section of:

$$\bigwedge^{2n} T^*M$$

More generally, if M is any closed smooth orientable n-dimensional manifold then:

$$\chi(M) = (e(TM), [M])$$

where $[M]$ is the fundamental class and $(-,-)$ is the Kronecker (evaluation) pairing.

1.2.1 Key takeaway: Euler Class and Top Chern Class

The Euler characteristic χ is already strongly linked to characteristic class theory as it can be seen as the pairing of the Euler class with the fundamental class. The details of the definition of the Euler class can be seen in [12], the important point to remember is that, for a complex manifold M, it is given by taking the *top Chern class* of its tangent bundle.

This definition does not consider non-trivial vector bundles, in particular when M is closed then $\chi(M)$ can be considered as the Euler characteristic over the trivial bundle $\mathbb{1}$, viewed as the constant sheaf $\underline{\mathbb{R}}$, as:

$$H^i(M, \mathbb{1}) = H^i(M, \mathbb{R})$$

(For the holomorphic theory below, $\mathbb{1}$ is instead read as the structure sheaf $\mathcal{O}_M$, which changes the value: see chapter 3.) By considering different bundles E, using sheaf cohomology we can define:

$$\chi(M, E) = \sum_{i=0}^{\infty} (-1)^i \dim H^i(M, \mathcal{O}_M(E))$$

The extra non-trivial algebraic information within the cohomology groups introduced by E motivates the focus on Chern classes for the generalization.

Thus, let us motivate what the Chern classes represent. Let $K_{\mathbb{C}}(X)$ be the collection of isomorphism classes of $\mathbb{C}$-vector bundles over X, and $H^*(X)$ the cohomology ring of X. First off, a Chern class c is a natural transformation $K_{\mathbb{C}}(-) \Rightarrow H^*$, showing that the Chern class c is a sum of ith Chern classes, one for each H^{2i}. Next, as H^i vanishes when $i > \dim(X)$, there can be at most finitely many ith Chern classes. As we are mapping into the cohomology ring, the Chern class will represent some obstruction information. Given a complex vector bundle E of dimension n, it is natural to ask whether it is possible to find k linearly independent sections of E (a k-frame), $1 \leq k \leq n$. The existence of a nontrivial ith Chern class shall represent an obstruction to having $n - i + 1$ globally linearly independent sections, in particular it will not be possible to extend an $(n-i)$-frame to an

$(n-i+1)$-frame. As this is a vector-bundle dependent invariant, the Chern classes represent some algebraic complexity of E in terms of the cohomology of X.

Given a Chern class $c_i \in H^{2i}(M,\mathbb{Z})$ (the $2i$ shall be explained in chapter 2), what would evaluating c_i represent? Given a representative T of a homology class in $H_{2i}(M,\mathbb{Z})$, we can define the *Chern number* $c_i(E)$ at T to be:

$$\langle c_i(E), [T]\rangle = \int_T c_i(E)$$

For the top Chern class, the Chern number counts the number of zeros/poles of a generic section of the vector bundle E, restricted to the submanifold T (see [12]). In terms of the earlier curvature interpretation, it is around zeros or poles (where poles can be considered as zeros in the reverse orientation) that we accumulate non-trivial curling/divergence. Let us apply this intuition where M is a Riemann surface and E a line bundle on M. Then $i=1$ so that $c_1(\mathcal{L}(D)) \in H^2(M,\mathbb{Z})$, for a line bundle $\mathcal{L}(D)$ associated to D. Then by the above intuition, we see that:

$$\int_M c_1(\mathcal{L}(D)) = \deg(D)$$

In a sense, this is the algebraic part of the Riemann-Roch theorem. If we let $D=0$, then:

$$\chi(M,\mathcal{O}_M) = 1-g$$

To get "$1-g$" from Chern classes, notice that as we can always define the tangent bundle TM of M, we can compute $c_1(TM)$. Then by the *Poincaré-Hopf Theorem* ([13] or [31]) we get that $\int_M c_1(TM) = 2-2g$. Thus we get:

$$\chi(M,\mathcal{O}_M) = \frac{1}{2}\int_M c_1(TM)$$

It should be noted that in this case, the Euler class is equal to the top Chern class, $c_1(TM) = e(TM)$, and so in this low-dimension case we could have just applied the Gauss-Bonnet Theorem.

When moving up to higher dimensions, it is important to limit to smooth (projective) varieties, namely because the higher cohomological information on closed manifolds of complex dimension ≥ 2 holds more information than there are geometric sub-manifolds; limiting to smooth varieties (later generalized by Grothendieck to proper morphisms) shall restrict the cohomology groups enough to use Chern classes in higher dimensions. To tie the geometric properties of manifolds to smooth varieties, we shall show in section 3.2 what conditions must be added on a manifold M so that it has to be a smooth variety. For now, we shall simply assume M is a smooth variety.

Note that the 1-dimensional case is exceptionally simple. For those who know the Riemann-Roch theorem for surfaces (e.g. see Hartshorne [35, chapter 5.1, theorem 1.6])

$$\chi(D) = \chi(0) + \frac{1}{2}D.(D-K)$$

they can verify that:

$$\chi(M,\mathcal{L}(D)) = \int_M \left[\frac{1}{2}c_1(\mathcal{L}(D))^2 + \frac{1}{2}c_1(\mathcal{L}(D))c_1(TM) + \frac{1}{12}\left(c_1(TM)^2 + c_2(TM)\right)\right]$$

showing that the ability to write χ in terms of TM and $\mathcal{L}(D)$ does extend in a non-trivial way to higher dimensions.

So the goal now is to find a connection between the Euler characteristic and some equation of Chern classes that works in higher dimensions. Two important decompositions of the Euler characteristic are:

$$\chi(M \times M', E \boxtimes E') = \chi(M, E) \cdot \chi(M', E')$$

which has a standard cohomology proof when E and E' are trivial, and in theorem 3.3.6 we shall show that:

$$\chi(M, E \oplus E') = \chi(M, E) + \chi(M, E')$$

Thus, we shall need two invariants to take into account the different algebraic behaviour of the manifold and vector bundles over direct products. The *Chern character* shall be the right invariant that captures the algebraic information for any manifold M and vector bundle E: we shall show that:

$$\mathrm{ch}(E \oplus E') = \mathrm{ch}(E) + \mathrm{ch}(E')$$

For the geometric counterpart, it was *John Arthur Todd* who found the geometric idea before Chern classes were even discovered ([26]). Hirzebruch put them to good use connecting them to Chern classes in defining the *Todd class* which shall have the property that:

$$\mathrm{td}(M \times M') = \mathrm{td}(M) \cdot \mathrm{td}(M')$$

where inputting $M, M', M \times M'$ should be thought of as inputting the tangent bundle of these manifolds (using $T(M \times M') \cong \mathrm{pr}_1^* TM \oplus \mathrm{pr}_2^* TM'$ and multiplicativity of td on direct sums of bundles). If we were working with trivial bundles, then we can simply write:

$$\chi(M, \mathcal{O}_M) = \int_M \mathrm{td}(M)$$

in analogy with how the Euler characteristic is obtained by integrating the Euler class. Adding the "correction term" given by the Chern character $\mathrm{ch}(-)$ to take into account the algebraic information, we finally come up with:

$$\chi(M, E) = \int_M \mathrm{ch}(E)\mathrm{td}(M)$$

Notice how this can be seen as a great generalization of the Chern-Gauss-Bonnet Theorem, where we are not limited to just looking at the curvature form, but had to limit to smooth varieties in order for the "data" to be well-defined. The complex-manifold form of the Chern-Gauss-Bonnet theorem can be recovered from the Hirzebruch-Riemann-Roch Theorem (see this link for an overview).

2

Necessary Vector Bundle Theory

This chapter will cover the necessary vector/fiber bundle theory that is used in Hirzebruch-Riemann-Roch. For an in-depth reading, there are my notes [12] that also include the algebraic K-theory background necessary to generalize the results of this chapter to Fulton's Intersection Theory. For a different reference, I recommend Hatcher's textbook on vector bundles [32] where he also covers geometric K-theory. Furthermore, this chapter shall freely use the bijective correspondence between locally free sheaves and vector bundles and that they share the same section/sheaf cohomology. I may convert between the language of sheaf theory and bundle theory without much warning.

Let $E \to X$ be a vector bundle, where X is paracompact and finite dimensional (either as a manifold or topologically)[1]. A natural set of questions to ask is what are the possible (global) sections on E with respect to X. A classical example of this question is to ask if there are any non-vanishing global sections on TS^2 with respect to S^2; the *Hedgehog Theorem* (or infamously, the *Hairy Ball Theorem*) asserts that no such global section can exist [13]. As local sections are easy to produce by taking advantage of local trivialization, the question is what type of *obstructions* are there in producing global sections. This is thus a cohomological problem.

By the work of mathematicians such as Chern, Thom, Whitney, and many others, there are natural transformations from the vector bundle functor to the cohomology functor of the underlying space:

$$c : K_{\mathbb{F}}(-) \Rightarrow H^*$$

where $K_{\mathbb{F}}(X)$ is the set of isomorphism classes of $\mathbb{F}$-vector bundles over X, and $\mathbb{F}$ is a field (usually $\mathbb{R}$ or $\mathbb{C}$). Without much trouble, this can be generalized to principal G-bundles; this shall be done for the important case of flag bundles and Grassmannians that will allow for both classification results and computational results. In this case, a *characteristic class* would be a natural transformation:

$$c : b_G(-) \Rightarrow H^*$$

[1] these conditions ensure we get maps between the cohomological groups and vector bundles while avoiding pathologies

where $b_G(-)$ is the functor taking X to the isomorphism classes of principal G-bundles over X. Note that there can be more than one characteristic class. Uncurling the categorical language, these natural transformations associate (up to isomorphism[2]) to each principal G bundle $P \to X \in b_G(X)$ an element $c(P) \in H^*$ such that

$$c(f^*(P)) = f^*(c(P))$$

that is, the pullback of the principal bundle commutes with the pullback of cohomology classes. The characteristic classes will be the right notion to capture the cohomological information, in particular when working over real and complex vector bundles, the cohomology of the classifying space (BO for real, BU for complex) is given by the *Stiefel-Whitney* and *Chern* classes (see [12]):

$$H^*(BO(n); \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})[w_1, ..., w_n] \qquad \text{and} \qquad H^*(BU(n); \mathbb{Z}) \cong \mathbb{Z}[c_1, ..., c_n] \tag{2.1}$$

In this paper, our base spaces are (complete) varieties[3], so we shall mainly focus on complex vector bundles and principal $\mathrm{GL}_n(\mathbb{C})$-bundles. The complex case works with *Chern class* and the *Pontryagin class*: the Chern class is the natural characteristic class of complex vector bundles (as seen in equation (2.1)), while the Pontryagin classes can be thought of as the characteristic classes of the complexification of a real vector bundle.

The Chern classes are central to the Hirzebruch-Riemann-Roch theorem as they are directly used in defining the Chern character and the Todd class. On the other hand, the Pontryagin classes shall be useful in that they will allow us to classify oriented manifolds up to cobordism, a result that will be necessary for only requiring checking χ agrees with $\int_X \mathrm{ch}(E)\mathrm{td}(T_X)$ on complex projective space.

Another characteristic class that shall not be covered but is important to know as it is one of the generalizations of the Euler characteristic is the *Euler class*. As the name suggests, the Euler class is linked to the Euler characteristic, in particular if M is an oriented closed n-manifold, TM the tangent bundle, $e(TM) \in H^n(M; \mathbb{Z})$, and $[M] \in H_n(M; \mathbb{Z})$ the fundamental class of M, then:

$$e(TM) \frown [M] = \chi(M)$$

or, given the de Rham isomorphism (see [31]), re-labeling the notation $e(TM)$ to the corresponding *Euler form*:

$$\int_M e(TM) = \chi(M)$$

If M were given a Riemannian metric, then the classical generalization of the Gauss-Bonnet theorem for manifolds without boundary (the *Chern Theorem*) lets us replace TM with Ω, the curvature form, giving:

$$\int_M e(\Omega) = \chi(M)$$

In other words, the Chern Theorem finds an alternative way of evaluating the Euler class at the fundamental class that relies on the curvature form Ω. The Hirzebruch-Riemann-Roch Theorem can be seen as finding another alternative evaluation of the Euler class when the underlying space M has enough structure for its characterization to be done using the Chern classes.

Finally, the definition of Chern class and Pontryagin class taken here can be thought of as the "classical" definition. In algebraic geometry, there is a natural "intersection theory" perspective of the Chern class that maps the Chow group $A_n(-)$ to $A_{n-k}(-)$. There is a natural way in which this definition is linked to the classical definition of Chern class via the Chern character, and we shall briefly bring it up in the context of the generalization of the Hirzebruch-Riemann-Roch to the *Grothendieck-Riemann-Roch theorem*; see section 6.3.

[2] even up to homotopy, see [12]
[3] in section 6.3 we shall extend to proper schemes

2.1 Definitions and Key Results

For reference, here is the definition of a vector bundle and principal bundle we shall use (this differs from the definition Hirzebruch gives via gluing of a vector bundle with structure groups, but matches contemporary textbooks such as those by Hatcher and John/Jeffrey M. Lee):

Definition 2.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \to B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber, is called a *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is homeomorphic to F, or in particular

$$\pi^{-1}(b) \cong \{b\} \times F$$

2. (local trivialization) For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes

$$\begin{tikzcd} \pi^{-1}(U) \arrow[r, "\sim"] \arrow[d, "\pi"'] & U \times F \arrow[dl, "\pi_1"] \\ U & \end{tikzcd}$$

When clear from context, the fiber bundle is usually denoted as E.

Definition 2.1.2: G-bundle

Let $E \to X$ be a fiber bundle and G a topological group that acts faithfully on the fibers. Then a G-*atlas* is a collection of local trivializations $\{U_i, \varphi_i\}_i$, where $\{U_i\}_i$ covers X, such that for any index i, j on the overlap $U_i \cap U_j$ (if it is non-empty):

$$\varphi_j \circ \varphi_i^{-1} : (U_i \cap U_j) \times F \to (U_i \cap U_j) \times F$$
$$\varphi_j \circ \varphi_i^{-1}(p, f) = (p, g_{ij}(p)f)$$

with the added conditions (cocycle conditions):

1. $g_{ii} = \text{id}$
2. $g_{ij} = g_{ji}^{-1}$
3. $g_{ik} = g_{jk}g_{ij}$

Two G-atlases are equivalent if their union is a G-atlas. A G-*bundle* is a fiber bundle with an equivalence class of G-atlases.

Definition 2.1.3: Principal G-Bundle

If $F = G$ is a topological group and $P \to X$ is a fiber bundle over X where there is a continuous right group action $P \times G \to P$ that preserves the fibers of P and is free and transitive on each fiber, then it is called a *principal G-bundle* (or *principal bundle* if G is clear from context). In other words, a principal G-bundle is a G-bundle where the fibers are G.

Each fiber is called a *G-torsor* (which is a space where the action $G \curvearrowright F_x$ is free and transitive, hence "is" the group G but does not intrinsically have a notion of identity element)

Definition 2.1.4: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then it is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle* or *complex vector bundle.*

The following fact shall freely be used (proven in [12]):

Theorem 2.1.5: Homotopy Invariance Vector Bundles

Let $p : E \to B$ be a vector bundle and $f_0, f_1 : A \to B$ be homotopic maps. Then if A is paracompact,

$$f_0^*(E) \cong f_1^*(E)$$

***Proof*:**
[12, section 1.2]

Theorem 2.1.6: Classifying Vector Bundles

Let X be a paracompact space. Then $\mathrm{Vect}^n_{\mathbb{F}}(X)$, the set of isomorphism classes of n-dimensional $\mathbb{F}$-vector bundles over X, is in correspondence with the set of homotopy classes of maps $X \to \mathrm{Grass}_n$, denoted $[X, \mathrm{Grass}_n]$. In other words:

$$\mathrm{Vect}^n_{\mathbb{F}}(X) \cong [X, \mathrm{Grass}_n]$$

In particular, if $E \to X$ is an n-dimensional vector bundle, it is isomorphic to $f^*(E_n)$, where $E_n \to \mathrm{Grass}_n$ is the tautological bundle, for some $f : X \to \mathrm{Grass}_n$ where E determines f up to homotopy.

***Proof*:**
[12, section 1.2]

This motivates the following natural definition:

Definition 2.1.7: Universal Bundle

The infinite Grassmannians Grass_k are called the *classifying spaces*; for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted $BO(k), BU(k)$, and $BSp(k)$ respectively. The tautological bundle $E_k \to \mathrm{Grass}_k$ is called the *universal bundle*.

2.2 The Splitting Principle

One of the most important results about vector bundles is that there is a pullback map which allows them to split into a direct sum of line bundles. As a consequence, the total Chern class (which is representable as a polynomial) can be split into a product of Chern classes of line bundles (which can be represented by linear terms). The key cohomological result is the following fiber-local to global theorem proved by Leray and Hirsch:

Theorem 2.2.1: Leray-Hirsch Theorem

Let R be a commutative ring and $p : E \to X$ be a fiber bundle. In [13] it was shown $H^*(E; R)$ is a $H^*(X; R)$-module where

$$\alpha \cdot \beta := p^*(\alpha) \smile \beta$$

with $\alpha \in H^*(X)$ and $\beta \in H^*(E)$. If

1. for each fiber F the inclusion $F \hookrightarrow E$ induces a surjection on $H^*(-; R)$
2. $H^n(F; R)$ is a free R-module of finite rank for each n.

Then $H^*(E; R)$ is a free $H^*(X; R)$-module. A basis for the $H^*(X; R)$-module $H^*(E; R)$ can be obtained by choosing any set of elements in $H^*(E; R)$ that map to a basis for $H^*(F; R)$ under the map induced by the inclusion.

***Proof*:**
see [13].

Theorem 2.2.2: Splitting Principle

Let $\pi : E \to B$ be a complex vector bundle. Then there exists a space $F(E)$ and a map $p : F(E) \to B$ such that the pullback $p^*(E) \to F(E)$ splits as a direct sum of line bundles, and

$$p^* : H^*(B; \mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z})$$

is injective

***Proof*:**
Consider the pullback $\mathbb{P}(\pi)^*(E)$ of E via the map $\mathbb{P}(\pi) : \mathbb{P}(E) \to B$ where $\mathbb{P}(E)$ is the space of lines through the origin in all fibers of E. This pullback contains a natural one-dimensional subbundle

$$L = \{(\ell, v) \in \mathbb{P}(E) \times E \mid v \in \ell\}$$

An inner product on E pulls back to an inner product on the pullback bundle, and thus a splitting of the pullback as a sum $L \oplus L^{\perp}$ with the orthogonal bundle $L^{\perp}$ having dimension one less than E. Then by applying the Leray-Hirsch theorem to $\mathbb{P}(E) \to B$, $H^*(\mathbb{P}(E); \mathbb{Z})$ is the free $H^*(B; \mathbb{Z})$-module with basis $1, x, \ldots, x^{n-1}$, where $x = c_1(L)$ is the first Chern class of the tautological line bundle L (obtained from the classifying map to $\mathbb{C}\mathrm{P}^{\infty}$ and the chosen generator of $H^2(\mathbb{C}\mathrm{P}^{\infty}; \mathbb{Z})$, as in the next section) and $n = \operatorname{rank} E$, and in particular the induced map $H^*(B; \mathbb{Z}) \to H^*(\mathbb{P}(E); \mathbb{Z})$ is injective since one of the basis elements is 1.

This construction can be repeated with $L^{\perp} \to \mathbb{P}(E)$ in place of $E \to B$, until we get the splitting, completing the proof.

The following will be important when applying the Euler characteristic to $F(E)$ for the Hirzebruch-Riemann-Roch:

Theorem 2.2.3: Splitting Principle Over Varieties

Let $E \to B$ be an algebraic vector bundle where B is a projective variety. Then $F(E)$ is also a projective variety.

***Proof*:**
[4]. A weaker version of this theorem is cited later in the paper (see theorem 3.2.14).

2.3 Chern and Pontryagin Classes

The axiomatic approach due to Grothendieck shall be taken:

Theorem 2.3.1: Existence and Uniqueness of Chern Classes

Let $E \to X$ be a complex vector bundle. Then there exists a sequence of functions $c_1, c_2, \ldots$ where $c_i(E) \in H^{2i}(X; \mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $c_i(f^*(E)) = f^*(c_i(E))$ for a pullback $f^*(E)$
2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \cdots \in H^*(X; \mathbb{Z})$
3. $c_i(E) = 0$ if $i > \dim E$
4. Given $L \to \mathbb{C}\mathrm{P}^{\infty}$ is the canonical line bundle, $c_1(L)$ is a generator of $H^2(\mathbb{C}\mathrm{P}^{\infty}; \mathbb{Z})$ specified in advance.

***Proof*:**
see [12] for a full proof. The key idea is to invoke the Leray-Hirsch Theorem for existence and the splitting principle for uniqueness. In fact, we get a bit more out of the Leray-Hirsch theorem: by the classification of line bundles (theorem 2.1.6) there is a map $f : \mathbb{P}(E) \to \mathbb{C}\mathrm{P}^{\infty}$ classifying the tautological line bundle on $\mathbb{P}(E)$, which induces a cohomology pullback $f^* : H^*(\mathbb{C}\mathrm{P}^{\infty}; \mathbb{Z}) \to H^*(\mathbb{P}(E); \mathbb{Z})$, in particular $f^* : H^2(\mathbb{C}\mathrm{P}^{\infty}; \mathbb{Z}) \to H^2(\mathbb{P}(E); \mathbb{Z})$ which pulls back the generator $\alpha \in H^2(\mathbb{C}\mathrm{P}^{\infty}; \mathbb{Z})$ to an element $f^*(\alpha) = x$. Then the elements $1, x, \ldots, x^{n-1}$ are the generators of

$H^*(\mathbb{P}(E);\mathbb{Z})$ as a free $H^*(X;\mathbb{Z})$-module, and if $\alpha\beta := \alpha \smile \beta$ then we get a unique relationship:

$$x^n - c_1(E)x^{n-1} + \cdots + (-1)^n c_n(E) \cdot 1 = 0$$

which gives a *relationship* between the Chern classes.

Uniqueness is given by the splitting principle, namely if there were another such relationship, then as the above equation must pullback and split, the Chern classes must match.

For the choice of normalization, note that for property (4), the canonical line bundle over $\mathbb{C}\mathrm{P}^1$ could have been used instead of $\mathbb{C}\mathrm{P}^\infty$ as $\mathbb{C}\mathrm{P}^1 \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ induces an isomorphism

$$H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong H^2(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$$

and we know by some cohomology theory that we can choose any hyperplane for the generator of $H^2(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$, hence $c_1(L) \in H^2(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$ becomes a choice of hyperplane.

Note that this definition differs slightly from what Hirzebruch presents: he replaces the second axiom with the following:

(2) If $\xi_1, \ldots, \xi_q$ are continuous $\mathbf{U}(1)$-bundles over X then

$$c(\xi_1 \oplus \cdots \oplus \xi_q) = c(\xi_1) \smile \cdots \smile c(\xi_q).$$

The real equivalent has more subtlety due to the fact that real manifolds and vector bundles have a choice of orientation, and hence the natural cohomological setting to work in is over $\mathbb{Z}/2\mathbb{Z}$-coefficients and classify $H^*(BO;\mathbb{Z}/2\mathbb{Z})$ where we take $\mathbb{Z}/2\mathbb{Z}$ to get rid of the choice of orientation. This uses *Stiefel-Whitney* classes which we did not touch on in this paper. To classify $H^*(BO;\mathbb{Z})$, we essentially take the complexification of a real vector bundle, which gives us the following best possible result:

Definition 2.3.2: Pontryagin Class

Let $E \to B$ be a real vector bundle. Complexify it by adding the almost complex structure to $E \oplus E$:

$$E^{\mathbb{C}} = E \oplus E \qquad i(x,y) = (-y,x)$$

Then a *Pontryagin class* $p_i(E) \in H^{4i}(B;\mathbb{Z})$ is given by:

$$p_i(E) = (-1)^i c_{2i}(E^{\mathbb{C}}) \in H^{4i}(B;\mathbb{Z})$$

To characterize Pontryagin classes, we state the following:

Proposition 2.3.3: Characterizing Grassmannian Cohomology

All torsion of $H^*(BO(n);\mathbb{Z})$ is 2-torsion; if T is the torsion subgroup then:

$$\frac{H^*(BO(n);\mathbb{Z})}{T} \cong \mathbb{Z}[p_1, \ldots, p_k] \qquad k = \lfloor n/2 \rfloor$$

***Proof*:**
This is not important for our current work, so the proof is deferred to [12] or [32]

What shall be important is the use of Pontryagin classes in the theory of cobordism in chapter 5.

Pontryagin and Chern classes relate through the following relation:

Theorem 2.3.4: Chern and Pontryagin Relation

The Chern classes c_i of the almost complex manifold X are related to the Pontryagin classes p_i of X (regarded as a differentiable manifold) by:

$$p = \sum_{i=0}^{\infty} (-1)^i p_i = \sum_{i=0}^{\infty} c_i \sum_{j=0}^{\infty} (-1)^j c_j.$$

***Proof*:**
untangle and use definition 2.3.2.

2.4 Chern Character and K-Theory

Fix a base space X and let $K(X) = K_{\mathbb{C}}(X)$ be the set of isomorphism classes of complex vector bundles on X. Let $c(-) : K(X) \to H^*(X; \mathbb{Z})$ be the Chern class. Recall that

$$c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$$

In particular, it brings the addition in $K(X)$ to multiplication in H^*. The collection $K(X)$ can be completed to form a group, also denoted $K(X)$, and it is known as the *Grothendieck K-group*. The tensor product gives it a natural ring structure. By the above construction, we see that $c(-)$ is *not* a ring homomorphism for it is not the case that $c_1(E \otimes F) = c_1(E) + c_1(F)$ (hint: at least one of E, F must have rank ≥ 2 for the counter-example, see [12]). From the Chern class $c(-)$, we can construct a new map known as the *Chern character*

$$\mathrm{ch}(-) : K(X) \to H^*(X; \mathbb{Z}) \otimes \mathbb{Q}$$

which *will* be a ring homomorphism, provided we add rational coefficients.

By the splitting principle, it suffices to work with line bundles. The total Chern class is:

$$c(L) = 1 + c_1(L)$$

where $c_1(L) \in H^2(X; \mathbb{Z})$. Then it is a worthwhile exercise to show that for *line bundles*[4] (not for general vector bundles!):

$$c_1(L_1 \otimes L_2) = c_1(L_1) + c_1(L_2)$$

which is addition. To turn this into multiplication, we can take:

$$\mathrm{ch}(L) = e^{c_1(L)} = 1 + c_1(L) + \frac{c_1(L)^2}{2!} + \frac{c_1(L)^3}{3!} + \cdots$$

where the power should be taken as repeated cup product. This fixes the multiplicative problem as:

$$\mathrm{ch}(L_1 \otimes L_2) = e^{c_1(L_1) + c_1(L_2)} = e^{c_1(L_1)} \cdot e^{c_1(L_2)}$$

[4] use the correspondence between line bundles and cohomology classes, see [12]

But this introduces rational coefficients: the division by $k!$ is not possible in $H^*(X;\mathbb{Z})$, and so the expression is not well-defined over the integers. Thus, we must define:

$$H^*(X;\mathbb{Q}) := H^*(X;\mathbb{Z}) \otimes \mathbb{Q}$$

in which division by $k!$ is possible (and torsion is eliminated, see [30]), and hence $\mathrm{ch}(L) \in H^*(X;\mathbb{Q})$ is well-defined. Then, for an arbitrary vector bundle E, the splitting principle gives a map $p : F(E) \to X$ such that the pullback splits as

$$p^*(E) = L_1 \oplus L_2 \oplus \cdots \oplus L_n$$

and we define:

$$\mathrm{ch}(E) = \mathrm{ch}(L_1) + \mathrm{ch}(L_2) + \cdots + \mathrm{ch}(L_n)$$

where the $c_1(L_i) = t_i$ are the *Chern roots* of $c(E)$; since $p^* : H^*(X;\mathbb{Q}) \to H^*(F(E);\mathbb{Q})$ is injective, this uniquely determines $\mathrm{ch}(E)$ as an element of $H^*(X;\mathbb{Q})$. By construction:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2)$$

Giving the desired algebraic properties. However, $\mathrm{ch}(E)$ is dependent on the Chern classes $c_1(L_i)$ where i goes over the Chern roots, as opposed to $c_j(E)$ where j increases in dimension. This is easily fixed with a bit of combinatorics. Namely, $c_j(E)$ is given by:

$$c_j(E) = \sigma_j(c_1(L_1), ..., c_1(L_n))$$

where σ_j is the jth elementary-symmetric polynomial. Define the *power sums* to be:

$$s_k(x_1, ..., x_n) = \sum_{i=1}^{n} x_i^k$$

Then the *Newton identities* (see this link) show that (via elementary computations):

$$\begin{aligned}
s_1 &= \sigma_1 \\
s_2 &= \sigma_1^2 - 2\sigma_2 \\
s_3 &= \sigma_1^3 - 3\sigma_2\sigma_1 + 3\sigma_3 \\
s_4 &= \sigma_1^4 - 4\sigma_2\sigma_1^2 + 4\sigma_3\sigma_1 + 2\sigma_2^2 - 4\sigma_4 \\
&\vdots
\end{aligned}$$

From these we get:

$$\mathrm{ch}(E) = \sum_{i=1}^{n} \left(\sum_{k=0}^{\infty} \frac{t_i^k}{k!} \right) = \sum_{k=0}^{\infty} \frac{1}{k!} s_k(t_1, ..., t_n)$$

where, by the Newton identities, each $s_k(t_1, ..., t_n)$ can be written as a polynomial in the elementary symmetric polynomials $\sigma_j(t_1, ..., t_n) = c_j(E)$. Writing out the first few terms we get:

$$\mathrm{ch}(-) = \dim_{\mathbb{C}}(-) + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3) + \cdots$$

giving us a power series purely in terms of the Chern classes of E! Note that the Chern character captures additional information the Chern classes miss (though, having rational coefficients, it in turn loses torsion information that the integral Chern classes retain): the Chern classes are *stable*, meaning they are invariant when taking the direct sum with trivial bundles, however $\mathrm{ch}(-)$ has $\dim_{\mathbb{C}}(-)$ in its 0th position, in particular it remembers the rank of the bundle[5]. To connect this to the K-group, we require the following:

Proposition 2.4.1: Extension and Tensors With Chern Character

1. let E'', E, E' be vector bundles such that:
$$0 \to E'' \to E \to E' \to 0$$
then:
$$\mathrm{ch}(E) = \mathrm{ch}(E') + \mathrm{ch}(E'')$$
2. Let E_1, E_2 be vector bundles. Then:
$$\mathrm{ch}(E_1 \otimes E_2) = \mathrm{ch}(E_1) \cdot \mathrm{ch}(E_2)$$
where $\cdot$ is the cup product.

***Proof*:**
exercise.

Thus, there is a natural map:

$$\mathrm{ch} : K(X) \to H^*(X; \mathbb{Q})$$

This shows that $\mathrm{ch}(-)$ is the right notion for capturing the cohomological information of vector bundles which respects the algebraic structure of $K(X)$. In chapter 4, we shall show that $\mathrm{ch}(E)$ captures the bundle information, while $\mathrm{td}(X)$ (defined in chapter 4) shall capture the relevant information from the underlying space.

A natural question to ask is whether, given a continuous map $f : X \to Y$, the Chern character $\mathrm{ch}(-)$ commutes with the pullback f^*, namely whether the following diagram commutes:

$$\begin{tikzcd} K(Y) \arrow[r, "f^*"] \arrow[d, "\mathrm{ch}"'] & K(X) \arrow[d, "\mathrm{ch}"] \\ H^*(Y; \mathbb{Q}) \arrow[r, "f^*"] & H^*(X; \mathbb{Q}) \end{tikzcd}$$

This is indeed the case: $\mathrm{ch}(-)$ is natural with respect to pullbacks. The situation changes when we instead take coherent sheaves and work with the proper pushforward rather than the pullback: the analogous diagram fails to commute, and what fixes the commutativity is the introduction of the Todd class. The details are left for section 6.3:

[5]As a fun aside, if $X = \mathbb{C}\mathrm{P}^n$, then the Hilbert polynomial can be recovered by the Chern character, and vice-versa; see [6]

$$\begin{tikzcd}
K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(Y) \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TY)"] \\
A_*(X;\mathbb{Q}) \arrow[r, "f_*"] & A_*(Y;\mathbb{Q})
\end{tikzcd}$$

where $A_*(-;\mathbb{Q})$ is the *Chow group* with rational coefficients.

3

Cohomology of Varieties: New Euler Characteristics

The Euler characteristic is in a sense “too rough” of an invariant for varieties. We need to capture some of the complex structure to find the right invariants. In particular, we shall refine the usual de Rham/singular cohomology into *Dolbeault* cohomology which shall capture the obstruction to a function/local-section being *holomorphic*. This shall form a new collection of cohomology groups (depending on the degree) that will allow us to define new Euler-characteristics that generalize the topological Euler characteristic.

We start by developing some basic Hodge theory. Some key facts that shall be shown are that the Dolbeault cohomology groups are finite dimensional by using the fact that elliptic operators have finite-dimensional kernels (see theorem 3.1.8), that the de Rham theorem has a natural generalization to the Dolbeault Theorem, that all forms have a harmonic representative, and that the cohomology classes of line bundles are precisely the cohomology classes of type $(1, 1)$; the reader who is comfortable with this result can skip over most of the technical aspects of this section.

Then, as every smooth projective manifold embeds into $\mathbb{CP}^n$ and hence is defined by algebraic equations by Chow’s theorem ([34]), there is a natural compatibility between the Riemann metric, symplectic form, and complex structure. This shall define Kähler manifolds, which are almost varieties; after defining what it means to be positive (essentially, ensuring some line bundle is very ample, think of $\mathcal{O}(n)$ vs $\mathcal{O}(-n)$ for $n \in \mathbb{N}$) we shall show that all Hodge manifolds are (projective) smooth varieties. This will let us translate the results of Dolbeault cohomology (ex. the finiteness of the cohomology groups) over to algebraic geometry. Note that in the proof of Grothendieck-Riemann-Roch, this is replaced with the fact that $H^i(M, \mathcal{F})$ is finite-dimensional for M projective and $\mathcal{F}$ coherent (see my notes [3] or Hartshorne [35]).

We then define new Euler-characteristics, and generalize to defining the *virtual characteristic*, which plays the role of the Euler characteristic of an intersection of submanifolds even when that intersection does not exist as a submanifold. We shall end off by mentioning some important vanishing theorems that would allow us to compute $H^0(M, \mathcal{F})$.

3.1 Dolbeault Cohomology

We shall assume the basic knowledge of differential geometry that can be found in [31] or [7], and of complex analysis that can be found in [33] or [24]. This section shall be very rapid to cover the key ideas necessary to define the Euler characteristics, sometimes omitting some proofs that are better covered in a full text in Hodge Theory.

Let M be a real manifold. Then we can take the fiber-wise tensor product $TM \otimes \mathbb{C}$ to complexify the tangent bundle. If $J : TM \to TM$ is an endomorphism such that $J^2 = -1$[1], then $TM \otimes \mathbb{C}$ can be decomposed into:

$$TM \otimes \mathbb{C} \cong T^{1,0}M \oplus T^{0,1}M$$

where

$$T^{1,0}M = \{x - iJx \mid x \in TM\} \qquad \text{and} \qquad T^{0,1}M = \{x + iJx \mid x \in TM\}$$

Thus, there is a natural decomposition of tangent bundles into holomorphic and anti-holomorphic components. The choice of J on TM makes (M, J) an *almost complex manifold*. All complex manifolds are almost complex, but the converse is not the case (see the Newlander-Nirenberg theorem in [4]). Much (but not all) of the work in this section works just as well on almost-complex manifolds; this paper will stick to complex manifolds but the reader should feel free to work with almost-complex manifolds.

For a moment, let M be a one-dimensional complex manifold. As holomorphic mappings respect complex and anti-complex structure, any 1-form with complex coefficients can be written uniquely as a sum:

$$f dz + g d\overline{z}$$

Let $\Omega^{1,0}$ represent 1-forms of type $(1, 0)$, and $\Omega^{0,1}$ represent 1-forms of type $(0, 1)$. The spaces $\Omega^{1,0}$ and $\Omega^{0,1}$ are both stable under holomorphic coordinate changes (use Cauchy-Riemann). A similar pattern takes shape in higher dimensions. Let M be n-dimensional. Like before, any 1-form with complex coefficients can be written as:

$$\sum_{j=1}^{n} (f_j dz^j + g_j d\overline{z}^j)$$

Define:

$$\Omega^{p,q} = \underbrace{\Omega^{1,0} \wedge \cdots \wedge \Omega^{1,0}}_{p \text{ times}} \wedge \underbrace{\Omega^{0,1} \wedge \cdots \wedge \Omega^{0,1}}_{q \text{ times}}$$

Then we know that for every $r \in \mathbb{N}$, every smooth r-form on M (with $\mathbb{C}$-coefficients) can be written as the sum of (p, q)-forms where $p + q = r$ (as the bases of these forms are wedge products of the lower dimensions, see [31], [13]).

Now, let E^k be the space of all complex differential forms of total degree k. Then

$$E^k = \Omega^{k,0} \oplus \Omega^{k-1,1} \oplus \cdots \oplus \Omega^{1,k-1} \oplus \Omega^{0,k} = \bigoplus_{p+q=k} \Omega^{p,q}$$

[1] Note this forces M to be even-dimensional

If M is a complex manifold, then:

$$\Omega^k(M, \mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}$$

Note that the direct sum decomposition is stable under holomorphic coordinate changes, hence it determines a vector bundle decomposition. From this direct sum, there are natural projections for each direct summand:

$$\pi^{p,q} : E^k \to \Omega^{p,q}$$

Now, define $d : \Omega^r \to \Omega^{r+1}$ in the natural way where:

$$d(\Omega^{p,q}) \subseteq \bigoplus_{r+s=p+q+1} \Omega^{r,s}$$

Definition 3.1.1: Dolbeault Operators

$$\partial = \pi^{p+1,q} \circ d \qquad \text{and} \qquad \overline{\partial} = \pi^{p,q+1} \circ d$$

In coordinates, let:

$$\alpha = \sum_{|I|=p, |J|=q} f_{IJ} dz^I \wedge d\bar{z}^J \in \Omega^{p,q}$$

where I and J are multi-indices. Then:

$$\partial\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial z^\ell} dz^\ell \wedge dz^I \wedge d\bar{z}^J$$

and

$$\bar{\partial}\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial \bar{z}^\ell} d\bar{z}^\ell \wedge dz^I \wedge d\bar{z}^J.$$

When M is a complex manifold (equiv. it has a complex structure), then:

$$d = \partial + \bar{\partial}$$

$$\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

These identities can fail on a general almost complex manifold; the *Newlander-Nirenberg theorem* says that an almost complex structure for which they hold is induced by a complex structure.

3.1.2 For those who know Hodge Theory Note that these operators require M to be complex, not just almost complex. There are similar results for almost-complex manifolds, see [4] or [17].

Thus, we see there is some interesting cohomology that can occur; this defines *Dolbeault cohomology*. The symbol ∂ suggests that they are analogous to d, and indeed the Poincaré lemma generalizes to complex manifolds.

Let us define the Dolbeault cohomology. Notice that $\overline{\partial} f = 0$ if f is holomorphic, therefore it is in some sense the natural choice; the other choice (∂) keeps track of the anti-holomorphic information. Thus, define:

Definition 3.1.3: Dolbeault Cohomology

$$H^{p,q}_{\overline{\partial}}(M) = \frac{\ker(\overline{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\mathrm{im}(\overline{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

Where the $\overline{\partial}$ is dropped if it is clear from context: $H^{p,q}(M)$.

As

$$H^{p,q}_{\overline{\partial}}(M) = \overline{H^{q,p}_{\partial}(M)}$$

we see that the difference of choice is a difference in conjugation, and hence we stick to $\overline{\partial}$ for the holomorphic relation.

For our purposes, we will naturally want to generalize this to the Dolbeault cohomology of vector bundles. Let E be a holomorphic vector bundle on a complex manifold M. Then take the map:

$$\overline{\partial}_E : \Gamma(E) \to \Omega^{0,1}(E)$$

taking sections of E to $(0,1)$-forms with values in E. This will satisfy the Leibniz rule with respect to $\overline{\partial}$, and is thus a $(0,1)$-connection on E. Then we can do the same trick from Riemannian geometry and extend it to the exterior covariant derivative, namely;

$$\overline{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E)$$

which acts on a section $\alpha \otimes s \in \Omega^{p,q}(E)$ by:

$$\overline{\partial}_E(\alpha \otimes s) = (\overline{\partial}\alpha) \otimes s + (-1)^{p+q} \alpha \wedge \overline{\partial}_E s$$

This naturally extends linearly to any section of $\Omega^{p,q}(E)$ and $\overline{\partial}_E^2 = 0$, hence it defines the Dolbeault cohomology with coefficients in E. It is standard to check that the Dolbeault cohomology groups do not depend on the choice of Dolbeault operator $\overline{\partial}_E$ compatible with the holomorphic structure of E [21], and so:

Definition 3.1.4: Dolbeault Cohomology For Vector Bundles

$$H^{p,q}(M, (E, \bar{\partial}_E)) = \frac{\ker(\bar{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E))}{\mathrm{im}(\bar{\partial}_E : \Omega^{p,q-1}(E) \to \Omega^{p,q}(E))}$$

which is unambiguously denoted as:

$$H^{p,q}(M, E)$$

If $\mathbb{1}$ is the trivial bundle, then we get back the usual Dolbeault cohomology.

Definition 3.1.5: Hodge Number

$$h^{p,q}(M,E) = \dim H^{p,q}(M,E)$$

If $E = \mathbb{1}$ then:

$$h^{p,q}(M) = \dim H^{p,q}(M,\mathbb{1})$$

Let Ω^p be the sheaf of holomorphic p-forms on M. Then the following is the natural generalization of de Rham's theorem for Dolbeault cohomology:

Theorem 3.1.6: Dolbeault's Theorem

$$H^{p,q}(M) \cong H^q(M,\Omega^p)$$

and if we are working over a vector bundle:

$$H^{p,q}(M,E) \cong H^q(M,\Omega^p \otimes E)$$

Furthermore:

$$H^{0,q}(M,E) \cong H^q(M,\Omega^0 \otimes E) = H^q(M,E)$$

***Proof*:**
This proof is very similar to the de Rham case. Note it can also be proven by taking a resolution of Ω^p by *fine sheaves* (sheaves that admit partitions of unity)[a] and applying the $\overline{\partial}$-lemma (see [21] or [16, section 15.4]).

[a]flasque (flabby) sheaves are not fine, but are the next best thing: they are *soft*, meaning every section over a closed subset can be extended to the whole space.

For future reference, if E is a complex vector bundle, we shall denote its conjugate by $\overline{E}$ (defined in the natural way) which is anti-complex and anti-isomorphic to E. We can also define the dual bundle E^* in the natural way. Note that, per fiber, the transition matrices of the dual bundle are the inverse transposes of those of E; for a unitary structure group these are the complex conjugates, so that $E^* \cong \overline{E}$.

Hodge Theorem: Finiteness of Cohomology Groups

Let M now be compact and complex. Let us build up to showing that each class in $H^{p,q}(M,\mathbb{C})$ is representable by a harmonic form. This will allow us to quote results from the theory of elliptic operators that show these vector spaces are finite dimensional, and this, in theorem 6.2.1, will be key in showing the χ_1-characteristic is equal to the signature. For notational simplicity, let:

$$A^{p,q}(W) = \Gamma(M,\Omega^{p,q}(W))$$

and:

$$A^{p,q} = A^{p,q}(\mathbb{1})$$

where $\mathbb{1}$ is the trivial bundle representing holomorphic functions. We shall define the Hodge operator that will allow us to relate the different $A^{p,q}$. If W is a complex vector bundle, we can define W^*

and a (perfect) pairing:

$$A^{p,q}(W) \times A^{r,s}(W^*) \to A^{p+r,q+s}(\mathbb{1}) \qquad (\alpha, \beta) \mapsto \alpha \wedge \beta$$

If $W = \mathbb{1}$, this is the usual wedge (or exterior) product. The product is anti-commutative in the sense that:

$$\alpha \wedge \beta = (-1)^{(p+q)(r+s)} \beta \wedge \alpha$$

Choosing $r = n - p$, $s = n - q$ we get the top form $\alpha \wedge \beta \in A^{n,n}(\mathbb{1})$. We can thus define:

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge \beta$$

By Stokes theorem, if $\alpha = \overline{\partial}\gamma$ for $\gamma \in A^{p,q-1}(W)$ and $\overline{\partial}\beta = 0$, (or if $\overline{\partial}\alpha = 0$ and $\beta = \overline{\partial}\gamma$) then:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta = \int_M \overline{\partial}(\gamma \wedge \beta) = \int_M d\,(\gamma \wedge \beta) = 0$$

Thus, we get a well-defined pairing:

$$H^{p,q}(M, W) \times H^{n-p,n-q}(M, W^*) \to \mathbb{C} \qquad (\alpha, \beta) \mapsto \langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta$$

Now, by some linear algebra and differential geometry we see that $\overline{TM} \cong T^*M$ (a conjugate-linear identification depending on the metric), and so:

$$\begin{aligned} \bigwedge^p TM \otimes \overline{\bigwedge^q TM} &\cong \bigwedge^p TM \otimes \bigwedge^n T^*M \otimes \bigwedge^n TM \otimes \bigwedge^q T^*M \\ &\cong \bigwedge^{n-p} T^*M \otimes \bigwedge^{n-q} TM \\ &\cong \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM} \end{aligned}$$

There is thus a natural duality operator:

$$* : \bigwedge^p TM \otimes \overline{\bigwedge^q TM} \to \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}$$

and $* \circ *$ becomes multiplication by $(-1)^{p+q}$.

Now, by vector-bundle theory we know we can without loss of generality reduce W to a unitary group bundle ($U(r)$-bundle); see [12]. There is then the natural Hermitian anti-isomorphism:

$$\psi : W \to W^* \qquad \psi^{-1} : W^* \to W$$

If κ is the anti-isomorphism from $\bigwedge^r TM \otimes \overline{\bigwedge^s TM}$ to its conjugate via conjugation, then define:

$$\star = \psi \otimes (\kappa \circ *) \qquad \widetilde{\star} = \psi^{-1} \otimes (\kappa \circ *)$$

This gives:

$$\begin{aligned} \star : W \otimes \bigwedge^p TM \otimes \overline{\bigwedge^q TM} &\to W^* \otimes \bigwedge^{n-p} TM \otimes \overline{\bigwedge^{n-q} TM} \\ \widetilde{\star} : W^* \otimes \bigwedge^r TM \otimes \overline{\bigwedge^s TM} &\to W \otimes \bigwedge^{n-r} TM \otimes \overline{\bigwedge^{n-s} TM}. \end{aligned}$$

If $r = n - p$, $s = n - q$, then $\widetilde{\star} \circ \star$ is multiplication by $(-1)^{p+q}$. Now, these produce natural anti-isomorphisms of:

$$\star : \Omega^{p,q}(W) \to \Omega^{n-p,n-q}(W^*)$$
$$\widetilde{\star} : \Omega^{r,s}(W^*) \to \Omega^{n-r,n-s}(W)$$

Using these, we can define the "co-" of $\overline{\partial}$:

$$\delta : \Omega^{p,q}(W) \to \Omega^{p,q-1}(W) \qquad \delta = -\widetilde{\star} \circ \overline{\partial} \circ \star$$

With this, if $\alpha, \beta \in A^{p,q}(W)$ then we can define:

$$(\alpha, \beta) = \langle \alpha, \star\beta \rangle = \int_M \alpha \wedge \star\beta$$

It is easy to see that $(\alpha, \alpha) \geq 0$ and $(\alpha, \alpha) = 0$ if and only if $\alpha = 0$. Moreover, for $\alpha \in A^{p,q}(W)$ and $\beta \in A^{p,q+1}(W)$ we get the natural adjoint identity:

$$(\alpha, \delta\beta) = \left(\overline{\partial}\alpha, \beta\right)$$

Indeed given:

$$(\alpha, \delta\beta) = -\int_M \alpha \wedge \star\widetilde{\star}\overline{\partial} \star \beta = (-1)^{p+q+1} \int_M \alpha \wedge \overline{\partial} \star \beta$$

Then:

$$\begin{aligned}
(\overline{\partial}\alpha, \beta) - (\alpha, \delta\beta) &= \int_M (\overline{\partial}\alpha \wedge \star\beta + (-1)^{p+q}\alpha \wedge \overline{\partial} \star \beta) \\
&= \int_M \overline{\partial}(\alpha \wedge \star\beta) \\
&= \int_M d(\alpha \wedge \star\beta) \\
&= 0 && \text{Stokes Thm.}
\end{aligned}$$

We now reach one of the central results in Hodge theory. Define the *Hodge Laplacian* (or Laplace-Beltrami operator):

$$\Delta = \delta\overline{\partial} + \overline{\partial}\delta$$

This defines an operator:

$$\Delta : A^{p,q}(W) \to A^{p,q}(W)$$

Let

$$\ker \Delta = B^{p,q}(W)$$

which could be interpreted as all complex harmonic forms, namely forms that satisfy:

$$\delta(\alpha) = \overline{\partial}(\alpha) = 0$$

The use of the elliptic operator is that it gives rise to the following decomposition:

$$A^{p,q}(W) = \overline{\partial}A^{p,q-1}(W) \oplus \delta A^{p,q+1}(W) \oplus B^{p,q}(W)$$

Notice that if $Z^{p,q}(W) = \ker(\overline{\partial} : A^{p,q}(W) \to A^{p,q+1}(W))$ denotes the space of $\overline{\partial}$-closed forms, we naturally get:

$$Z^{p,q}(W) = \overline{\partial} A^{p,q-1}(W) \oplus B^{p,q}(W)$$

Thus, when taking the Dolbeault cohomology group:

Theorem 3.1.7: Hodge Theorem

$$H^{p,q}(M,W) \cong \frac{Z^{p,q}(W)}{\overline{\partial} A^{p,q-1}(W)} \cong B^{p,q}(W)$$

In other words, these forms are naturally harmonic! Note that in general $h^{p,q}(M) \neq h^{q,p}(M)$; if M is a Kähler manifold then equality holds.

As Δ is an elliptic operator, we get by elliptic operator theory that each of these spaces is *finite dimensional* (see [4]). If M is moreover a compact Kähler manifold, we also get:

$$H^r(M, \mathbb{C}) \cong \bigoplus_{p+q=r} B^{p,q}$$

and the Betti number is the sum of Hodge numbers:

$$b_r(M) = \sum_{p+q=r} h^{p,q}(M)$$

For future reference, an element of $H^{p+q}(M, \mathbb{Z})$ or $H^{p+q}(M, \mathbb{R})$ is said to be of type (p, q) if it is of type (p, q) when regarded as an element of $H^{p+q}(M, \mathbb{C})$.

All of this is summarized in the following two theorems:

Theorem 3.1.8: Kodaira Finite Dimension Theorem

The Dolbeault cohomology group $H^{p,q}(M, E)$ is finite dimensional and isomorphic to the vector space of complex harmonic forms of type (p, q) with coefficients in E. Furthermore:

$$H^p(M, E) = H^{0,p}(M, E)$$

and $H^{p,q}(M, E) = 0$ whenever $p > n$ or $q > n$.

***Proof*:**
For more details, see for example [4] or [25].

Theorem 3.1.9: Serre-Hodge Duality Theorem

Let E be a holomorphic vector bundle over the compact complex manifold M of dimension n. Then the bilinear form $\langle -, - \rangle$ is a dual pairing of $H^{p,q}(M, E)$ with $H^{n-p,n-q}(M, E^*)$. If $K = \bigwedge^n T^*M$ is the canonical line bundle, then $H^p(M, E)$ and $H^{n-p}(M, K \otimes E^*)$ are dual vector spaces.

***Proof*:**
The work above is most of it, but the details can be found in [4] or [25].

3.2 Geometric-Algebraic Bridge: Kähler Manifold

Let M be a compact complex manifold. A Hermitian metric on M is defined as:

$$ds^2 = 2\sum g_{\alpha\beta}(z,\overline{z})(dz^\alpha \cdot d\overline{z}^\beta)$$

where $\overline{g_{\alpha\beta}} = g_{\beta\alpha}$ and the matrix $(g_{\alpha\beta})$ is positive definite. From a Hermitian metric ds^2, we get a natural exterior differential form:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

This form can be re-written as a real differential form by taking real coordinates x^α, $1 \le \alpha \le 2n$ and

$$z^\alpha = x^{2\alpha-1} + ix^{2\alpha}$$

Definition 3.2.1: Kähler Metric

Let ds^2 be a Hermitian metric on the compact complex manifold M and let:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

Then if $d\omega = 0$, ds^2 is called a *Kähler metric.*

If ds^2 is a Kähler metric, ω represents an element of $H^2(M,\mathbb{R})$ (using the de Rham isomorphism); this is called the *fundamental class of the Kähler metric.*

Definition 3.2.2: Kähler Manifold

Let M be a compact complex manifold with Hermitian metric ds^2, and let ω be the associated exterior form. If ds^2 is a Kähler metric (that is, $d\omega = 0$), then (M,ω) is called a *Kähler manifold.*

On a Kähler manifold, Δ commutes with the conjugation $\alpha \mapsto \overline{\alpha}$ between $B^{q,p}$ and $B^{p,q}$ which induces an isomorphism between these groups, and so:

Theorem 3.2.3: Symmetry of Hodge Number On Kähler Manifold

Let M be a Kähler manifold. Then:

$$h^{p,q}(M) = h^{q,p}(M)$$

***Proof*:**
Details in [4] or [25].

Note that these numbers do not depend on the choice of Kähler metric (the Dolbeault cohomology groups $H^{p,q}(M)$ are defined without reference to any metric) and hence it is enough for a Kähler metric to exist.

Now, being a Kähler manifold is very close to being a projective variety. Recall from Hartshorne that an invertible sheaf on a scheme is very ample[2] if it has enough global sections to embed the scheme into projective space (see Hartshorne's exposition at [35] or my exposition at [3]). Not all Kähler manifolds admit such a sheaf, but there is a surprisingly simple condition that ensures that a Kähler manifold can embed into projective space.

Definition 3.2.4: Real Positivity

Let $x \in H^{1,1}(M, \mathbb{R})$. Then x is said to be *positive* if x can be chosen as the fundamental class of a Kähler metric on M.

Proposition 3.2.5: Properties of Real Positivity

Let M be a Kähler manifold. Then

1. At least one element of $H^{1,1}(M, \mathbb{R})$ is positive
2. The zero element of $H^{1,1}(M, \mathbb{R})$ is not positive
3. If $x > 0$ and $y > 0$ then $x + y > 0$ $(x, y \in H^{1,1}(M, \mathbb{R}))$
4. If $x > 0$ and $r > 0$ then $rx > 0$ $(x \in H^{1,1}(M, \mathbb{R})$, $r \in \mathbb{R})$
5. if $x, y \in H^{1,1}(M, \mathbb{R})$ and $x > 0$, then there exists a positive integer g (depending on x, y) such that $gx - y > 0$.

***Proof*:**
Exercise.

Definition 3.2.6: Integral Positivity

Let M be a Kähler manifold and $x \in H^{1,1}(M, \mathbb{Z})$. Then x is said to be *positive* if it is positive when regarded as an element of $H^{1,1}(M, \mathbb{R})$.

Definition 3.2.7: Hodge Manifold

A Kähler manifold M is called a *Hodge Manifold* if $H^{1,1}(M, \mathbb{Z})$ contains at least one positive element.

[2] or ample, meaning some tensor power of it is very ample

Another way of saying this is that M admits a Kähler metric whose fundamental class is in the image of the natural homomorphism $H^2(M,\mathbb{Z}) \to H^2(M,\mathbb{R})$.

Example 3.2.8: Complex Projective Space is a Hodge Manifold

Let $M = \mathbb{C}\mathrm{P}^n$. Then

$$H^{1,1}(\mathbb{C}\mathrm{P}^n,\mathbb{Z}) = H^2(\mathbb{C}\mathrm{P}^n,\mathbb{Z}) \cong \mathbb{Z}$$

and so if $x \in H^{1,1}(\mathbb{C}\mathrm{P}^n,\mathbb{R}) = H^2(\mathbb{C}\mathrm{P}^n,\mathbb{R})$ is nonzero, there is a real number $r > 0$ such that rx lies in the image of the aforementioned inclusion homomorphism. In particular, the positive elements of $H^2(\mathbb{C}\mathrm{P}^n,\mathbb{Z})$ are the positive integral multiples of the hyperplane class.

Theorem 3.2.9: Hodge Manifolds Are Algebraic

Let M be a Kähler manifold. Then M embeds into $\mathbb{C}\mathrm{P}^n$ for some n if and only if it is a Hodge Manifold.

***Proof*:**
see [25].

Similarly, if L is a line bundle over M, then it is *projectively induced* if for some embedding of M into $\mathbb{C}\mathrm{P}^n$, L is the restriction to M of a line bundle H with cohomology class h_n (h_n representing the cohomology class characterized by a hyperplane). Mirroring the result in divisor theory, a projectively induced line bundle can always be given by a divisor:

Theorem 3.2.10: Bertini Theorem

Let F be a projectively induced line bundle over the projective variety M. Then a divisor D induces F, namely $F = \mathcal{L}(D)$.

This is actually a special case of the general Bertini theorem that you find when googling "Bertini Theorem".

***Proof*:**
[25]

Next, just like in sheaf cohomology over schemes, higher-order cohomology shall vanish (given the vector bundle is sufficiently "twisted" to be very ample):

Theorem 3.2.11: Kodaira Higher Order Vanishing Theorem

Let F be a holomorphic line bundle over a Kähler manifold M, and let F' be a positive line bundle over M (a line bundle is *positive* if its cohomology class is positive). Then

$$H^i(M, F \otimes (F')^k)$$

vanishes for all $i > 0$ and k sufficiently large.

***Proof*:**
[25]

Theorem 3.2.12: Characterizing Line Bundles on Projective Varieties

Let M be a Hodge Manifold, L an analytic line bundle over M. Then there are projectively induced line bundles A, B where $L = A \otimes B^{-1}$. Thus, writing $A = \mathcal{L}(D_A)$ and $B = \mathcal{L}(D_B)$ for divisors D_A, D_B, we have:

$$L = \mathcal{L}(D_A) \otimes \mathcal{L}(D_B)^{-1}$$

***Proof*:**
see [16, p.141, thm 18.2.5].

Finally, the following results show when the total spaces of certain fibre bundles are algebraic:

Theorem 3.2.13: Projective Bundles Are Algebraic Over Projective Varieties

Let L be a fiber bundle over the projective variety M with $\mathbb{C}\mathrm{P}^r$ as fibers and $\mathrm{PGL}_{r+1}(\mathbb{C})$ as the structure group. Then it is a projective variety.

***Proof*:**
see [16, section 18.3].

Theorem 3.2.14: Borel Vanishing Theorem

Let L be a complex fibre bundle over the projective variety M with a projective variety as fibre and a connected structure group. Assume that the first Betti number of the fibre is zero. Then L is itself algebraic.

***Proof*:**
[16, p.141]

As Hodge manifolds and projective varieties are interchangeable, this paper shall freely use any analytic properties given by their correspondence to Hodge manifolds.

3.3 Euler Characteristic and Generalization

In the final proof, we shall work with smooth projective varieties and locally free coherent sheaves, but it will do no harm at the start of this section to let M be a compact complex manifold and E a holomorphic vector bundle; we shall re-introduce more assumptions as needed. As we have a double complex, we can define more than one Euler characteristic. First, recall the natural Euler characteristic:

$$\chi(M, E) = \sum_{i=0}^{\infty} (-1)^i \dim H^i(M, E) = \sum_{i=0}^{n} (-1)^i \dim H^i(M, E)$$

If $E = \mathbb{1}$, then $\chi(M, \mathbb{1}) = \chi(M, \mathcal{O}_M)$ is the *arithmetic genus* of M, not the topological Euler characteristic; the latter, denoted χ_{Top}, will instead arise below as the specialization χ_{-1}. By theorem 3.1.6 and theorem 3.1.8 there is a natural grading so that we can define:

$$\chi^p(M, E) = \chi\left(M, E \otimes \bigwedge^p T^*M\right) = \sum_{q=0}^{n} (-1)^q h^{p,q}(M, E)$$

Immediately we get:

$$\chi^0(M, E) = \chi(M, E) \qquad \text{and} \qquad \chi^p(M, E) = 0 \text{ when } p < 0 \text{ or } p > n$$

and if $E = \mathbb{1}$ then:

$$\chi^p(M, \mathbb{1}) = \chi^p(M) = \sum_{q=0}^{n} (-1)^q h^{p,q}(M)$$

We can collect these characteristics in a generating function:

Definition 3.3.1: χ_y-characteristic

$$\chi_y(M, E) = \sum_{p=0}^{n} \chi^p(M, E) y^p \qquad \text{(sim.)} \qquad \chi_y(M) = \sum_{p=0}^{n} \chi^p(M) y^p$$

The term $\chi_y(M, E)$ is the χ_y-characteristic of E and $\chi_y(M)$ is called the χ_y-genus of M.

By construction:

$$\chi_0(M, E) = \chi^0(M, E) = \chi(M, E) \qquad \text{(sim.)} \qquad \chi_0(M) = \chi^0(M) = \chi(M)$$

Following Hirzebruch's convention (Hartshorne [35] instead uses $p_a = (-1)^n(\chi(M, \mathcal{O}_M) - 1)$), recall that we call:

$$\chi(M) = \sum_{q=0}^{n} (-1)^q h^{0,q}(M)$$

the *arithmetic genus* of M. By Serre Duality:

Proposition 3.3.2: χ^p-characteristic and Serre Duality

$$\chi^p(M, E) = (-1)^n \chi^{n-p}(M, E^*)$$
$$\chi(M, E) = (-1)^n \chi(M, K \otimes E^*)$$

***Proof*:**
Apply Serre duality.

Let us now assume M is Kähler. Let $g_q = h^{q,0}$. Then by the symmetry, the arithmetic genus $\chi(M)$ of a compact Kähler manifold is equal to:

$$\chi(M) = \sum_{i=0}^{n} (-1)^i g_i$$

Proposition 3.3.3: Euler Characteristic is Multiplicative

Let M, N be Kähler. Then:

$$\chi_y(M \times N) = \chi_y(M) \cdot \chi_y(N).$$

***Proof*:**
If M and N are Kähler manifolds then by the Künneth formula:

$$h^{p,q}(M \times N) = \sum_{\substack{r+u=p \\ s+v=q}} h^{r,s}(M) h^{u,v}(N).$$

Let y, z be indeterminates and associate to each Kähler manifold M the polynomial $\prod_{y,z}(M) = \sum_{p,q} h^{p,q} y^p z^q$. Then the above is equivalent to:

$$\prod_{y,z}(M \times N) = \prod_{y,z}(M) \cdot \prod_{y,z}(N).$$

Let $z = -1$ in the above. Then $\prod_{y,-1}(-) = \chi_y(-)$ and so:

$$\chi_y(M \times N) = \chi_y(M) \cdot \chi_y(N)$$

Next, let us understand $\chi_y(M)$ for two specific values of y. Let us start with $y = -1$. Then:

$$\chi_{-1}(M) = \sum_{p=0}^{n} (-1)^p \chi^p(M) = \sum_{p,q} (-1)^{p+q} h^{p,q}(M) = \sum_{i=0}^{2n} (-1)^i \dim H^i(M) = \chi_{\mathrm{Top}}(M)$$

is equal to the usual Euler characteristic.

The most important value to prove will be for when $y = 1$. In this case, the χ_1-characteristic will be equal to a value called the *signature* of M[3]. When defining a generating function T_y that will hold the Chern character and Todd class information, $y = 0$ shall represent $\mathrm{ch}(E) \smile \mathrm{td}(M)$, and when $y = 1$ it shall *also* be equal to the signature of M, and the proof of the Hirzebruch-Riemann-Roch boils down to showing that this forces the coefficients of T_y and χ_y to be equal and hence they must also agree when $y = 0$. The signature results shall be proven in chapter 5 and the specific theorem for the character when $y = 1$ is proven in theorem 6.2.1.

Let us finish with the special case M a Kähler manifold with $h^{p,q} = 0$ when $p \neq q$. Then $\chi_y(M)$ is (essentially) equal to the Poincaré polynomial:

$$P(t; M) = \sum b_r t^r$$

where b_r is the rth Betti number of M. Indeed, we get:

$$\chi^p(M) = \sum_q (-1)^q h^{p,q} = (-1)^p h^{p,p} = (-1)^p b_{2p}$$

[3]Called the *index* in Hirzebruch's book (see the terminology remark in section 5.2).

therefore:

$$\chi_{-t^2}(M) = P(t; M) = \sum_r b_r t^r$$

3.3.4 Why do we care about this special case? Because complex projective spaces $\mathbb{C}\mathrm{P}^n$ and flag manifolds $F(n)$ both are Kähler manifolds with this property. The case for $\mathbb{C}\mathrm{P}^n$ comes from elementary algebraic topology. For the flag manifold $F(n)$, recall that $H^*(F(n); \mathbb{Z})$ by the splitting principle is generated by elements $\gamma_i \in H^2(F(n); \mathbb{Z})$. Then by characteristic class theory there are complex $\mathbb{C}^*$-bundles E_i over $F(n)$ where $c_1(E_i) = \gamma_i$, and the E_i can be taken holomorphic (the tautological quotient line bundles of the flag manifold).

The correspondence in the above remark is in fact an "if and only if": an element of $H^2(F(n); \mathbb{Z})$ is the cohomology class of a holomorphic line bundle if and only if it is of type $(1, 1)$ (theorem 3.3.5). The "only if" direction gives that each γ_i is of type $(1, 1)$, and therefore the cohomology classes of $F(n)$ are of type (p, p). Let us build up to proving this theorem. Take the usual exact sequence:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_M \to \mathcal{O}_M^* \to 0$$

where $\underline{\mathbb{Z}}$ is the constant sheaf, $\mathcal{O}_M$ is the sheaf of holomorphic functions, and $\mathcal{O}_M^*$ is the sheaf of never-zero holomorphic functions (invertible holomorphic functions). Then we get an exact cohomology sequence:

$$H^1(M, \mathcal{O}_M^*) \xrightarrow{\delta_*^1} H^2(M, \mathbb{Z}) \to H^2(M, \mathcal{O}_M)$$

Then on a Kähler manifold we have $H^2(M, \mathcal{O}_M) = H^2(M, \mathbb{1}) \cong B^{0,2}$. Thus, an element $a \in H^2(M, \mathbb{Z})$ is mapped to zero in $H^2(M, \mathcal{O}_M)$ if and only if a is of type $(1, 1)$. Then from characteristic class theory for vector bundles we know that if $\xi \in H^1(M, \mathcal{O}_M^*)$ is associated to a holomorphic $\mathbb{C}^*$-bundle, then:

$$\delta_*^1(\xi) = c_1(\xi)$$

If L is a holomorphic line bundle over M and ξ is the associated $\mathbb{C}^*$-bundle, then $c_1(\xi)$ is the *cohomology class* of L. Then we get:

Theorem 3.3.5: Characterizing Line bundles on Kähler Manifolds

Let M be a compact Kähler manifold. Then an element $a \in H^2(M, \mathbb{Z})$ is the cohomology class of a holomorphic line bundle over M if and only if it is of type $(1, 1)$.

***Proof*:**
Hirzebruch references [16, p.127, Thm 15.9.1]. The reader can also prove the more general correspondence between cohomology classes of H^2 and line bundles (see [12] for the details) and then apply the harmonic theory developed in this section.

3.3.1 Euler Characteristic on Intersections

The Hirzebruch-Riemann-Roch will reduce to being proven on line bundles. As line bundles on projective varieties are associated to codimension 1 subvarieties/submanifolds, we can take a filtering of M of codimension 1 subvarieties to represent the line bundles. This will lead to decomposing χ

in terms of the characteristics of all possible filterings. To that end, we shall develop some more properties of χ_y. Let:

$$0 \to E' \to E \to E'' \to 0$$

be a short exact sequence of holomorphic vector bundles. This naturally induces a short exact sequence of sheaves [3]:

$$0 \to \mathcal{O}_M(E') \to \mathcal{O}_M(E) \to \mathcal{O}_M(E'') \to 0$$

Thus, without loss of generality we may use common results from sheaf theory, in particular the results from Hartshorne [35] shall be quoted freely (for my own notes with proofs see [3]). What is important is that we get that each new Euler-characteristic is additive on exact sequences:

Theorem 3.3.6: Generalized Euler Characteristic Additive

Let

$$0 \to \mathcal{O}_M(E') \to \mathcal{O}_M(E) \to \mathcal{O}_M(E'') \to 0$$

be a short exact sequence of holomorphic vector bundles over M. Then:

$$\chi(M, E) = \chi(M, E') + \chi(M, E'')$$
$$\chi^p(M, E) = \chi^p(M, E') + \chi^p(M, E'')$$
$$\chi_y(M, E) = \chi_y(M, E') + \chi_y(M, E'')$$

***Proof*:**
Proved in [3], also an exercise in Hartshorne, see [35, section 2.5].

Theorem 3.3.7: Splitting Principle and Euler Characteristic

Let E be a complex vector bundle over a compact complex manifold M, and suppose E splits into $A_1, A_2, ..., A_q$. Let E' be another complex vector bundle over M. Then:

$$\chi(M, E' \otimes E) = \sum_{i=1}^{q} \chi(M, E' \otimes A_i)$$

***Proof*:**
Use induction and the above theorem

Thus, χ on vector bundles splits additively, similarly to how the Chern character $\mathrm{ch}(-)$ behaves.

As Weil divisors are in correspondence to line bundles on projective varieties (for algebraic line bundles see [35], [3], and recall theorem 3.2.12 for the right generalization to analytic line bundles), and Weil divisors are codimension 1 sub-(varieties/manifolds), they shall be the ideal object for the aforementioned filtering. From now on, all divisors are smooth/regular Weil divisors. As usual let E be a complex vector bundle over a complex manifold M, and let D be a divisor of M. Then it is associated to a line bundle $\mathcal{O}_M(D) = \mathcal{L}(D)$. We can now tensor $E \otimes \mathcal{L}(D)$, and consider $(E \otimes \mathcal{L}(D))_D$ to be the restriction to D of the vector bundle $E \otimes \mathcal{L}(D)$. We can take the sheaf $\mathcal{O}_M((E \otimes \mathcal{L}(D))_D)$ over D. By extending by zero, this is a sheaf on M (in Hartshorne, this is denoted by the pushforward j_* along the inclusion map $j : D \hookrightarrow M$ [35], where j_* is the proper pushforward in the sense of

Fulton [20][4]). By the correspondence of vector bundles and locally free sheaves, we may without loss of generality consider $\mathcal{O}_M(E)|_D$ a sheaf or a vector bundle.

Theorem 3.3.8: Extending via Divisors

Let M be a complex manifold, D a Cartier divisor of M, E a complex vector bundle over M. Then there is a short exact sequence:

$$0 \to \mathcal{O}_M(E) \to \mathcal{O}_M(E \otimes \mathcal{L}(D)) \to \mathcal{O}_M(E \otimes \mathcal{L}(D))|_D \to 0$$

***Proof*:**
This is a standard proof; it can be found in pure-sheaf form in Hartshorne [35, chapter 2] or my notes [3, chapter 5].

When it is not confusing, we shall denote by E_D the restriction $E|_D$, a complex vector bundle over D. We shall also write:

$$\chi(D, E) := \chi(D, E_D)$$
$$\chi^p(D, E) := \chi^p(D, E_D)$$
$$\chi_y(D, E) := \chi_y(D, E_D)$$

Theorem 3.3.9: Splitting Euler Characteristic Using Divisor

Let M be a compact complex manifold, D a divisor, E a complex vector bundle over M. Then:

$$\chi(M, E) = \chi(M, E \otimes \mathcal{L}(D)^{-1}) + \chi(D, E)$$

In particular, if $E = \mathbb{1}$, then:

$$\chi(M) = \chi(M, \mathcal{L}(D)^{-1}) + \chi(D)$$

***Proof*:**
Combine above results

We want to extend this inductively so that χ_y can be written in terms of χ^p. First, recall from Riemannian geometry that we get the following short exact sequence:

$$0 \to T(D) \to T(M)|_D \to N_{D/M} \to 0$$

where $N_{D/M}$ is the normal bundle of D with respect to M. It is not hard to see its fibers are of dimension 1, and a little more work shows it is isomorphic to $\mathcal{L}(D)_D$, and so:

$$0 \to T(D) \to T(M)|_D \to \mathcal{L}(D)_D \to 0$$

We then get a natural exact sequence:

$$0 \to \bigwedge^p (T(D)) \to \bigwedge^p (T(M)|_D) \to \bigwedge^{p-1} (T(D)) \otimes \mathcal{L}(D)_D \to 0$$

[4] This will be important for section 6.3

Dualizing we get:

$$0 \to \bigwedge^{p-1}(T^*(D)) \otimes \mathcal{L}(D)_D^{-1} \to \bigwedge^{p}(T^*(M)|_D) \to \bigwedge^{p}(T^*(D)) \to 0$$

Now if E is a complex vector bundle over M, we can tensor the sequence. Then by the above theorems:

$$\chi\left(D, E \otimes \bigwedge^{p}(T^*(M))\right) = \chi^{p-1}(D, E \otimes \mathcal{L}(D)^{-1}) + \chi^p(D, E)$$

Replacing E with $E \otimes \bigwedge^p(T^*(M))$, and manipulating, we get:

Definition 3.3.10: Four-term Formula

$$\chi^p(M, E) = \chi^p(M, E \otimes \mathcal{L}(D)^{-1}) + \chi^p(D, E) + \chi^{p-1}(D, E \otimes \mathcal{L}(D)^{-1})$$

This is true for $p > 0$, and for $p = 0$ interpret the last term on the right hand side as being 0. Furthermore, $\chi^p(D, E) = 0$ for $p = n = \dim M$ and if $p > n$ then all four terms are 0. Letting y be an indeterminate, the above equation holds with each term multiplied by y^p. Summing over $p \geq 0$ gives:

Definition 3.3.11: Four-term Formula: Generating Function

$$\chi_y(M, E) = \chi_y(M, E \otimes \mathcal{L}(D)^{-1}) + \chi_y(D, E) + y\chi_y(D, E \otimes \mathcal{L}(D)^{-1})$$

Now, by repeating the application of equation in definition 3.3.10, we can write $\chi^p(D, E)$ as a linear combination of integers of the form $\chi^q(M, A)$ where A will be an appropriate complex vector bundle over M. For example:

$$\chi^0(D, E) = \chi^0(M, E) - \chi^0(M, E \otimes \mathcal{L}(D)^{-1}) \tag{3.1}$$

Next, apply the four-term formula with $p = 1$ (its last term is then computed by the analogue of equation (3.1)):

$$\chi^1(M, E) = \chi^1(M, E \otimes \mathcal{L}(D)^{-1}) + \chi^1(D, E) + \chi^0(D, E \otimes \mathcal{L}(D)^{-1})$$

and replace E with $E \otimes \mathcal{L}(D)^{-1}$ in equation (3.1) to get:

$$\chi^1(D, E) = \chi^1(M, E) - \chi^1(M, E \otimes \mathcal{L}(D)^{-1}) - \chi^0(M, E \otimes \mathcal{L}(D)^{-1}) + \chi^0(M, E \otimes \mathcal{L}(D)^{-2})$$

Recursively continuing this, we get:

$$\chi^p(D, E) = \sum_{i=0}^{p}(-1)^i[\chi^{p-i}(M, E \otimes \mathcal{L}(D)^{-i}) - \chi^{p-i}(M, E \otimes \mathcal{L}(D)^{-(i+1)})]$$

This gives an important equation. For an honest divisor D these relations hold on any compact complex manifold, but in the virtual generalization of the next section (where $\mathcal{L}(D)$ is replaced by an arbitrary line bundle carrying no divisor) they are not automatic: it is not guaranteed the right hand side cancels out, while the left hand side will always be zero for $p \geq n$. If M is Kähler, or can be embedded into projective space, the relations hold, as shall be seen in the next section.

Combining these together in a generating function, we get:

$$\chi_y(D, E) = \sum_{i=0}^{\infty}(-y)^i[\chi_y(M, E \otimes \mathcal{L}(D)^{-i}) - \chi_y(M, E \otimes \mathcal{L}(D)^{-(i+1)})] \tag{3.2}$$

3.4 Virtual χ_y-characteristic

In equation (3.2), the right hand side is a complicated expression. The goal of this section is to show that the formula can be represented in terms of manifolds given by line bundles. By the splitting principle and the correspondence between divisors and line bundles, this goal is well-defined.

The idea behind the following section goes as follows. Let's say $V \hookrightarrow M$ is $\mathbb{C}$-codimension 1. Then what is the relation between $\chi(M, E)$ and $\chi(V, E)$? What if we have a sequence $V_r \hookrightarrow V_{r-1} \hookrightarrow \cdots \hookrightarrow V_1 \hookrightarrow M$ of codimension 1 at each inclusion? Can we relate $\chi(M, E)$ to V_r in terms of the information given by $V_1, ..., V_r$? Ideally, we can define some:

$$\chi((V_1, ..., V_r), E) = \cdots$$

which will give us the Euler characteristic of V_r given the information of the intersection. Conversely, it would be good if there exist functions F_i such that:

$$\chi(M, E) = \sum_i F_i(V_1, ..., V_r, E)$$

which allows us to link the properties of $\chi(M, E)$ to those of simpler submanifolds. We shall do exactly this in this section.

The greatest advantage of this is that each of these manifolds is codimension one with respect to the next one, so we can associate to them line bundles, and the line bundles are in bijective correspondence with divisor classes (as M is projective). Thus, given a set of line bundles $\{L_1, ..., L_r\}$, we get a filtration $V_r \subseteq V_{r-1} \subseteq \cdots \subseteq V_1 \subseteq M$ allowing us to reduce the eventual proof of Hirzebruch-Riemann-Roch to a proof on line bundles; we can also think of each V_i as the intersection of V_{i-1} with some hyperplane (use theorem 3.2.10 and some divisor theory). As it will turn out, the result is independent of order, and hence it is well-defined to take the set $\{L_1, ..., L_r\}$ instead of the tuple $(L_1, ..., L_r)$.

By using the splitting principle, given that TM splits into line bundles $A_1, ..., A_m$, we aim to find an equation for $\chi(M, E)$ in terms of the line bundles.

A quick word must be given in terms of the reformulation of this in the Grothendieck-Riemann-Roch. The inclusion $j : V \to M$ will be used to pull back a cohomology class $j^* : H^*(M) \to H^*(V)$ giving a contra-variant theory. However, using Fulton's intersection theory we instead would take $j_* : A_*(V) \to A_*(M)$; this is the *proper pushforward* (corresponding, under Poincaré duality, to the cohomological Gysin map). As the proof of the Grothendieck-Riemann-Roch is purely algebraic, the pushforward will be the "correct" generalization to scheme theory. Note that the Grothendieck-Riemann-Roch will not require us to keep track of the set of line bundles $\{A_1, ..., A_m\}$ as the proof technique hinges on a very different approach (see section 6.3).

3.4.1 Arithmetic

Let us represent the arithmetic of the intersection. Let M be a compact complex manifold, $\{L_1, ..., L_r\}$ be associated line bundles. Define:

$$R = \mathbb{Z}[e, \ell_1, \ell_2, ..., \ell_r, \ell_1^{-1}, \ell_2^{-1}, ..., \ell_r^{-1}]$$

where each ℓ_i is a free variable that should be thought of as corresponding to a line bundle L_i of M. Define two additive homomorphisms $h^M, h_y^M : R \to \mathbb{Z}[[y]]$ by defining them on the monomial basis:

$$h^M(e^m \cdot \ell_1^{m_1} \cdots \ell_r^{m_r}) = \chi(M, E^m \otimes L_1^{m_1} \otimes \cdots \otimes L_r^{m_r})$$
$$h_y^M(e^m \cdot \ell_1^{m_1} \cdots \ell_r^{m_r}) = \chi_y(M, E^m \otimes L_1^{m_1} \otimes \cdots \otimes L_r^{m_r})$$

where the powers should be taken as taking the tensor product. Then by some elementary algebra, both homomorphisms extend to the $\mathbb{Z}[[y]]$-module $R[[y]]$; label the extensions h^M and h_y^M as well. By definition:

$$h^M(1) = \chi(M) \qquad \text{and} \qquad h_y^M(1) = \chi_y(M)$$

By using the results of the previous section, given a divisor D, adjoin to R a further variable d (together with its inverse d^{-1}) corresponding to the line bundle $\mathcal{L}(D)$, and define a $\mathbb{Z}[[y]]$-homomorphism

$$h_y^D(e^m \cdot \ell_1^{m_1} \cdots \ell_r^{m_r}) = \chi_y(D, E^m \otimes L_1^{m_1} \otimes \cdots \otimes L_r^{m_r})$$

where like before the line bundles have a natural restriction to D. By equation (3.2):

$$\chi_y(D, E) = h_y^M\left(e\frac{1 - d^{-1}}{1 + yd^{-1}}\right)$$

and by theorem 3.3.9:

$$\chi(D) = h^M(1 - d^{-1}) \qquad \text{and} \qquad \chi(D, E) = h^M(e(1 - d^{-1}))$$

By taking successive divisors, we multiply by further factors of $\frac{1-d^{-1}}{1+yd^{-1}}$, motivating the following definition:

Definition 3.4.1: Virtual Characteristic

$$\chi_y(\{L_1, ..., L_r\}, E)_M := h_y^M\left(e\prod_{i=1}^{r}\frac{1 - \ell_i^{-1}}{1 + y\ell_i^{-1}}\right)$$

If $E = \mathbb{1}$, it shall simply be denoted

$$\chi_y(\{L_1, ..., L_r\})_M := \chi_y(\{L_1, ..., L_r\}, \mathbb{1})_M$$

By the construction of the virtual characteristic, it is clear it is not dependent on the order of the filtering but on the choice of line bundles associated to them, and that it respects the same property as the usual χ_y-characteristic:

$$\chi_y(\{L_1, ..., L_r\}, E)_M = \sum_{p=0}^{\infty} \chi^p(\{L_1, ..., L_r\}, E)_M y^p$$

By taking $\chi_M = \chi_M^0$ we can write:

$$\chi(\{L_1, ..., L_r\}, E)_M = h^M\left(e\prod_{i=1}^{r}(1 - \ell_i^{-1})\right)$$

An important concrete example would then be:

Proposition 3.4.2: Euler Characteristic: Split Manifold and Bundle

$$\chi(\{L\})_M = \chi(M) - \chi(M, L^{-1})$$

Notice that we are subtracting; this motivates the term "virtual", as the formal difference of bundle classes being taken is well-defined in K-theory (the dual line bundle L^{-1} itself is an honest bundle).

Let us now find χ_y in terms of virtual characteristics. By definition of h, we immediately get:

$$\chi_y(D, E) = \chi_y(\{\mathcal{L}(D)\}, E)_M \tag{3.3}$$

As $\chi_y(D, E)$ is a polynomial of degree at most $n-1$ in y, so is the right hand side. We shall extend this to intersections $D_1 \cap \cdots \cap D_r$, which shall give a polynomial of degree at most $n-r$ when M is a nonsingular variety (theorem 3.4.8). First, the restriction to a divisor D of the virtual characteristic of any set $\{L_1, ..., L_r\}$ must be computed:

Theorem 3.4.3: Virtual Characteristic Divisor Restriction

Let $M, E, L_1, ..., L_r$ be as usual, and D a divisor of M, $\mathcal{L}(D) = L_1$. Then:

$$\chi_y(\{L_1, ..., L_r\}, E)_M = \chi_y(\{(L_2)_D, ..., (L_r)_D\}, E_D)_D$$

***Proof*:**
Let:

$$R(x) = \frac{1 - x^{-1}}{1 + yx^{-1}}$$

Then by definition

$$\chi_y(\{(L_2)_D, \ldots, (L_r)_D\}, E_D)_D = h_y^D\left(e\prod_{i=2}^{r} R(\ell_i)\right)$$

It follows from our work above that:

$$h_y^D(e^m \ell_1^{h_1} \ldots \ell_r^{h_r}) = h_y^M(e^m \ell_1^{h_1} \ldots \ell_r^{h_r} R(\ell_1))$$

Then we get:

$$h_y^D\left(e\prod_{i=2}^{r} R(\ell_i)\right) = h_y^M\left(e\prod_{i=1}^{r} R(\ell_i)\right) = \chi_y(\{L_1, \ldots, L_r\}, E)_M$$

Corollary 3.4.4: Virtual Characteristic and Trivial Bundle

Let $M, E, L_1, ..., L_r$ be as usual, and assume one $L_i = \mathbb{1}$ is the trivial bundle. Then:

$$\chi_y(\{L_1, ..., L_r\}, E)_M = 0$$

***Proof*:**
Immediate consequence, since $R(1) = 0$.

Theorem 3.4.5: Algebra of Product of Line Bundles

Let M be a compact complex manifold, E a complex vector bundle, $L_1, ..., L_r, A, B$ complex line bundles. Then:

$$\begin{aligned}\chi_y(\{L_1, \cdots, L_r, A \otimes B\}, E)_M = &\\ &\chi_y(\{L_1, \cdots, L_r, A\}, E)_M \\ &+ \chi_y(\{L_1, \cdots, L_r, B\}, E)_M \\ &+ (y-1)\chi_y(\{L_1, \cdots, L_r, A, B\}, E)_M \\ &- y\chi_y(\{L_1, \cdots, L_r, A, B, A \otimes B\}, E)_M\end{aligned}$$

***Proof*:**
By definition:

$$\chi_y(\{L_1, \ldots, L_r\}, E)_M = h_y^M(e \cdot \prod_{i=1}^{r} R(\ell_i))$$

where the function $R(x)$ is given by:

$$R(x) = \frac{x-1}{x+y}$$

Let the term u correspond to the common part of the bundles in the equation:

$$u = e \cdot \prod_{i=1}^{r} R(\ell_i)$$

Now, we can express each term in the theorem's main equation using this notation:

1. $\chi_y(\{L_1, \ldots, L_r, A \otimes B\}, E)_M = h_y^M(u \cdot R(ab))$
2. $\chi_y(\{L_1, \ldots, L_r, A\}, E)_M = h_y^M(u \cdot R(a))$
3. $\chi_y(\{L_1, \ldots, L_r, B\}, E)_M = h_y^M(u \cdot R(b))$
4. $\chi_y(\{L_1, \ldots, L_r, A, B\}, E)_M = h_y^M(u \cdot R(a)R(b))$
5. $\chi_y(\{L_1, \ldots, L_r, A, B, A \otimes B\}, E)_M = h_y^M(u \cdot R(a)R(b)R(ab))$

The theorem states:

$$h_y^M(u \cdot R(ab)) = h_y^M(u \cdot R(a)) + h_y^M(u \cdot R(b)) + (y-1)h_y^M(u \cdot R(a)R(b)) - y \cdot h_y^M(u \cdot R(a)R(b)R(ab))$$

Since h_y^M is a homomorphism, it is linear with respect to polynomials in y. We can thus factor out $h_y^M(u \cdot \ldots)$ from the right-hand side:

$$h_y^M(u \cdot R(ab)) = h_y^M\left(u \cdot [R(a) + R(b) + (y-1)R(a)R(b) - yR(a)R(b)R(ab)]\right)$$

Therefore, the theorem is proven if the following algebraic identity holds:

$$R(ab) = R(a) + R(b) + (y-1)R(a)R(b) - yR(a)R(b)R(ab)$$

and indeed:

$$\begin{aligned}
&R(a) + R(b) + (y-1)R(a)R(b) - yR(a)R(b)R(ab) \\
&= \frac{a-1}{a+y} + \frac{b-1}{b+y} + (y-1)\frac{(a-1)(b-1)}{(a+y)(b+y)} - y\frac{(a-1)(b-1)}{(a+y)(b+y)}\frac{ab-1}{ab+y} \\
&= \frac{(a-1)(b+y) + (b-1)(a+y) + (y-1)(a-1)(b-1)}{(a+y)(b+y)} - y\frac{(a-1)(b-1)(ab-1)}{(a+y)(b+y)(ab+y)} \\
&= \frac{ab + ay - b - y + ab + by - a - y + (y-1)(ab-a-b+1)}{(a+y)(b+y)} - \dots \\
&= \frac{2ab + a(y-1) + b(y-1) - 2y + (y-1)(ab-a-b+1)}{(a+y)(b+y)} - \dots \\
&= \frac{2ab + ay - a + by - b - 2y + yab - ya - yb + y - ab + a + b - 1}{(a+y)(b+y)} - \dots \\
&= \vdots \\
&= \frac{ab-1}{(a+y)(b+y)(ab+y)}\left[(a+y)(b+y)\right] \\
&= \frac{ab-1}{ab+y} \\
&= R(ab)
\end{aligned}$$

This establishes the identity, and hence the functional equation for the virtual χ_y-characteristic, as we sought to show. In particular, specializing to $y = 0$ we get the equation:

$$\begin{aligned}
\chi(\{L_1, ..., L_r, A \otimes B\}, E)_M = \chi(\{L_1, ..., L_r, A\}, E)_M &+ \chi(\{L_1, ..., L_r, B\}, E)_M \\
&- \chi(\{L_1, ..., L_r, A, B\}, E)_M
\end{aligned}$$

Let us now assume that every vector bundle on M splits. Then TM splits into line bundles $A_1 \oplus \cdots \oplus A_m$. Then $\bigwedge^p T^*M$ splits into $\binom{m}{p}$ line bundles of the form:

$$A_{i_1}^{-1} \otimes \cdots \otimes A_{i_p}^{-1} \qquad (i_1 < \cdots < i_p)$$

By theorem 3.3.7:

$$\begin{aligned}
\chi^p(M, E) &= \chi(M, E \otimes \bigwedge^p T^*M) \\
&= \sum_{i_1 < \cdots < i_p} \chi(M, E \otimes A_{i_1}^{-1} \otimes \cdots \otimes A_{i_p}^{-1})
\end{aligned}$$

Thus, by the definition of h^M:

$$\chi_y(M, E) = h^M\left(e \prod_{i=1}^{m} (1 + y a_i^{-1})\right)$$

This shall now give us a decomposition of χ in terms of the virtual characteristic.

Theorem 3.4.6: Virtual Characteristic Decomposition

Let M be a compact complex manifold where TM splits completely into line bundles $A_1, ..., A_m$. Let E be a complex vector bundle. Then:

$$(1+y)^m \chi(M,E) = \sum_{l=0}^{m} y^l \sum_{i_1<\cdots<i_l} \chi_y(\{A_{i_1}, ..., A_{i_l}\}, E)_M$$

***Proof*:**
Writing in terms of h_y^M:

$$\begin{aligned}\sum_{l=0}^{m} y^l \sum_{i_1<i_2<\cdots<i_l} h_y^M(eR(a_{i_1})\dots R(a_{i_l})) &= h_y^M\left(\sum_{l=0}^{m} y^l \sum_{i_1<i_2<\cdots<i_l} eR(a_{i_1})\dots R(a_{i_l})\right)\\ &= h_y^M\left(e\prod_{i=1}^{m}(1+yR(a_i))\right)\\ &= h_y^M\left(e\prod_{i=1}^{m}(1+y)(1+ya_i^{-1})^{-1}\right)\\ &= (1+y)^m h_y^M\left(e\prod_{i=1}^{m}(1+ya_i^{-1})^{-1}\right)\end{aligned}$$

Note that, independently of the above, by definition we have the relation:

$$h_y^M(e^{h_0}a_1^{h_1}a_2^{h_2}\dots a_m^{h_m}) = h^M\left(e^{h_0}a_1^{h_1}a_2^{h_2}\dots a_m^{h_m}\prod_{i=1}^{m}(1+ya_i^{-1})\right)$$

Therefore:

$$\begin{aligned}(1+y)^m h_y^M\left(e\prod_{i=1}^{m}(1+ya_i^{-1})^{-1}\right) &= (1+y)^m h^M\left(e\prod_{i=1}^{m}(1+ya_i^{-1})^{-1}(1+ya_i^{-1})\right)\\ &= (1+y)^m h^M(e) = (1+y)^m\chi(M,E)\end{aligned}$$

as we sought to show.

3.4.2 Relating Euler Characteristic to Chern Numbers

Finally, let us show that the virtual characteristic with r line bundles is a polynomial of degree at most $n-r$, and that evaluating when $n=r$ gives $q = \mathrm{rk}E$ times the integral over M of the cup product of the respective classes $l_i \in H^2(M,\mathbb{Z})$.

Lemma 3.4.7: 0-dimensional Case

Let $M = \{p_1, ..., p_k\}$ be a 0-dimensional manifold, E a q-dimensional vector bundle over M, $L_1, ..., L_r$ line bundles over M. Then:

1. $\chi_y(M, E) = qk$
2. $\chi_y(\{L_1, ..., L_r\}, E) = 0$ for $r \geq 1$

***Proof*:**

1. $\chi_y(M, E) = \chi(M, E) = \dim H^0(M, E) = qk$
2. By Serre's splitting theorem and Bass's cancellation theorem (see [12, section 5.1][a]), every line bundle over a finite set of points is trivial, so by corollary 3.4.4

$$\chi_y(\{L_1, ..., L_r\}, E) = 0$$

[a] A special case is proven in theorem 6.1.2, which is a geometric proof as compared to the algebraic proof in the cited result

In general, recall that $\chi_y(M, E)$ is a polynomial, as $\chi^p(M, E) = 0$ for $p > n$ on any compact complex manifold and so the power series must terminate. Let us prove by induction on the dimension n of M that the virtual χ_y-characteristic is also a polynomial.

Theorem 3.4.8: Induction Case

Let M be a smooth projective variety of complex dimension n. Let E be a q-dimensional complex vector bundle, and let $L_1, ..., L_r$ be analytic line bundles over M. Then:

1. $\chi_y(\{L_1, ..., L_r\}, E)_M = 0$ for $r > n$.
2. For $r \leq n$, $\chi_y(\{L_1, ..., L_r\}, E)_M$ is a polynomial of degree $\leq n - r$ in y with integer coefficients
3. If $r = n$, then

$$\chi_y(\{L_1, ..., L_n\}, E)_M = \chi(\{L_1, ..., L_n\}, E)_M = q \int_M l_1 \smile l_2 \smile \cdots \smile l_n$$

where $l_i \in H^2(M, \mathbb{Z})$ are the cohomology classes of L_i.

We shall often use $\cdot$ instead of $\smile$ to de-clutter notation.

***Proof*:**

1/2. lemma 3.4.7 proves it for $\dim M = 0$. Suppose it is true for $\dim M < n$. By theorem 3.2.12 and theorem 3.2.10 there are divisors D and S of M such that $\mathcal{L}(D) = L_1 \otimes \mathcal{L}(S)$. Then by theorem 3.4.5:

$$
\begin{aligned}
&\chi_y(\{\mathcal{L}(D), L_2, \cdots, L_r\}, E)_M \\
&\quad = \chi_y(\{L_1, \cdots, L_r\}, E)_M + \chi_y(\{\mathcal{L}(S), L_2, \cdots, L_r\}, E)_M \\
&\quad + (y-1)\chi_y(\{\mathcal{L}(S), L_1, \cdots, L_r\}, E)_M - y\chi_y(\{\mathcal{L}(D), \mathcal{L}(S), L_1, \cdots, L_r\}, E)_M
\end{aligned}
$$

We must show the 2nd term has degree $\leq n-r$. Note that induction hypothesis, together with theorem 3.4.3 shows 1st, 3rd, 4th and 5th term all have degree $\leq n-r$ and vanish for $r > n$. Note that if $r = 1$ then 1st term is $\chi_y(D, E_D)$ and 3rd term is $\chi_y(S, E_S)$; these terms are polynomials of degree $\leq n-1$ by the definition of the (non-virtual) χ_y-characteristic. Therefore 2nd term is a polynomial of degree $\leq n-r$ and zero for $r > n$.

3. By theorem 3.2.12 and theorem 3.2.10 there are divisors D and S of M such that $\mathcal{L}(D) = L_1 \otimes \mathcal{L}(S)$.

   Then for $n \geq 2$, the $y = 0$ specialization of theorem 3.4.5 together with (1) gives:

$$
\chi(\{\mathcal{L}(D), L_2, \ldots, L_n\}, E) = \chi(\{L_1, L_2, \ldots, L_n\}, E) + \chi(\{\mathcal{L}(S), L_2, \ldots, L_n\}, E) \tag{3.4}
$$

   and for $n = 1$

$$
\chi(\{\mathcal{L}(D)\}, E) = \chi(\{L_1\}, E) + \chi(\{\mathcal{L}(S)\}, E)
$$

   Then by lemma 3.4.7 the $n = 1$ case gives

$$
\chi(\{L_1\}, E) = qd - qs = ql_1[M]
$$

   where d, s are the number of points of D, S. Therefore (3) is true for $\dim M = 1$. Now for the sake of induction suppose it is proved for $1 \leq \dim M < n$. We now apply the induction hypothesis to equation (3.4). First, recall that if $h \in H^2(M, \mathbb{Z})$ is the cohomology class representing the divisor D then $c_1([D]) = h$. If $j : D \hookrightarrow M$ is an inclusion (treating D as a codimension 1 submanifold), then for any $x \in H^{2n-2}(M, \mathbb{Z})$:

$$
\int_D j^*(x) = \int_M h \smile x = \int_M hx
$$

   Then by theorem 3.4.3, we get:

$$
\begin{aligned}
\chi(\{L_1, L_2, \ldots, L_n\}, E) &= \int_D q \cdot (l_2 \ldots l_n)_D - \int_S q \cdot (l_2 \ldots l_n)_S \\
&= \int_M q \cdot c_1(\mathcal{L}(D)) l_2 \ldots l_n - \int_M q \cdot c_1(\mathcal{L}(S)) l_2 \ldots l_n \\
&= \int_M q \cdot l_1 l_2 \ldots l_n
\end{aligned}
$$

   as we sought to show.

When $n = 1$, this is in fact Riemann-Roch for curves! Let $M = C$ be a curve, L a line bundle with cohomology class $l \in H^2(C, \mathbb{Z})$. Then the virtual characteristic of L^{-1} is:

$$
\chi(\{L^{-1}\})_C = -\int_C l
$$

Then: $\chi(\{L^{-1}\})_C = \chi(C) - \chi(C, L)$ and so we get:

$$\chi(C, L) = \chi(C) + \int_C l$$

Next by using theorem 3.2.3 we get:

$$\chi(C) = 1 - g$$

By classical algebraic topology, $\int_C l$ is called the *degree* of L; it agrees with the degree of any divisor associated to L. Using Serre duality we get:

$$\chi(C, L) = \dim H^0(C, L) - \dim H^1(C, L) = \dim H^0(C, L) - \dim H^0(C, K \otimes L^{-1})$$

Combining it all together, we obtain:

$$\dim H^0(C, L) - \dim H^0(C, K \otimes L^{-1}) = 1 - g + \deg(L)$$

i.e., classical Riemann-Roch! This result does not generalize as the correspondence between codimension 2 objects and elements of H^4 is not well-defined in the necessary way, which motivates inducting on line bundles.

Finally, an important result that will be used is that the virtual Euler characteristic characterizes these relations:

Theorem 3.4.9: Uniqueness Result

Let $M, E, L_1, ..., L_r$ be as usual, $\mathbf{L} = \{L_1, ..., L_r\}$. Let G be a function associating each $(\mathbf{L}, E)_M$ to a $\mathbb{Q}$-power series in y that is independent of the order of the L_i. Then if:

1. $G(M, E) = \chi_y(M, E)$
2. $$\begin{aligned} G(\{L_1, \cdots, L_r, A \otimes B\}, E)_M = \\ & G(\{L_1, \cdots, L_r, A\}, E)_M \\ & + G(\{L_1, \cdots, L_r, B\}, E)_M \\ & + (y-1)G(\{L_1, \cdots, L_r, A, B\}, E)_M \\ & - yG(\{L_1, \cdots, L_r, A, B, A \otimes B\}, E)_M \end{aligned}$$
3. If D is a (nonsingular) divisor of M and $L_1 = \mathcal{L}(D)$, then:

$$G(\mathcal{L}, E)_M = G(\{(L_2)_D, ..., (L_r)_D\}, E_D)_D$$

Then for all $(\mathcal{L}, E)_M$ with $r \geq 1$:

$$\chi_y(\mathbf{L}, E)_M = G(\mathbf{L}, E)_M$$

The analogous result holds with the virtual characteristic χ in place of the virtual χ_y-characteristic, provided condition (2) is replaced by its $y = 0$ specialization (derived in the proof of theorem 3.4.5) and condition (1) by $G(M, E) = \chi(M, E)$.

***Proof*:**
We outline the result for the virtual χ_y-characteristic, leaving the case of the virtual characteristic χ as an exercise. First, the virtual χ_y-characteristic has properties (2) and (3) and therefore the function $\chi_y - G$ has properties (2) and (3). It is therefore sufficient to show that any function G' which satisfies (2), (3) and the additional property that:

$$G'(M, E) = 0$$

must be identically zero.

We proceed by induction on $\dim M$. In the base case $\dim M = 0$ every line bundle is trivial, so property (3) together with $G'(M, E) = 0$ already gives $G' = 0$ there. For the inductive step, by theorem 3.2.12 and theorem 3.2.10 there are divisors D and S of M such that $\mathcal{L}(D) = L_1 \otimes \mathcal{L}(S)$. Then (2) implies the equation

$$\begin{aligned} &G'(\{\mathcal{L}(D), L_2, \cdots, L_r\}, E)_M \\ &\quad = G'(\{L_1, \cdots, L_r\}, E)_M + G'(\{\mathcal{L}(S), L_2, \cdots, L_r\}, E)_M \\ &\quad + (y-1)G'(\{\mathcal{L}(S), L_1, \cdots, L_r\}, E)_M - yG'(\{\mathcal{L}(D), \mathcal{L}(S), L_1, \cdots, L_r\}, E)_M \end{aligned}$$

which G' satisfies by property (2) (it is the functional equation of theorem 3.4.5). This equation has five terms, and we have to prove that the second term is zero. The induction hypothesis and (3) imply that terms 1, 3, 4, 5 are zero. For $r = 1$, it is necessary to use $G' = 0$ to prove that terms 1, 3 are zero. Therefore term 2 is zero.

This shall be important when we show that the T-characteristic, which holds information about the Chern character and Todd class, satisfies the same properties, and hence will have to be equal to it (comes back in theorem 6.2.4).

3.5 *Vanishing Theorems

In this section, we mention but do not prove two important vanishing results that will let us sometimes reduce χ to H^0, therefore making it particularly simple to answer the Riemann-Roch Problem:

Theorem 3.5.1: Kodaira Vanishing Theorem

Let L be a complex line bundle over a compact complex manifold M. If L^{-1} is positive, then the cohomology groups $H^i(M, L)$ vanish for all $i \neq n$, where $n = \dim_{\mathbb{C}} M$.

***Proof*:**
[27]

Using Serre duality, we get

Theorem 3.5.2: Kodaira Vanishing Theorem: Euler Characteristic

Let L, M be as before and let K be the canonical bundle. If $L \otimes K^{-1}$ is positive, then the groups $H^i(M, L)$ vanish for all $i > 0$, hence:

$$\dim H^0(M, L) = \chi(M, L)$$

***Proof*:**
[27]

4

Todd Class

In section 2.4, we defined the Chern character which gives:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2) \in H^*(M, \mathbb{Q})$$

In section 3.3, we saw that we can define the Euler characteristic for vector bundles and that:

$$\chi(M, E_1 \oplus E_2) = \chi(M, E_1) + \chi(M, E_2)$$

As the Euler class is given by the top Chern class (recall remark 1.2.1), this algebraic property gives the hope that we can directly generalize Chern's Theorem to:

$$\chi(M, E) = \int_M \mathrm{ch}(E)$$

However, this does not give the expected result:

Example 4.0.1: Incorrect Generalization

Let $E = \mathbb{1}$ be the trivial bundle. Then $\mathrm{ch}(\mathbb{1}) = 1$. Now consider $M = S^2$. Then:

$$2 = \chi_{\mathrm{Top}}(S^2) \neq \int_{S^2} \mathrm{ch}(\mathbb{1}) = 0$$

showing we do not recover the Euler characteristic. (Note that $\chi(S^2) = 2$ here is the topological Euler characteristic; this is in contrast with the arithmetic genus $\chi(\mathbb{C}\mathrm{P}^1) = 1$ used later in this chapter.)

The problem is that we are not taking into consideration the geometric information of M[1]. What would be desired is a "correction class" $C(TM)$ (C for correction), where TM is the tangent bundle,

[1] recall the Euler class is not defined directly as the top Chern class on TM to take into account the geometric information of the given manifold, see [12]

such that:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \smile C(TM)$$

This shall be the *Todd class* $\mathrm{td}(M)$ where $\mathrm{td}(M) = C(TM)$. What properties should $\mathrm{td}(-)$ require? For one, we must have:

$$\chi(M, \mathbb{1}) = \chi(M) = \int_M \mathrm{td}(M)$$

For another, as $\chi(M \times Y) = \chi(M) \cdot \chi(Y)$, $\mathrm{td}(M \times Y) = \mathrm{td}(M) \smile \mathrm{td}(Y)$, namely it is *multiplicative* on direct products. Finally, we shall be able to reduce to proving the result on the complex projective spaces (see theorem 5.1.10), hence it is necessary to have $\mathrm{td}(\mathbb{C}\mathrm{P}^k)$ agree with $\chi(\mathbb{C}\mathrm{P}^k)$. In particular, as we know the line bundles over $\mathbb{C}\mathrm{P}^k$, we can deduce $\mathrm{td}(-)$ by making it agree with the values of $\chi(-)$.

Let us show the strategy in the one-dimensional case. First, we shall show that $\mathrm{td}(-)$ will correspond to a power series:

$$Q(x) = 1 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$$

which will return the appropriate polynomial of Chern classes given the dimension of $\mathbb{C}\mathrm{P}^n$ (exactly how is explained in section 4.1). The first term of this series gives the information for $\mathbb{C}\mathrm{P}^1$, hence let's find it by ensuring $\mathrm{td}(\mathbb{C}\mathrm{P}^1) = \chi(\mathbb{C}\mathrm{P}^1)$. Recall that $T\mathbb{C}\mathrm{P}^1 = \mathcal{O}(2)$, the twisted line bundle, and $\chi(\mathbb{C}\mathrm{P}^1) = 1$. Thus we want: $\mathrm{td}(\mathbb{C}\mathrm{P}^1) = Q\left(c_1(\mathcal{O}(2))\right)$. It is easy to compute that

$$c_1(\mathcal{O}(2)) = 2h \in H^*(\mathbb{C}\mathrm{P}^1; \mathbb{Z}) \cong \mathbb{Z}[h]/(h^2)$$

as $h^2 = 0$, we get:

$$\begin{aligned}
\chi(\mathbb{C}\mathrm{P}^1) = 1 &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{td}(\mathbb{C}\mathrm{P}^1) \\
&= \int_{\mathbb{C}\mathrm{P}^1} Q(c_1(T\mathbb{C}\mathrm{P}^1)) \\
&= \int_{\mathbb{C}\mathrm{P}^1} Q(2h) \\
&= \int_{\mathbb{C}\mathrm{P}^1} (1 + 2a_1 h) = 2a_1
\end{aligned}$$

Thus, it must be that $a_1 = \frac{1}{2}$, giving us:

$$Q(x) = 1 + \frac{1}{2}x + a_2 \frac{x^2}{2!} + \cdots$$

Notice that if we consider $\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d))$, then we get:

$$\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d)) = d + 1$$

and we compute

$$\mathrm{ch}(\mathcal{O}(d)) = e^{dh} = 1 + dh$$

as $h^2 = 0$. Thus:

$$\begin{aligned}
\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d)) = d+1 &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile \mathrm{td}(\mathbb{C}\mathrm{P}^1) \\
&= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile Q(c_1(T\mathbb{C}\mathrm{P}^1)) \\
&= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile Q(2h) \\
&= \int_{\mathbb{C}\mathrm{P}^1} (1+dh) \smile (1+h) \\
&= \int_{\mathbb{C}\mathrm{P}^1} [1+(1+d)h] = d+1
\end{aligned}$$

This gives equality in this special case. Repeating this process again for $\mathbb{C}\mathrm{P}^k$ for $k \in \mathbb{N}$, we shall get the next coefficients. In section 4.1, we go over the general algebraic method for finding such a power series, and in lemma 4.1.7 we shall show how we deduce the rest of this Todd power series. Overall, repeating this process will define $\mathrm{td}(-)$ and by construction it agrees on at least the Euler characteristic of $\mathbb{C}\mathrm{P}^n$ with the standard line bundles. This shall be enough in the proof of Hirzebruch-Riemann-Roch by taking advantage of the results on Cobordism and the splitting principle.

After defining the Todd class, it shall be combined with the Chern character to create the *T-characteristic* which shall be shown to behave well algebraically and to have the desired properties for any vector bundle $E \to M$ over a space M.

4.1 Algebraic Build-up: Multiplicative sequences

The direct sum of bundles translates to a cup product of characteristic classes. This gives a natural algebraic relation that we now codify for future reference. Let B be a commutative ring, and define:

$$\mathcal{B} = B[p_1, p_2, ...]$$

Define a grading given by the following:

$$p_{j_1} p_{j_2} \cdots p_{j_r} \text{ has weight } j_1 + j_2 + \cdots + j_r$$

Then let $\mathcal{B}_k$ be the weight-k groups; note $\mathcal{B}_0 = B$ and $\mathcal{B}_r \mathcal{B}_s = \mathcal{B}_{r+s}$. Let $\pi(k)$ be the number of partitions of k. Then $\mathcal{B}_k$ is a B-module of rank $\pi(k)$.

Definition 4.1.1: Multiplicative Sequence

Let $\{K_j\}$ be a sequence of polynomials in indeterminates p_i where $K_0 = 1$ and $K_j \in \mathcal{B}_j$. Then the sequence $\{K_j\}$ is called a *multiplicative sequence* or *m-sequence* if whenever a power series is the product of two power series:

$$1 + p_1 z + p_2 z^2 + \cdots = (1 + p'_1 z + p'_2 z^2 + \cdots)(1 + p''_1 z + p''_2 z^2 + \cdots)$$

with indeterminates z, p'_i, p''_i, then we get the identity:

$$\sum_{j=0}^{\infty} K_j(p_1, p_2, ...p_j) z^j = \left(\sum_{i=0}^{\infty} K_i(p'_1, p'_2, ...p'_i) z^i \right) \left(\sum_{j=0}^{\infty} K_j(p''_1, p''_2, ...p''_j) z^j \right)$$

that is:

$$K\left(\sum_{i=0}^{\infty} q_i z^i \right) = \sum_{i=0}^{\infty} K_i(q_1, ..., q_i) z^i$$

where $q_i \in \mathcal{B}$ of weight i. In other words a sequence $\{K_j\}$ is called multiplicative if it induces an endomorphism on

$$B[p_1, p_2, ...]\,[[z]]$$

Given a multiplicative sequence $\{K_j\}$, we associate to it its *characteristic power series* as the image of $K(1+z)$:

$$K(1+z) = \sum_{j=0}^{\infty} K_j(1, 0, ..., 0) z^j = \sum_{j=0}^{\infty} b_j z^j$$

An important property of vector bundles is that they can be pulled back over a flag manifold so that they split into line bundles. This strategy shall be important to us, and hence we will be considering:

$$1 + p_1 z + \cdots + p_n z^n = \prod_{i=1}^{n} (1 + \beta_i z)$$

where the p_i are the elementary symmetric polynomials in the β_i's. By some elementary ring theory and theory of symmetric functions, we can interpret $\mathcal{B}$ as the ring of all symmetric functions in $\beta_1, ..., \beta_n$.

We can characterize m-sequences via 1-unit power series:

Proposition 4.1.2: Correspondence 1-unit Power Series and m-sequences

There is a bijective correspondence between m-sequences and formal power series with constant term 1.

***Proof*:**
First, let's show any m-sequence $\{K_j\}$ is completely determined by its characteristic power series

$$Q(z) = K(1+z)$$

Proof. By decomposing:

$$1 + p_1 z + \cdots + p_m z^m = \prod_{i=1}^{m} (1 + \beta_i z)$$

We get:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

Therefore any polynomial K_j with $j \leq m$ is determined as a symmetric polynomial in the β_i and hence as a polynomial in the p_i. This holds for arbitrary m, completing the proof. $\square$

Next, let

$$Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i \qquad b_i \in B$$

be a formal power series. Then there is an associated m-sequence $\{K_j\}$ where $K(1 + z) = Q(z)$.

Proof. Consider the usual decomposition:

$$1 + p_1 z + \cdots + p_n z^n = \prod_{i=1}^{n} (1 + \beta_i z)$$

and consider the product $\prod_{i=1}^{m} Q(\beta_i z)$, where the p_i are the elementary symmetric polynomials in the β_i. Then the coefficients of z^j are homogeneous of weight j. Thus, it is a polynomial $K_j^{(m)}(p_1, \ldots, p_j)$. It follows that $K_j^{(m)}$ does not depend on m for $m \geq j$.

Now, define $K_j = K_j^{(m)}$ for $m \geq j$. I claim $\{K_j\}$ is the required m-sequence. Indeed, the earlier proof shows the multiplicative property is true if the indeterminates p_i', p_i'' are replaced by 0 for large values of i, and so $\{K_j\}$ is an m-sequence. Finally applying $m = 1$ in equation:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

shows that $K(1 + z) = Q(z)$, completing the proof. $\square$

Combining these, we get the correspondence.

The simplest example would be $\{p_i\}$ which has the characteristic series $1 + z$. The notation p_i is due to the connection to Pontryagin classes. Recall Pontryagin classes are given in the $4i$th cohomology group. The HRR theorem shall work with Chern classes which are in the $2i$th cohomology groups, and so the following will be a useful substitution. Let $c_0 = p_0 = 1$, $z = x^2$, and γ_i be such that $\beta_i = \gamma_i^2$. Then:

$$\sum_{i=0}^{\infty} p_i (-z)^i = \left(\sum_{j=0}^{\infty} c_j (-x)^j \right) \left(\sum_{i=0}^{\infty} c_i x^i \right)$$

Naturally, if $\{K_j(p_1, ..., p_j)\}$ is an m-sequence with $Q(z)$ as the characteristic power series, then let

$\{\widetilde{K}_j(c_1, ...c_j)\}$ be an m-sequence where $\widetilde{Q}(x) = Q(x^2)$. Then:

$$K_j(p_1, ..., p_j) = \widetilde{K}_{2j}(c_1, ..., c_{2j})$$
and
$$0 = \widetilde{K}_{2j+1}(c_1, ..., c_{2j+1})$$

Thus, the m-sequence in c_i with the characteristic power series $1 + x^2$ is given by

$$1, 0, p_1, 0, p_2, ...$$

From these relations, we get:

$$\begin{aligned} p_1 &= -2c_2 + c_1^2 \\ p_2 &= 2c_4 - 2c_3c_1 + c_2^2 \\ p_3 &= -2c_6 + 2c_5c_1 - 2c_4c_2 + c_3^2, ... \end{aligned}$$

Let's now find a way to explicitly calculate the polynomials of m-sequences. Given $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$, consider the factorization:

$$1 + b_1 z + b_2 z^2 + \cdots + b_m z^m = (1 + \beta_1' z) \cdots (1 + \beta_m' z)$$

Then define the sum:

$$\sum (\beta_1')^{j_1} (\beta_2')^{j_2} \cdots (\beta_r')^{j_r} \qquad \text{where} \qquad j_1 \geq j_2 \geq \cdots \geq j_r \geq 1, \ \sum_{s=1}^{r} j_s = k \leq m$$

which denotes the symmetric function in the β_i' which is the sum of all pairwise distinct monomials obtained by permuting $(\beta_1', ..., \beta_m')$ to the monomial $(\beta_1')^{j_1} (\beta_2')^{j_2} \cdots (\beta_r')^{j_r}$. The number of monomials in the sum is $m!/h$, h being the number of permutations leaving the aforementioned monomial fixed. Now:

Lemma 4.1.3: Computation of m-sequence

Let $\{K_j(p_1, ..., p_j)\}$ be the m-sequence corresponding to $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$. Then the coefficient of $p_{j_1} p_{j_2} \cdots p_{j_r}$ in K_k where $j_1 \geq j_2 \geq \cdots \geq j_r \geq 1$ and $\sum_{s=1}^{r} j_s = k$ is equal to:

$$\sum (j_1, j_2, ..., j_r)$$

***Proof*:**
It is a computation that will be omitted.

Define:

$$Q(z) = \frac{\sqrt{z}}{\tanh \sqrt{z}} = 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \frac{2^{2k}}{(2k)!} B_k z^k$$

where the B_k are the Bernoulli numbers. Note that in this case the coefficient ring is $B = \mathbb{Q}$. Furthermore, the B_k here are the Bernoulli numbers with signs removed and with the value $\frac{1}{2}$ omitted from the indexing (so all $B_k > 0$). Thus we get:

$$B_1 = \frac{1}{6},\ B_2 = \frac{1}{30},\ B_3 = \frac{1}{42},\ B_4 = \frac{1}{30},\ B_5 = \frac{5}{66},\ B_6 = \frac{691}{2730},\ B_7 = \frac{7}{6},\ B_8 = \frac{3617}{510}, ...$$

We shall denote the m-sequence of $Q(z)$ as $\{L_j(p_1, ..., p_j)\}$. Computing we get:

$$\begin{aligned}
L_1 &= \frac{1}{3}p_1 \\
L_2 &= \frac{1}{3^2 \cdot 5}(7p_2 - p_1^2) \\
L_3 &= \frac{1}{3^3 \cdot 5 \cdot 7}(62p_3 - 13p_2p_1 + 2p_1^3) \\
&\vdots
\end{aligned}$$

We will want the following normalization: $L_k(p_1, ..., p_k) = 1$. To that end, if we let $p_i = \binom{2k+1}{i}$, then we get:

$$1 + p_1 z + p_2 z^2 + \cdots + p_k z^k = (1+z)^{2k+1} \bmod z^{k+1}$$

From this we get:

Proposition 4.1.4: Coefficient of L-characteristic Series

Let $Q(z) = \frac{\sqrt{z}}{\tanh\sqrt{z}}$. Then for every k, the coefficient J_k of z^k in $(Q(z))^{2k+1}$ is equal to 1, and $Q(z)$ is the *only* power series with rational coefficients with this property.

***Proof*:**
Apply Cauchy's integral formula and compute (see [16, p.12] to see the exact computations)

The above is useful for Pontryagin classes. Let us find the equivalent for Chern classes. Let:

$$Q(x) = \frac{x}{1-e^{-x}} = 1 + \frac{1}{2}x + \sum_{k=1}^{\infty}(-1)^{k-1}\frac{B_k}{(2k)!}x^{2k}$$

be the characteristic power series. Then we get an m-sequence of polynomials $\{T_k\}$ called *Todd polynomials*. For future reference, we have the identity:

$$\frac{x}{1-\exp(-x)} = \exp\left(\frac{1}{2}x\right)\frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$$

The m-sequence for $Q(z) = \frac{2\sqrt{z}}{\sinh(2\sqrt{z})}$ is given by:

$$A_1 = \frac{-2}{3}p_1,\ A_2 = \frac{2}{45}(-4p_2 + 7p_1^2),\ A_3 = \frac{-4}{3^3 \cdot 5 \cdot 7}(16p_3 - 44p_2p_1 + 31p_1^3),\ \cdots$$

By using the translation from Pontryagin to Chern class lemma, we get:

$$T_k(c_1, ..., c_k) = \sum_{\substack{r,s \\ r+2s=k}} \frac{1}{2^{4s}r!}\left(\frac{1}{2}c_1\right)^r A_s(p_1, ..., p_s)$$

We can compute the first few values:

$$\begin{aligned}
T_1 &= \frac{1}{2}c_1 \\
T_2 &= \frac{1}{12}(c_2 + c_1^2) \\
T_3 &= \frac{1}{24}c_2c_1 \\
T_4 &= \frac{1}{720}(-c_4 + c_3c_1 + 3c_2^2 + 4c_2c_1^2 - c_1^4) \\
T_5 &= \frac{1}{1440}\left(-c_4c_1 + c_3c_1^2 + 3c_2^2c_1 - c_2c_1^3\right) \\
&\vdots
\end{aligned}$$

The reader should already notice how the Riemann-Roch for curves and surfaces has part of the geometric component captured within the 1st and 2nd Todd polynomials.

Again, by substituting $c_i = \binom{n+1}{i}$ defined by:

$$1 + c_1x + \cdots + c_nx^n = (1+x)^{n+1} \bmod x^{n+1}$$

gives

$$T_n(c_1, ..., c_n) = 1$$

Just like before, $Q(x)$ is the only power series with rational coefficients with the property that the coefficient of x^k in $(Q(x))^{k+1}$ is 1 (see [16, p.14], or just repeat the above results with the appropriate modifications).

Recall the Euler characteristic is called the *arithmetic genus* (following Hartshorne's vocabulary). This will motivate how evaluating characteristic series on tangent bundles by selecting their Pontryagin/Chern classes shall be called the *genus* with respect to the characteristic series. Depending on the characteristic series, we shall give a name to the genus.

Definition 4.1.5: K-genus

Let M be an oriented n-dimensional manifold and $\{K_j(p_1, ..., p_j)\}$ be an m-sequence. Then:

$$\begin{cases} 0 & \dim_{\mathbb{R}} M \not\equiv 0 \bmod 4 \\ \int_M K_k(p_1, ..., p_k) & \dim_{\mathbb{R}} M = 4k \end{cases}$$

is called the *K-genus* of M, where $\int_M K_k(p_1, ..., p_k)$ represents taking the tangent bundle TM, finding the Pontryagin classes, and evaluating at the fundamental class $[M]$.

If we are working with Chern classes, then we would have $n = 2k$.

By what we've discussed earlier, the K-genus is multiplicative.

Proposition 4.1.6: K-genus is Multiplicative

$$K(M \times N) = K(M)K(N)$$

***Proof*:**
immediate from the properties of characteristic series.

4.1.1 Generating Function

As seen in chapter 3, we defined a natural generating function χ_y combining the Euler characteristics χ^p. The idea is that we can figure out the coefficients of this generating function when plugging in different values of the dummy variable y. Let us do the same for the T-characteristic. Let $B = \mathbb{Q}[y]$. Then consider the m-sequence $T_j(y; c_1, ..., c_j)$ with characteristic power series:

$$Q(y; x) = \frac{x(y+1)}{1 - e^{-x(y+1)}} - yx = \frac{x(y+1)}{e^{x(y+1)} - 1} + x$$

Lemma 4.1.7

For every n the coefficient of x^n in $(Q(y; x))^{n+1}$ is equal to $\sum_{i=0}^{n}(-1)^i y^i$, and $Q(y; x)$ is the only power series with coefficients in $\mathbb{Q}[y]$ which has this property.

***Proof*:**
Take the substitution

$$c_i = \binom{n+1}{i}$$

(the Chern classes of $\mathbb{C}\mathrm{P}^n$). The coefficient of x^n in $(Q(y; x))^{n+1}$ is then computed via the Cauchy integral formula in [16, lemma 1.8.1], yielding:

$$T_n(y; c_1, \ldots, c_n) = 1 - y + y^2 - \cdots + (-1)^n y^n.$$

Consistently, as $Q(\frac{1}{y}; yx) = Q(y; -x)$:

$$y^n T_n(\frac{1}{y}; c_1, \ldots, c_n) = (-1)^n T_n(y; c_1, \ldots, c_n)$$

as we sought to show. Uniqueness follows since these coefficient conditions determine the m-sequence, and hence its characteristic power series.

The polynomial $T_n(y; c_1, \ldots, c_n)$ can be written in a unique way in the form

$$T_n(y; c_1, \ldots, c_n) = \sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) y^p$$

satisfying the same grading as χ_y, and the polynomials $T_n^p(c_1, \ldots, c_n)$ satisfy

$$T_n^p(c_1, \ldots, c_n) = (-1)^n T_n^{n-p}(c_1, \ldots, c_n)$$

that is the "Poincaré duality". Now, consider the factorization:

$$1 + c_1 x + \cdots + c_n x^n = \prod_{i=1}^{n}(1 + \gamma_i x)$$

Then we have:

Proposition 4.1.8: Computing T_y-Polynomial

$$T_n^p(c_1,...,c_n) = \kappa_n\left[\sum \exp(-\gamma_{j_1} - \cdots - \gamma_{j_p}) \prod_{i=1}^n \frac{\gamma_i}{1-\exp(-\gamma_i)}\right]$$

where the sum is over the $\binom{n}{p}$ combinations $j_1 < \cdots < j_p$, and $\kappa_n[-]$ denotes the degree-n homogeneous component (mirroring evaluating at the fundamental class).

***Proof*:**
Indeed:

$$\begin{aligned}\sum_{p=0}^n T_n^p y^p &= \kappa_n\left[\prod_{i=1}^n (1+y\exp(-\gamma_i))\frac{\gamma_i}{1-\exp(-\gamma_i)}\right] \\ &= \kappa_n\left[\prod_{i=1}^n \frac{(1+y\exp(-(1+y)\gamma_i))}{1+y}\cdot\frac{(1+y)\gamma_i}{1-\exp(-(1+y)\gamma_i)}\right] \\ &= \kappa_n\left[\prod_{i=1}^n Q(y;\gamma_i)\right] = \sum_{p=0}^n T_n^p(c_1,\dots,c_n)y^p\end{aligned}$$

where the middle equality is not a formal identity: it holds after applying κ_n, via the substitution $\gamma_i \mapsto (1+y)\gamma_i$, which scales the degree-n component by $(1+y)^n$. This is what we sought to show

Finally note that $Q(0;x) = \frac{x}{1-e^{-x}}$, $Q(-1;x) = 1+x$ and $Q(1;x) = \frac{x}{\tanh x}$. Therefore:

$$T_n^0(c_1,\dots,c_n) = T_n(c_1,\dots,c_n) \text{ (Todd polynomial).}$$

Proposition 4.1.9: $y=-1$ and $y=1$ Values

1. $\sum_{p=0}^n (-1)^p T_n^p(c_1,\dots,c_n) = c_n$
2. $\sum_{p=0}^n T_n^p(c_1,\dots,c_n) = \tilde{L}_n(c_1,\dots,c_n)$, that is:

$$\sum_{p=0}^n T_n^p(c_1,\dots,c_n) = \begin{cases} 0 \text{ if } n \text{ is odd} \\ L_k(p_1,\dots,p_k) \text{ if } n = 2k. \end{cases}$$

***Proof*:**
exercise

The first equation we see links to the Euler-characteristic as the Euler class is given by the top Chern class, while the second equation will be equal to the signature as will be defined in chapter 5.

4.2 Todd Class

With the algebraic preliminaries established, we can formally define the Todd class:

Definition 4.2.1: Todd Class

Let M be a complex smooth manifold, and E a continuous q-vector bundle on M. Let $c_i \in H^{2i}(M, \mathbb{Z})$ be its Chern classes. Then we can define the *total Todd class* of E by taking the Todd polynomials and taking:

$$\operatorname{td}(E) = \sum_{j=0}^{\infty} T_j(c_1, ..., c_j)$$

where $\{T_j(c_1, ..., c_j)\}$ is given by the characteristic series:

$$Q(x) = \frac{x}{1 - e^{-x}} = 1 + \frac{1}{2}x + \sum_{j=1}^{\infty}(-1)^{j-1}\frac{B_j}{(2j)!}x^{2j}$$

where the B_j are the Bernoulli numbers.

The reader can verify that the Todd class is multiplicative on direct products of manifolds $M \times Y$. Separately, if there is a short exact sequence

$$0 \to E' \to E \to E'' \to 0$$

then:

$$(1 + c_1' t + \cdots + c_{q'}' t^{q'})(1 + c_1'' t + \cdots + c_{q''}'' t^{q''}) = 1 + c_1 t + \cdots + c_q t^q$$

and so:

$$\operatorname{td}(E) = \operatorname{td}(E') \cdot \operatorname{td}(E'')$$

This mirrors the property of the Euler characteristic.

In the special case where $q = 1$ and $c_1(E) = d \in H^2(M, \mathbb{Z})$:

$$\operatorname{td}(E) = \frac{d}{1 - e^{-d}}$$

Then for a general q-bundle we can apply the splitting principle. In terms of the Todd class, we define:

$$\sum_{j=0}^{q} c_j x^j = \prod_{i=1}^{q}(1 + \gamma_i x)$$

Thus, when dealing with a general complex vector bundle E and taking its Chern roots γ_i:

$$\operatorname{td}(E) = \prod_{i=1}^{q} \frac{\gamma_i}{1 - e^{-\gamma_i}}$$

Let us compare this to the usual results we get about the Chern character: it is the natural "dual" of the Todd class in the following way:

Theorem 4.2.2: Borel-Serre Identity

Let E be a continuous $\mathrm{GL}_q(\mathbb{C})$-bundle over M. Then:

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch}\left(\bigwedge^r E^*\right) = (\mathrm{td}(E))^{-1} c_q(E)$$

***Proof*:**
If $\sum_{j=0}^{q} c_j(E)x^j = \prod_{i=1}^{q}(1+\gamma_i x)$ by the properties of Chern classes

$$\mathrm{ch}\left(\bigwedge^r E^*\right) = \sum e^{-(\gamma_{i_1}+\cdots+\gamma_{i_r})}$$

where the sum is over all combinations $i_1,\ldots,i_r$ with $1 \le i_1 < \cdots < i_r \le q$. Therefore

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch}\bigwedge^r E^* = \prod_{i=1}^{q}(1-e^{-\gamma_i}) = (\gamma_1\cdots\gamma_q)\prod_{i=1}^{q}\frac{1-e^{-\gamma_i}}{\gamma_i} = (\mathrm{td}(E))^{-1}c_q(E).$$

Recall that if we are talking about the Chern class of a complex manifold M, we are talking about the Chern class of its tangent bundle, and the K-genus is defined for such structures:

Definition 4.2.3: Todd Genus

Let M be an n-dimensional almost complex manifold. The *Todd genus* (or *T-genus*) is the number:

$$T(M_n) = T_n(c_1,...,c_n)[M_n] = \kappa_n[\mathrm{td}(TM)] = \int_M \mathrm{td}(TM)$$

where $[M_n]$ represents the fundamental class of M_n.

The *generalized Todd Genus* is given by $\{T_n(y;c_1,...,c_n)\}$ and:

$$T_y(M_n) = \sum_{p=0}^{n} T^p(M_n)y^p$$

The generalized Todd genus is a natural generalization of the Todd genus as:

$$T_0(M_n) = T^0(M_n) = T(M_n)$$

Two important terms that we must evaluate are at $y = -1$ and $y = 1$. For $y = -1$:

$$\begin{aligned} T_{-1}(M_n) &= \sum_{p=0}^{n} (-1)^p T^p(M_n) \\ &= \prod_i Q(-1; \gamma_i)[M_n] \\ &= \prod_i (1 + \gamma_i)[M_n] \\ &\stackrel{!}{=} c_n[M_n] \\ &= \chi(M_n) \end{aligned}$$

where $\stackrel{!}{=}$ comes from ignoring the lower order terms because of the perfect pairing. Evaluating at 1, by proposition 4.1.9 we get:

$$T_1(M_n) = \sum_{p=0}^{n} T^p(M_n) = \tau(M_n)$$

where τ is shorthand for the signature of M_n. By lemma 4.1.7, the T-genus is the only genus associated to an m-sequence with rational coefficients which takes the value 1 on every complex projective space $\mathbb{CP}^n$, and the T_y-genus is the only genus associated to an m-sequence with coefficients in $\mathbb{Q}[y]$ which takes the value $1 - y + y^2 - \cdots + (-1)^n y^n$ on $\mathbb{CP}^n$.

Proposition 4.2.4: Properties of Todd Genus

Let B be a commutative ring where $\mathbb{Q} \subseteq B$. Then

1. $\sum_{l=0}^{n} (-1)^l T_n^{(l)}(c_1, \ldots, c_n) = c_n$, for all n.
2. $T_n^{(0)}(c_1, \ldots, c_n) = \mathrm{td}(c_1, \ldots, c_n) (= T_n(c_1, \ldots, c_n))$.
3. $$T_n^{(l)}(c_1, ..., c_n) = (-1)^n T_n^{(n-l)}(c_1, ..., c_n)$$
4. $$\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ L_{\frac{n}{2}}(p_1, \ldots, p_{\frac{n}{2}}) & \text{if } n \text{ is even.} \end{cases}$$

***Proof*:**
exercise

4.2.1 Virtual Todd Genus

Let us now translate the results over for splitting manifolds. Let $E \to M$ be a complex vector bundle over a manifold, M^Δ the associated flag manifold such that the pullback along $f : M^\Delta \to M$ makes E split into linear bundles. The same applies even if $M = V$ is a variety with singularities. We can break down the pullback of E into a filtration:

$$f^*(E) = E_r \supseteq E_{r-1} \supseteq \cdots \supseteq E_1 \supseteq E_0 = 0$$

This shall afford us the ability to calculate the Todd class, and hence the Todd genus, using line bundles.

Let us start by relating the Chern class to a sub-manifold. Let $V_{n-k} \subseteq M_n$ be a submanifold of codimension k and let $j : V_{n-k} \to M_n$ be an embedding. If M_n is almost complex, then we have the pullback of vector bundles:

$$j^*(TM_n) \cong TV_{n-k} \oplus N_{V/M} \tag{4.1}$$

where $N_{V/M}$ is the normal bundle of V with respect to M; let us label it ν. If $k = 1$, then we simply get:

$$c(\nu) = 1 + j^*(v)$$

where $v \in H^2(M_n; \mathbb{Z})$ is the cohomology class determined by the fundamental class of V_{n-1}. By equation (4.1):

$$\begin{aligned} &1 + c_1(V_{n-1}) + c_2(V_{n-1}) + \cdots \\ &= j^*\left((1 + c_1(M_n) + c_2(M_n) + \cdots)(1 + v)^{-1}\right) \end{aligned}$$

Then we can define the T_y-genus of V_{n-1}. To that end, define the characteristic power series:

$$Q(y; x) = \frac{x}{R(y; x)}$$

where $R(y; x)$ is a shorthand for $\frac{e^{x(y+1)}-1}{e^{x(y+1)}+y}$. Then equation (4.1) implies:

$$\sum_{i=0}^{\infty} T_i(y; c_1(V_{n-1}), ..., c_i(V_{n-1})) = j^*\left(\frac{R(y; v)}{v} \sum_{i=0}^{\infty} T_i(y; c_1(M_n), ...)\right)$$

and so the T_y-genus is:

$$T_y(V_{n-1}) = \kappa_n\left[R(y; v) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ..., c_j(M_n))\right]$$

which at $y = 0$ specializes to:

$$T(V_{n-1}) = \kappa_n[(1 - e^{-v})\mathrm{td}(TM)]$$

both being immediate applications of equation (4.1). Now, by the splitting principle, we can filter TM, or any vector bundle E, where each intermediate step is linear. This motivates the following definition:

Definition 4.2.5: Virtual Todd Genus

Let $v_1, ..., v_r \in H^2(M_n; \mathbb{Z})$. Then:

$$T_y(\{v_1, ..., v_r\})_M = \kappa_n\left[R(y; v_1) \cdots R(y; v_r) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ...)\right]$$

Define $T^p(\{v_1, ..., v_r\})_M$ to be the coefficient of y^p in T_y when expanded:

$$T_y(\{v_1, ..., v_r\})_M = \sum_{p=0}^{n-r} T^p(\{v_1, ..., v_r\})_M y^p$$

By proposition 4.1.8, the virtual Todd genus is a polynomial of degree $n-r$ in y with rational coefficients. If $r > n$, it is 0. If $r = n$:

$$T_y(\{v_1, ..., v_n\})_M = v_1 v_2 \cdots v_n[M_n]$$

The virtual Todd genus respects the same properties as the Todd genus, namely:

$$T(\{v_1, ..., v_r\})_M = T_0(\{v_1, ..., v_r\})_M = T^0(\{v_1, ..., v_r\})_M$$

and

$$T^p(\{v_1, ..., v_r\})_M = (-1)^{n-r} T^{n-r-p}(\{v_1, ..., v_r\})_M$$

And by definition:

$$T_y(V_{n-1}) = T_y(\{v\})_M$$

for appropriate v and where $V_{n-1} \hookrightarrow M_n$ is a codimension 1 manifold. Let us now work towards showing that the virtual Todd genus satisfies some of the algebraic properties of theorem 3.4.9. Let

$$R(x) = \frac{e^{ax} - 1}{e^{ax} + y}$$

for some indeterminates a, y. Then:

$$R(u+v) = R(u) + R(v) + (y-1)R(u)R(v) - yR(u)R(v)R(u+v)$$

Letting $a = 1 + y$, we get an equation for $R(y; x)$. Then as T_y decomposes into products of such equations we get:

Theorem 4.2.6: Functional Equation of T_y

Let $v_1, ..., v_r, u, v \in H^2(M_n; \mathbb{Z})$. Then:

$T_y(\{v_1, ..., v_r, u+v\}) =$
$T_y(\{v_1, ..., v_r, u\}) + T_y(\{v_1, ..., v_r, v\}) + (y-1)T_y(\{v_1, ..., v_r, u, v\}) - yT_y(\{v_1, ..., v_r, u, v, u+v\})$

If $r = 0$:

$$T_y(\{u+v\}) = T_y(\{u\}) + T_y(\{v\}) + (y-1)T_y(\{u, v\}) - yT_y(\{u, v, u+v\})$$

As $y = -1$ gives the Euler characteristic, we see that it satisfies:

$$T_{-1}(\{u+v\}) = T_{-1}(\{u\}) + T_{-1}(\{v\}) - 2T_{-1}(\{u, v\}) + T_{-1}(\{u, v, u+v\})$$

When $y = 0$, we get the Todd genus satisfies:

$$T(\{u+v\}) = T(\{u\}) + T(\{v\}) - T(\{u, v\})$$

and when $y = 1$, we shall see that the signature will satisfy the equation:

$$T_1(\{u+v\}) = T_1(\{u\}) + T_1(\{v\}) - T_1(\{u, v, u+v\})$$

4.3 T-characteristic: Combining Algebraic and Geometric Information

So far, we saw that the Todd class captures some information about the underlying space M. We shall want to capture some information about the bundle that is given by the Chern character. Combining these two together, we get:

Definition 4.3.1: T-characteristic

Let M be a compact complex manifold, E a complex vector bundle. Then

$$T(M,E) = \kappa_n[\mathrm{ch}(E)\mathrm{td}(M)] = \int_M \mathrm{ch}(E)\cdot\mathrm{td}(M)$$

is called the *T-characteristic* of E over M.

Note that if $E = \mathbb{1}$, then $T(M,\mathbb{1}) = \kappa[\mathrm{ch}(\mathbb{1})\mathrm{td}(TM)] = \kappa[\mathrm{td}(TM)]$ showing that we reduce to only holding information about M. As $\mathbb{C}^*$-bundles over M_n are classified by elements $d \in H^2(M_n,\mathbb{Z})$, we shall sometimes write $T(M_n, d)$ for $T(M_n, E)$ where E is associated to d. By the results in section 4.2.1:

Proposition 4.3.2: Separating Manifold and Bundle

$$T(M,d) = T(M) - T(\{-d\})_M$$

***Proof*:**
this can also be proven directly:

$$\begin{aligned}
&= T(M_n) - T(\{-d\})_M\\
&= \kappa_n[\mathrm{td}(TM)] - \kappa_n[(1-e^d)\cdot\mathrm{td}(TM)]\\
&= \kappa_n[\mathrm{td}(TM) - (1-e^d)\cdot\mathrm{td}(TM)]\\
&= \kappa_n[\big(1-(1-e^d)\big)\cdot\mathrm{td}(TM)]\\
&= \kappa_n[\big(1-1+e^d\big)\cdot\mathrm{td}(TM)]\\
&= \kappa_n[e^d\cdot\mathrm{td}(TM)]\\
&= T(M,d)
\end{aligned}$$

Notice how this mirrors the same property as the Euler characteristic (proposition 3.4.2). If E is a complex vector bundle of dimension q, E' of dimension q', both over M_n, then using the above equation we get:

$$T(M_n, E\oplus E') = T(M_n,E) + T(M_n,E')$$

Then by the properties of the characters we get:

$$T(M\times N, \pi_M^*(E)\otimes\pi_N^*(\eta)) = T(M,E)T(N,\eta)$$

Let us extend this to T_y-characteristics and show that it is additive and multiplicative in the same way. First, by proposition 4.1.8:

$$T\left(M_n, \bigwedge^p (T^*M)\right) = T^p(M_n)$$

Then define $T^p(M_n, E) := T\left(M_n, \bigwedge^p (T^*M) \otimes E\right)$, and set:

$$T_y(M_n, E) = \sum_{p=0}^{n} T^p(M_n, E) y^p$$

Thus we can re-write the T-characteristic as:

$$T^p(M_n, E) = \kappa_n \left[\mathrm{ch}(E)\mathrm{ch}\left(\bigwedge^p (T^*M)\right) \mathrm{td}(TM)\right]$$

Then by applying the same proof techniques for proposition 4.1.8 we get:

$$T_y(M_n, E) = \kappa_n \left[\left(\sum_{i=1}^{q} e^{(1+y)d_i}\right)\left(\sum_{j=0}^{n} T_j(y; c_1, \ldots, c_j)\right)\right]$$

where the d_i are the Chern roots of E. Notice that:

$$T_{-1}(M_n, E) = q \cdot \chi(M_n)$$

recovering the Euler-Poincaré characteristic of M_n, and multiplying by q as it is the dimension of E. Similarly, generalizing proposition 4.2.4 we get:

$$T^p(M_n, E) = (-1)^n T^{n-p}(M_n, E^*)$$

and if $p = 0$:

$$T(M_n, E) = (-1)^n T\left(M_n, \bigwedge^n (T^*M) \otimes E^*\right)$$

Recovering the duality of χ^p. Next, by extending $\mathrm{ch}(-)$ so that it works on generating functions:

$$\mathrm{ch}_{(y)}(E) = e^{(1+y)\gamma_1} + \cdots + e^{(1+y)\gamma_q}$$

where γ_i are the Chern roots of E. Then:

$$\mathrm{ch}_{(y)}(E \oplus E') = \mathrm{ch}_{(y)}(E) + \mathrm{ch}_{(y)}(E') \qquad \text{and} \qquad \mathrm{ch}_{(y)}(E \otimes E') = \mathrm{ch}_{(y)}(E) \cdot \mathrm{ch}_{(y)}(E')$$

and so combining with the previous results we get:

$$T_y(M_n, E \oplus E') = T_y(M_n, E) + T_y(M_n, E')$$

and

$$T_y(M \times V, \pi_M^*(E) \otimes \pi_V^*(\eta)) = T_y(M, E)T_y(V, \eta)$$

Let us now develop this to define the virtual T-characteristic. Motivated by the case for the Euler characteristic (definition 3.4.1), we define the following:

Definition 4.3.3: Virtual T-characteristic

$$T_y(\{v_1, ..., v_r\}, E)_M = \kappa_n \left[\mathrm{ch}_{(y)}(E) \prod_{i=1}^{r} R(y; v_i) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ...) \right]$$

Proposition 4.3.4: Properties of Virtual T-characteristic and Submanifolds

1. Let V_{n-1} be an almost complex submanifold of M_n, $v \in H^2(M_n, \mathbb{Z})$ the cohomology class determined by V_{n-1} and $j : V_{n-1} \hookrightarrow M_n$ be its embedding into M_n. Let $v_2, ..., v_r \in H^2(M_n, \mathbb{Z})$ and E a q-dimensional complex bundle over M_n. Then:
$$T_y(\{j^*v_2, ..., j^*v_r\}, j^*(E))_V = T_y(\{v, v_2, ..., v_r\}, E)_M$$
2. the functional equation in theorem 4.2.6 extends to the virtual T-characteristic.
3. $T_y(\{v_1, ..., v_r\}, E)$ is a degree-$(n - r)$ polynomial in y with rational coefficients, and if $r = n$:
$$T_y(\{v_1, ..., v_n\}, E) = q \cdot \int_M v_1 \cdots v_n$$
4. $y^{n-r}T_{\frac{1}{y}}(\{v_1, ..., v_r\}, E)_M = (-1)^{n-r}T_y(\{v_1, ..., v_r\}, E^*)_M$
5. If M is an n-dimensional smooth manifold, L is a line bundle over M with Chern class $1 + v$, $v \in H^2(M, \mathbb{Z})$, E is a q-dimensional vector bundle over M, and $v_1, ..., v_r \in H^2(M, \mathbb{Z})$, then:
$$T_y(\{v_1, ..., v_r\}, E)_M =$$
$$T_y(\{v_1, ..., v_r, v\}, E)_M + T_y(\{v_1, ..., v_r\}, E \otimes L^{-1})_M + yT_y(\{v_1, ..., v_r, v\}, E \otimes L^{-1})_M$$
6. following the above convention and letting $V \subseteq M$ be a codimension 1 submanifold:
$$T^p(M_n, E) = T^p(V_{n-1}, j^*(E)) + T^p(M_n, E \otimes L^{-1}) + T^{p-1}(V, j^*(E \otimes L^{-1}))$$

***Proof*:**
definition unpacking.

Let us apply this to a split manifold. Let M be a compact almost complex split manifold of complex dimension n so that $c(M) = \prod_{i=1}^{n}(1 + a_i)$ for appropriate a_i. Then the Todd genus $T(M)$ can be expressed in terms of virtual signatures:

Proposition 4.3.5: Todd Genus in Terms of Virtual T-characteristic

Let M be a complex split manifold, TM split into $A_1, ..., A_n$, and let $a_i = c_1(A_i)$. Then:

$$(1+y)^n T(M) = \sum_{l=0}^{n} y^l \sum_{1 \leq i_1 < \cdots < i_l \leq n} T_y(\{a_{i_1}, \ldots, a_{i_l}\})_M$$

***Proof*:**

$$\begin{aligned}
\kappa_n \left[\prod_{i=1}^{n} (1 + yR(y; a_i)) \prod_{i=1}^{n} Q(y; a_i) \right] &= \kappa_n \left[\prod_{i=1}^{n} (Q(y; a_i) + a_i y) \right] \\
&= \kappa_n \left[\prod_{i=1}^{n} \frac{(1+y)a_i}{1 - \exp(-(1+y)a_i)} \right] \\
&= (1+y)^n \kappa_n \left[\prod_{i=1}^{n} \frac{a_i}{1 - \exp(-a_i)} \right] \\
&= (1+y)^n T(M)
\end{aligned}$$

4.4 Todd Genus Invariant Under Splitting

Let $j : M^\Delta \to M$ be the flag-bundle projection, along which the pullback of TM splits. Let us now show that $T(M) = T(M^\Delta)$, that is the Todd genus is invariant under splitting. Essentially, if $\mathbf{F}(n)$ is the appropriate flag manifold, then $T(\mathbf{F}(n)) = 1$, and we shall show

$$T(M^\Delta) = T(M)T(\mathbf{F}(n)) = T(M)$$

Algebraic Preliminaries for Flag Manifold having Trivial Todd Genus

Let $\mathbb{F}$ be a field of characteristic 0, and let $c_1, \ldots, c_n$ be indeterminates over $\mathbb{F}$. Take the field extension $\mathbb{F}(c_1, \ldots, c_n)(\gamma_1, \ldots, \gamma_n)$ where the c_i are the elementary symmetric functions of the γ_j. The $n!$ elements $\gamma_1^{a_1}\gamma_2^{a_2} \cdots \gamma_{n-1}^{a_{n-1}}$ with $0 \leq a_i \leq n - i$ form a basis for this extension.

Any formal power series P in $\gamma_1, \ldots, \gamma_n$ with coefficients in $\mathbb{F}$ can be uniquely expressed as:

$$P = \sum_{a_1, \ldots, a_{n-1}} e_{a_1, \ldots, a_{n-1}} \gamma_1^{a_1} \cdots \gamma_{n-1}^{a_{n-1}}$$

where the coefficients $e_{a_1, \ldots, a_{n-1}}$ are power series in $c_1, \ldots, c_n$.

Definition 4.4.1: Indicator

The *indicator* of a formal power series P in the variables $\gamma_1, \dots, \gamma_n$ is defined as:

$$e(P) = (-1)^{n(n-1)/2} e_{n-1,n-2,\dots,1}$$

where $e_{n-1,n-2,\dots,1}$ is the coefficient of the basis element $\gamma_1^{n-1}\gamma_2^{n-2}\cdots\gamma_{n-1}^1$.

The indicator can be calculated with the formula:

$$e(P) = \frac{\sum_{s \in S_n} \operatorname{sign}(s) \cdot s(P)}{\prod_{i>j}(\gamma_i - \gamma_j)}$$

where the denominator is the Vandermonde determinant.

Lemma 4.4.2: Invariance of Indicator

If a polynomial P is invariant under the interchange of γ_i and γ_j for some $i \neq j$, then its indicator $e(P)$ is zero.

Lemma 4.4.3: Indicator of a Product

Let $P = \prod_{i>j} \frac{\gamma_i - \gamma_j}{1-e^{-(\gamma_i-\gamma_j)}}$. Then the indicator is $e(P) = 1$.

The Invariance

The flag manifold $\mathbf{F}(n) = GL(n, \mathbb{C})/\mathcal{T}(n, \mathbb{C})$, where $\mathcal{T}(n, \mathbb{C})$ is the subgroup of upper triangular matrices, is a complex analytic split manifold. Its cohomology ring is given by [12]:

$$H^*(\mathbf{F}(n), \mathbb{Z}) \cong \mathbb{Z}[\gamma_1, \dots, \gamma_n]/I^+(c_1, \dots, c_n)$$

where I^+ is the ideal generated by the elementary symmetric functions of the γ_i of positive degree. The total Chern class of its tangent bundle is:

$$c(T\mathbf{F}(n)) = \prod_{i>j}(1 + \gamma_i - \gamma_j)$$

Theorem 4.4.4: Todd Genus of the Flag Manifold

The Todd genus of the flag manifold $\mathbf{F}(n)$ is 1.

***Proof*:**
This result is derived from lemma 4.4.3 and the definition of the Todd genus:

$$T(\mathbf{F}(n)) = \kappa_{n(n-1)/2}\left[\prod_{i>j} \frac{\gamma_i - \gamma_j}{1 - e^{-(\gamma_i-\gamma_j)}}\right] = 1$$

where $\kappa_{n(n-1)/2}$ denotes evaluation on the fundamental cycle.

Thus:

Theorem 4.4.5: T-characteristic Pullback Invariant

Let E be a q-dimensional complex vector bundle over a compact n-dimensional complex manifold M. Let M^Δ be the associated fiber bundle with the flag manifold $\mathbf{F}(q)$ as fiber, and let $\varphi : M^\Delta \to M$ be the projection. For any complex line bundle C over M, the T-characteristic is multiplicative:

$$T(M^\Delta, \varphi^* C) = T(M, C) \cdot T(\mathbf{F}(q)) = T(M, C)$$

More generally given a collection $\{\ell_1, ..., \ell_r\} \subseteq H^2(M, \mathbb{Z})$:

$$T(\{\varphi^*\ell_1, \ldots, \varphi^*\ell_r\}, \varphi^* C)_{M^\Delta} = T(\{\ell_1, \ldots, \ell_r\}, C)_M$$

***Proof*:**
The proof relies on expressing the T-characteristic as an integral over the total space and using the indicator $e(P)$ from section 4.4 to evaluate the contribution from the fiber, which is shown to be 1 by lemma 4.4.3 (see [16] for the details)

theorem 4.4.5 is a crucial tool for relating the invariants of a manifold to those of its associated split manifold. By applying this result to the split manifold M^Δ associated with M, we arrive at the main integrality theorem.

Theorem 4.4.6: Integrality Theorem

Let M be a compact almost complex manifold of complex dimension n.

1. The Todd genus $T(M)$ multiplied by 2^n is an integer.
2. More generally, for any classes $b_1, \ldots, b_r \in H^2(M, \mathbb{Z})$, the virtual Todd genus $T(\{b_1, \ldots, b_r\})_M$ multiplied by 2^{n-r} is an integer.

***Proof*:**
Theorem 4.4.5 implies that $T(M) = T(M^\Delta)$. One then applies the virtual form of the split-manifold decomposition at $y = 1$, where every summand is a virtual signature and hence an integer (see [16, section 14.3] for the bookkeeping, which accounts for the dimension of M^Δ exceeding n).

Theorem 4.4.7: Multiplicative Property of the Generalized Todd Genus

The multiplicative property extends to the generalized T_y-genus. For a fiber bundle M^Δ over M with fiber $\mathbf{F}(q)$, we have:

$$T_y(M^\Delta) = T_y(M) \cdot T_y(\mathbf{F}(q))$$

This property also holds for fiber bundles where the fiber is a complex projective space $\mathbb{C}\mathrm{P}^{q-1}$. Specifically, if M^Δ is a fiber bundle over M with fiber $\mathbb{C}\mathrm{P}^{q-1}$ associated with a q-dimensional complex vector bundle, then:

$$T_y(M^\Delta) = T_y(M) \cdot T_y(\mathbb{C}\mathrm{P}^{q-1})$$

***Proof*:**
Combine the above results with section 4.3.

5

Reduction via Cobordism

In this section, the necessary cobordism concepts are covered to show the signature of a manifold M is a cobordism invariant and is defined using Pontryagin classes. Due to the length of this paper, the key proof showing the isomorphism between the geometric and algebraic cobordism rings will be omitted; the proof can be found in [16, section 7].

Note that the Hirzebruch-Riemann-Roch theorem was generalized by Atiyah and Singer to the *Atiyah-Singer Index Theorem*, see [11]. In later proofs (Atiyah and Singer, 1968), the use of Pontryagin classes was replaced with *Bott periodicity* (which is 2-periodic for complex K-theory; defined in my note in [12]) allowing for a broader application to topological spaces. This paper sticks to the cobordism method for its connections to recent developments in mathematics (ex. derived algebraic cobordism [14], connection to the generalized Poincaré Conjecture [10], etc.).

5.1 (Oriented) Cobordism Ring

By Stokes' Theorem, if ω is a form on an orientable manifold M with boundary ∂M, then:

$$\int_{\partial M} \omega = \int_M d\omega$$

Thus, figuring out the possible manifolds N such that $N = \partial M$ is a worthwhile endeavor. With a bit of elementary effort, it can be shown that $N = \mathbb{R}\mathrm{P}^2$ is not the boundary of any compact 3-manifold M, and so the answer is no in general. However, $\mathbb{R}\mathrm{P}^2$ is not orientable, so does the restriction to orientable manifolds guarantee the existence? The answer is still no: for example, $\mathbb{C}\mathrm{P}^2$ is not the boundary of any compact oriented 5-manifold, though this is harder to see (it follows from the signature theorem developed in this chapter). Thus this spurs the question of trying to classify manifolds "up to boundary". This leads to the following notion:

Definition 5.1.1: Cobordant

Let M, N be two closed oriented smooth n-manifolds. Then M, N are *cobordant* if there exists a compact oriented $(n+1)$-dimensional manifold C whose oriented boundary is

$$\partial C = M \sqcup (-N)$$

where $-N$ is N with reversed orientation. If $M = \partial C$ as oriented boundary, then M is said to be null-cobordant (cobordant to the empty set).

The cobordism operation defines a natural ring structure (addition by disjoint union, multiplication under direct product), inspiring us to ask which ring it is isomorphic to and to find generators of the ring. This culminates in two important results:

1. The cobordism ring tensored with the rationals, $\Omega \otimes \mathbb{Q}$, is isomorphic to $\mathbb{Q}[x_1, x_2, ..., x_n, ...]$ where $[\mathbb{C}\mathrm{P}^{2i}] \mapsto x_i$

2. The *signature* of an oriented manifold (defined in section 5.2) is a homomorphism from the cobordism ring to $\mathbb{Q}$

The key result this gives us is that it represents all cobordism equivalence classes using Pontryagin classes. Recall that the T_1-characteristic is equal to a combination of Pontryagin classes (proposition 4.1.9). Theorem 6.2.1 will show that χ_1 is equal to the signature which will be shown to be a cobordism invariant, thus the signature is a key part in connecting the two characteristics.

In this section, the manifolds shall be real manifolds. Let V^n be a connected oriented compact differentiable manifold where n is the dimension. As V is compact, connected, and orientable, it is a cycle, and by cohomology theory $H^n(V^n, \mathbb{Z})$ is generated by one element. The value of the n-dimensional cohomology class x on the fundamental cycle V^n is denoted $x[V^n]$. If the coefficients of the cohomology group are in A, then:

$$x \in H^n(V^n, A) \qquad x[V^n] \in A$$

We can extend this by tensoring by a group B so that:

$$x[V^n] \in A \otimes B \qquad x \in H^n(V^n, A) \otimes B \cong H^n(V^n, A \otimes B)$$

For our purposes, let $n = 4k$ and let $p_i \in H^{4i}(V^{4k}, \mathbb{Z})$ be the Pontryagin classes of V^{4k} (i.e. of its tangent bundle). We get a natural grading by the indices i so that given $k = j_1 + j_2 + \cdots + j_r$ we get the product $p_{j_1} \cdots p_{j_r}$ and an integer (as $A = \mathbb{Z}$):

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

Let $\pi(k)$ be the number of distinct partitions of k.

Definition 5.1.2: Pontryagin Numbers and Chern Numbers

The *Pontryagin numbers* are the $\pi(k)$ numbers:

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

where $n = 4k$. If n is not divisible by 4, then the Pontryagin numbers are 0.

Similarly the *Chern numbers* of a complex manifold V^{2k} of complex dimension k (with an almost complex structure so that we are given a particular orientation) are all:

$$c_{j_1} \cdots c_{j_r}[V^{2k}]$$

where $k = j_1 + j_2 + \cdots + j_r$ and the j_i run through all partitions of k.

By definition of the Euler class (Gauss-Bonnet), if M is an oriented almost complex manifold of complex dimension n, then $c_n(TM)[M] = e(TM)[M] = \chi(M)$, the topological Euler characteristic, showing that the Pontryagin/Chern numbers are "generalizations" of the Euler characteristic. We shall focus on Pontryagin numbers as we shall need them for the classification result.

We can define an equivalence relation where two manifolds $M \sim_p N$ if they have the same dimension and the same Pontryagin numbers. Note that if the dimension is not divisible by 4, all manifolds are in the same equivalence class given by $\sim_p$ as their Pontryagin numbers are all 0.

If M, N are disjoint, the Pontryagin numbers of $M \sqcup N$ are the sums of the Pontryagin numbers of M, N; this comes straight from the properties of characteristic classes. Let us show the same works for the product of manifolds. Now let M be a smooth n-dimensional manifold and N a smooth m-dimensional manifold. Then by classical differential geometry we have $T(M \times N) = \pi_M^*(TM) \oplus \pi_N^*(TN)$. Then:

$$(\pi_M^*(x)\pi_N^*(y))[M \times N] = x[M] \cdot y[N] \qquad \forall x \in H^n(M, \mathbb{Z}) \otimes B,\ y \in H^m(N, \mathbb{Z}) \otimes B$$

Thus, we can extend these results to characteristic sequences of Pontryagin classes. Considering the Pontryagin classes of M, N and $M \times N$:

$$\begin{aligned} &1, p_1, p_2, ... \\ &1, p_1', p_2', ... \\ &1, p_1'', p_2'', ... \end{aligned}$$

By the properties of characteristic classes we get:

$$1 + p_1'' + p_2'' + \cdots = (1 + p_1 + p_2 + \cdots)(1 + p_1' + p_2' + \cdots) \text{ mod torsion}$$

Or, as a generating function:

$$\sum_{k=0}^{\infty} p_k'' z^k = \sum_{i=0}^{\infty} \pi_M^*(p_i) z^i \sum_{j=0}^{\infty} \pi_N^*(p_j') z^j \text{ mod torsion.}$$

Thus, define the following:

Definition 5.1.3: Algebraic K-genus

Let $\{K_j\}$ be an m-sequence and let M be a closed oriented 4ℓ-manifold. Then the *algebraic K-genus* of M is:

$$K(M) = K_\ell(p_1, ..., p_\ell)[M] = \int_M K_\ell(p_1, ..., p_\ell)$$

where $p_1, ..., p_\ell$ are the Pontryagin classes of TM.

Proposition 5.1.4: Properties of Algebraic Genus

Let $\{K_j\}$ be an m-sequence, M an oriented 4ℓ-manifold, and N an oriented $4m$-manifold (take $m = \ell$ in (1)). Then:

1. $K(M \sqcup N) = K(M) + K(N)$
2. $K(M \times N) = K(M) \cdot K(N)$
3. $K(-M) = -K(M)$ where $-M$ is the reverse orientation put on M.

Thus, $K(-)$ respects the equivalence relation $\sim_p$.

***Proof*:**
(1) and (2) follow from the above discussion, (3) follows from elementary cohomology.

Definition 5.1.5: Algebraic Oriented Cobordism Ring

Let:

$$\widetilde{\Omega}^n = \left\{[M]_p \;\middle|\; M \text{ a closed oriented smooth } n\text{-manifold}\right\}$$

where the subscript p is dropped when it is clear. This ring is called the *algebraic cobordism ring*. By proposition 5.1.4, we can define the graded ring:

$$\widetilde{\Omega}^* = \bigoplus_{n=0}^{\infty} \widetilde{\Omega}^n$$

where:

$$\widetilde{\Omega}^n \widetilde{\Omega}^m \subseteq \widetilde{\Omega}^{n+m}$$

Note that $\widetilde{\Omega}^n = 0$ if $4 \nmid n$ and $\widetilde{\Omega}^*$ is a commutative graded (torsion-free) ring.

The ring is called "algebraic" as it was defined using the algebraic properties of Pontryagin classes, we shall soon define the "geometric" cobordism ring that is the equivalence classes of cobordant manifolds and show they are isomorphic. For now, let us characterize the algebraic structure of the algebraic cobordism ring: the goal is to show that

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, ...]$$

To that end, we must find which elements of $\widetilde{\Omega}$ can represent each x_i. There is no natural choice, hence we define candidate sequences of elements given by manifolds $V^4, V^8, V^{12}, ...$ that can be chosen to form this basis. We shall see that the complex projective spaces of even dimension satisfy these required conditions. Let us start by recalling that the Pontryagin class of V^{4k} can be written in indeterminate z and decomposed into:

$$1 + p_1 z + \cdots + p_k z^k = \prod_{i=1}^{k}(1 + \beta_i z)$$

Define:

$$s(V^{4k}) = (\beta_1^k + \cdots + \beta_k^k)[V^{4k}]$$

By some symmetric function theory, we can think of this as needing the appropriate homogeneity and dimension properties, and hence if they are nonzero for each V^{4k} they are a desired list of representatives:

Definition 5.1.6: Basis Sequence

A sequence of manifolds $V^4, V^8, V^{12}, \ldots$ is called a *basis sequence* if:

$$s(V^{4k}) \neq 0$$

Theorem 5.1.7: Basis Sequence Correspondence

Let $\{V^{4k}\}$ be a basis sequence and B a commutative ring containing $\mathbb{Q}$. Then the set of sequences $\{a_k\}_k$ where $a_k \in B$ are in bijective correspondence to the set of m-sequences $\{K_j(p_1, \ldots, p_j)\}$ with coefficients in B where $K(V^{4k}) = a_k$.

***Proof*:**
By proposition 4.1.2, it suffices to show that there is exactly one power series $Q(z)$ such that, for each V^{4k} in the sequence with Pontryagin roots $\beta_1, \ldots, \beta_k$:

$$a_k = K_k[V^{4k}],$$

where K_k is the coefficient of z^k in $\prod_{i=1}^{k} Q(\beta_i z)$. Writing $Q(z) = 1 + b_1 z + b_2 z^2 + \cdots$, this equation can be written

$$a_k = s(V^{4k}) b_k + \text{polynomial in } b_1, b_2, \ldots, b_{k-1} \text{ of weight } k$$

The above polynomial depends only on V^{4k} and has integer coefficients. Then the coefficients b_k can be deduced via induction.

Theorem 5.1.8: Complex Projective Forms Basis Sequence

Let $\mathbb{C}\mathrm{P}^{2k}$ be the $2k$-dimensional complex projective space. Then the spaces $\mathbb{C}\mathrm{P}^{2k}$, $k \geq 1$, form a basis sequence for which

$$s(\mathbb{C}\mathrm{P}^{2k}) = 2k + 1$$

***Proof*:**
Let $h \in H^2(\mathbb{C}\mathrm{P}^{2k}, \mathbb{Z})$ be a generator. Then the Pontryagin class of $\mathbb{C}\mathrm{P}^{2k}$ is $(1+h^2)^{2k+1}$. The m-sequence of the power series $1+z^k$ defines a genus which for V^{4k} has the value $s(V^{4k})$ and which takes the value $2k+1$ on $\mathbb{C}\mathrm{P}^{2k}$, completing the proof.

Let us now consider $\widetilde{\Omega} \otimes \mathbb{Q}$, so we have rational coefficients: $\frac{a}{b}[V^{4k}]$. Certainly the Pontryagin number, K-genus, and $s(V^{4k})$ are all still well-defined (where the K-genus is well-defined as we required B to contain $\mathbb{Q}$). Now, the Pontryagin number of an element of $\widetilde{\Omega} \otimes \mathbb{Q}$ can be a non-integer. This new ring can still be represented using a basis sequence:

Theorem 5.1.9: Basis Sequence With Rational Numbers Correspondence

Let $\{V^{4k}\}$ be a basis sequence of oriented manifolds, and let $(j) = (j_1, \ldots, j_r)$ be a partition of k. Let $V_{(j)} = V^{4j_1} \times \cdots \times V^{4j_r}$. Then every $\alpha \in \widetilde{\Omega}^{4k} \otimes \mathbb{Q}$ can be uniquely represented by the sum:

$$\alpha = \sum_{(j)} r_{(j)}[V_{(j)}] \qquad r_{(j)} \in \mathbb{Q}$$

where the sum goes over all partitions (j) of k. Furthermore, given a system $\{a_{(j)}\}$ of rational numbers, there exists an element $\alpha \in \widetilde{\Omega} \otimes \mathbb{Q}$ where

$$p_{(j)}[\alpha] = a_{(j)}$$

***Proof*:**
This is a linear algebra result. It suffices to prove that a sum $\sum r_{(j)}(V_{(j)})$ over all partitions (j) of k is zero if and only if $r_{(j)} = 0$ for each (j). Suppose that $\sum r_{(j)}(V_{(j)}) = 0$. Let $q_1, q_2, q_3, \ldots$ be a sequence of indeterminates. Then we can find an m-sequence which takes the value q_k on V^{4k}. This implies that

$$\sum_{(j)} r_{(j)} q_{(j)} = 0,$$

where $q_{(j)}$ denotes the product $q_{j_1} q_{j_2} \ldots q_{j_r}$ for $(j) = (j_1, j_2, \ldots, j_r)$. Since the $q_{(j)}$ are pairwise distinct monomials in the independent indeterminates $q_1, q_2, \ldots$, they are linearly independent, so each $r_{(j)}$ vanishes, completing the proof.

Theorem 5.1.10: Isomorphism of Algebraic Cobordism Ring

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, \ldots]$$

***Proof*:**
Let $\{V^{4k}\}$ be an arbitrary sequence of oriented manifolds. Consider the ring homomorphism $\mathbb{Q}[x_1, x_2, \ldots] \to \widetilde{\Omega} \otimes \mathbb{Q}$ defined by $x_k \mapsto [\mathbb{C}\mathrm{P}^{2k}]$. What's left to show is that:

1. Given $\alpha = \sum_{(j)} r_{(j)}[V_{(j)}]$ then $s(\alpha) = r_{(k)} s(V^{4k})$, where $r_{(k)}$ is the coefficient of the length-one partition (k). Let $\{K_j\}$ be the m-sequence of the power series $1+z^k$. This m-sequence takes the value $s(\alpha)$ on elements $\alpha \in \widetilde{\Omega}^{4k} \otimes \mathbb{Q}$ and the value 0 on elements of

$\widetilde{\Omega}^{4j} \otimes \mathbb{Q}$ with $1 \leq j < k$.

2. If for all k, every element of $\alpha \in \widetilde{\Omega}^{4k} \otimes \mathbb{Q}$ can be represented as a sum $\alpha = \sum_{(j)} r_{(j)}[V_{(j)}]$, then $\{V^{4k}\}$ is a basis sequence. For the sake of contradiction, suppose that for some k, $s(V^{4k}) = 0$. Then, by (1), $s(\alpha) = 0$ for all $\alpha \in \widetilde{\Omega}^{4k} \otimes \mathbb{Q}$. But by theorem 5.1.8, $s(\mathbb{C}\mathrm{P}^{2k}) = 2k + 1$, a contradiction.

Injectivity and surjectivity then follow by applying theorem 5.1.9 to the basis sequence $\{\mathbb{C}\mathrm{P}^{2k}\}$ of theorem 5.1.8, giving an isomorphism.

Lastly, the K-genus can be characterized as representing a homomorphism from the algebraic cobordism ring to $\mathbb{Q}$:

Theorem 5.1.11: Characterizing Rational Homomorphisms

Let $\varphi : \widetilde{\Omega} \otimes \mathbb{Q} \to \mathbb{Q}$ be a ring homomorphism. Then there exists an m-sequence K such that $\varphi(-) = K(-)$.

***Proof*:**
The proof in theorem 5.1.10 applies to any basis sequence with the appropriate properties; using this one can create an m-sequence such that $K(V^{4k}) = \varphi(x_k)$.

This fully encodes the algebraic information. Let us now define the geometric counterpart.

Definition 5.1.12: Geometric Oriented Cobordism Ring

Two closed oriented smooth n-dimensional manifolds M^n, N^n are *cobordant* if there exists a compact $(n+1)$-dimensional manifold C such that

$$\partial C \cong (M^n \sqcup -N^n)$$

where $-N^n$ is the reverse orientation; if such a C exists it defines an equivalence relation $M \sim_c N$ which respects disjoint union, reverse orientation, and direct product; $\sqcup, -, \times$. Then Ω^n is the collection of such equivalence classes and:

$$\Omega = \bigoplus_n \Omega^n$$

is the *geometric cobordism ring.*

Definition 5.1.13: Geometric Genus

Let Ω be the oriented geometric cobordism ring. Then a *Geometric genus* (with respect to B) is a ring homomorphism:

$$\varphi : \Omega \to B$$

After showing $\Omega/\mathrm{Torsion}$ and $\widetilde{\Omega}$ are isomorphic, we shall see that all geometric genera are representable by an m-sequence.

Certainly this ring is graded, and it is *anti-commutative*, namely:

$$\alpha \cdot \beta = (-1)^{nm} \beta \cdot \alpha \qquad \alpha \in \Omega^n, \ \beta \in \Omega^m$$

Theorem 5.1.14: Algebraic - Geometric Cobordism Ring Isomorphism

$$\Omega/\mathrm{Torsion} \cong \widetilde{\Omega}, \qquad \text{equivalently} \qquad \Omega \otimes \mathbb{Q} \cong \widetilde{\Omega} \otimes \mathbb{Q}$$

***Proof*:**
see [16, section 7]

5.2 Intersection Form: Signature and the Hirzebruch Signature Theorem

On any closed oriented $4k$-dimensional manifold, there is a natural symmetric bilinear form known as the *intersection form*: given a choice of $x, y \in H^{2k}(M, \mathbb{R})$, define:

$$Q(x, y) = x \smile y[M^{4k}]$$

As $x \smile y \in H^{4k}$:

$$(x \smile y) = (-1)^{(2k)^2}(y \smile x) = (y \smile x)$$

showing that the form is *symmetric* (rather than anti-symmetric), which is why we are working with $4k$-dimensional manifolds. Thus, the eigenvalues of the corresponding matrix representations are always *real* and there always exists an orthogonal basis. If e_p and e_n represent the number of positive and negative eigenvalues, then:

Definition 5.2.1: Signature

$$\tau(Q) = e_p - e_n$$

For a closed oriented $4k$-manifold M, write $\tau(M) := \tau(Q_M)$ for the intersection form Q_M on $H^{2k}(M, \mathbb{R})$, and set $\tau(M) = 0$ when $\dim_{\mathbb{R}} M \not\equiv 0 \bmod 4$.

5.2.2 Terminology: Hirzebruch's names vs. modern names This document uses the modern name *signature* for $\tau(M)$. In Hirzebruch's original treatment [16] the same invariant is called the *index*, and what is now universally known as the *Hirzebruch signature theorem* (theorem 5.2.4) appears there as the *index theorem* (not to be confused with the later Atiyah-Singer index theorem, which generalizes it). Likewise, Hirzebruch's *virtual index* is our *virtual signature*, and his *m-sequences* are the *multiplicative sequences* of section 4.1 (we keep "*m*-sequence" as a convenient shorthand).

The most important result about the signature is that it is a *genus*:

Theorem 5.2.3: The Signature is a Genus

Let τ be the signature function. Then:

1. $\tau(M \sqcup N) = \tau(M) + \tau(N)$
2. $\tau(M \times N) = \tau(M) \cdot \tau(N)$
3. $\tau(-M) = -\tau(M)$
4. If M is null-cobordant, $\tau(M) = 0$

***Proof*:**
The proofs of (2) and (4) are long; a summary is given (see [15] for the details).

(1) Follows by definition.

(3) Reversing the orientation negates the fundamental class, so $Q \mapsto -Q$ and e_p, e_n swap, giving $\tau(-M) = -\tau(M)$.

(2) Let $M^{4k} = V^n \times W^m$. Then

$$H^{2k}(M^{4k}, \mathbb{R}) \cong \sum_{s=0}^{2k} H^s(V^n, \mathbb{R}) \otimes H^{2k-s}(W^m, \mathbb{R}). \tag{5.1}$$

Let $x, y \in H^{2k}(M^{4k}, \mathbb{R})$ be *orthogonal* if $xy[M^{4k}] = 0$. Introduce bases $\{v_i\}$ for $H^s(V^n, \mathbb{R})$ and $\{w_j\}$ for $H^t(W^m, \mathbb{R})$ such that $v_i^s v_j^{n-s}[V^n] = \delta_{ij}$ for $s \neq \frac{n}{2}$ and $w_i^t w_j^{m-t}[W^m] = \delta_{ij}$ for $t \neq \frac{m}{2}$.
Now, consider the group $A = H^{\frac{n}{2}}(V^n, \mathbb{R}) \otimes H^{\frac{m}{2}}(W^m, \mathbb{R})$, taking $A = 0$ if n and m are odd. Then A is orthogonal to the subgroup B of $H^{2k}(M^{4k}, \mathbb{R})$, which consists of all elements of the summation in equation (5.1) in which no elements of A occur. As a basis for the group B, take:

$$\{v_i^s \otimes w_j^{2k-s}\}, (0 \leq s \leq n, s \neq \frac{n}{2})$$

then:

$$\begin{aligned}(v_i^s \otimes w_j^{2k-s})(v_{i'}^{s'} \otimes w_{j'}^{2k-s'})[M^{4k}] &= \pm 1 \text{ if } s + s' = n, i = i', j = j' \\ &= 0 \text{ otherwise.}\end{aligned}$$

It follows that, with respect to this basis, the restriction of the bilinear form $xy[M^{4k}]$ to B is represented by a matrix with blocks

$$\pm \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

down the diagonal and zero elsewhere. Therefore the signature of the restriction of $xy[M^{4k}]$ to B is 0. Since A and B are orthogonal, $\tau(M^{4k})$ is equal to the signature $\tau(A)$ of the restriction of the bilinear form $xy[M^{4k}]$ to A. There are now two cases to consider. If n and m are not divisible by 4 then $\tau(A) = 0$. If n and m are divisible by 4 then certainly: $\tau(A) = \tau(V^n) \cdot \tau(W^m)$.

(4) Suppose that V^{4k} is the oriented boundary of M^{4k+1}, and that $j : V^{4k} \to M^{4k+1}$ is the embedding. Consider the following commutative diagram (exercise to show it's commutative, use Poincaré duality):

$$\begin{tikzcd}
H^{2k}(M^{4k+1},\mathbb{R}) \arrow[r, "j^*"] \arrow[d, "\cong"] & H^{2k}(V^{4k},\mathbb{R}) \arrow[r] \arrow[d, "i"] & H^{2k+1}(M^{4k+1} \bmod V^{4k},\mathbb{R}) \arrow[d, "\cong"] \\
H_{2k+1}(M^{4k+1} \bmod V^{4k},\mathbb{R}) \arrow[r] & H_{2k}(V^{4k},\mathbb{R}) \arrow[r, "j_*"] & H_{2k}(M^{4k+1},\mathbb{R})
\end{tikzcd}$$

Let A^{2k} be the image of j^* in $H^{2k}(V^{4k},\mathbb{R})$ and let K_{2k} be the kernel of j_* in $H_{2k}(V^{4k},\mathbb{R})$. Then A^{2k} is dual to $H_{2k}(V^{4k},\mathbb{R})/K_{2k}$ under the duality between $H^{2k}(V^{4k},\mathbb{R})$ and $H_{2k}(V^{4k},\mathbb{R})$. On the other hand, the diagram implies that, for $x \in H^{2k}(V^{4k},\mathbb{R})$,

$$x \in A^{2k} \Leftrightarrow i(x) \in K_{2k}.$$

Therefore, if $b_{2k} = \dim H_{2k}(V^{4k},\mathbb{R})$ is the $2k$-th Betti number of V,

$$\dim A^{2k} = \dim K_{2k} = b_{2k} - \dim K_{2k}$$

and

$$\dim A^{2k} = \frac{1}{2} b_{2k}.$$

If $x = j^*y \in A^{2k}$, and v is the fundamental cycle of V^{4k} then $x^2[V^{4k}] = (j^*y^2)[v] = (y^2)[j_*v] = 0$. Therefore, the cone

$$\{x \in H^{2k}(V^{4k},\mathbb{R}); x^2[V^{4k}] = 0\}$$

contains the linear subspace A^{2k} of dimension $\frac{1}{2}b_{2k}$. It follows that the bilinear form $xy[V^{4k}]$ has $e_p = e_n$ and hence that $\tau(V^{4k}) = 0$, completing the proof.

Theorem 5.2.4: Hirzebruch Signature Theorem

Let M be a closed real differentiable oriented manifold, and let $\tau(M)$ be its signature. Then:

1. if $\dim_{\mathbb{R}} M \not\equiv 0 \bmod 4$, then
$$\tau(M) = 0$$
2. if $\dim_{\mathbb{R}} M \equiv 0 \bmod 4$, then:
$$\tau(M) = L_k(p_1, \ldots, p_k)[M]$$

***Proof*:**
We must show they match on the generators $\mathbb{C}\mathrm{P}^{2k}$. By theorem 5.1.11, it suffices to show that $\tau(\mathbb{C}\mathrm{P}^{2k}) = 1$ since by proposition 4.1.4:

$$L_k(p_1, \ldots, p_k)[\mathbb{C}\mathrm{P}^{2k}] = 1$$

Take as basis sequence the spaces $\mathbb{C}\mathrm{P}^{2k}$. Recall that the signature is defined by the intersection form $Q(x, y) = \int_{\mathbb{C}\mathrm{P}^{2k}} x \smile y$ on the middle cohomology group $H^{2k}(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z})$. The cohomology ring of complex projective space is:

$$H^*(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z}) \cong \mathbb{Z}[h]/\langle h^{2k+1}\rangle$$

where $h \in H^2(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z})$ is the Poincaré dual to a hyperplane. The middle cohomology group $H^{2k}(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z})$ is a free abelian group of rank one with the single basis element $\{h^k\}$.

The matrix representing the intersection form Q in this basis is a 1×1 matrix whose only entry is $Q(h^k, h^k)$. Computing its value:

$$Q(h^k, h^k) = \int_{\mathbb{C}\mathrm{P}^{2k}} h^k \smile h^k = \int_{\mathbb{C}\mathrm{P}^{2k}} h^{2k}$$

By the definition of the generator h, the integral of the top-degree class h^{2k} over the fundamental class of the manifold is 1. The matrix representation is therefore 1 by 1 and is simply (1). Thus:

$$\tau(\mathbb{C}\mathrm{P}^{2k}) = e_p - e_n = 1 - 0 = 1$$

as we sought to show.

Thus, by theorem 5.1.11, there is an m-sequence that represents this map:

Theorem 5.2.5: Signature Correspondence M-sequence

The signature τ is represented by the m-sequence $\{L_j\}$ associated to the power series:

$$\frac{\sqrt{z}}{\tanh\sqrt{z}}$$

***Proof*:**
Follows immediately from theorem 5.2.4 and proposition 4.1.4.

5.3 Signature and Splitting Principle

Just like the Euler Characteristic and T-characteristic, the signature $\tau(M)$ can be written in terms of codimension-2 submanifolds (those dual to classes in H^2) by considering a filtration $V_r \subseteq V_{r-1} \subseteq \cdots \subseteq V_1 \subseteq M$, each submanifold being codimension 2 with respect to the next one. For the following results, we require a bit more theory of Chern classes:

Theorem 5.3.1: First Chern Class of the Normal Bundle of a Codimension-2 Submanifold

Let $j : X \to Y$ be the embedding of an oriented closed real $(m-2)$-dimensional differentiable manifold X in an oriented closed m-dimensional differentiable manifold Y. Let $h_X \in H^2(Y, \mathbb{Z})$ be the cohomology class defined, with respect to the given orientations, by the $(m-2)$-dimensional homology class represented by X. Let $N_{X/Y}$ be the normal bundle of X in Y. Then

$$c_1(N_{X/Y}) = j^*(h_X)$$

***Proof*:**
[16, theorem 4.8.1]

Theorem 5.3.2: Relating Chern and Pontryagin Classes

Let ξ be a $\mathbf{U}(q)$-bundle over M. Then

$$\begin{aligned}\tilde{p}(\varrho(\xi)) &= 1 - p_1(\varrho(\xi)) + p_2(\varrho(\xi)) - p_3(\varrho(\xi)) + \cdots \\ &= (1 + c_1(\xi) + c_2(\xi) + \cdots)(1 - c_1(\xi) + c_2(\xi) - \cdots).\end{aligned}$$

***Proof*:**
[16, theorem 4.5.1]

Theorem 5.3.3: Relation For Almost Complex Manifolds

The Chern classes c_i of the almost complex manifold M are related to the Pontryagin classes p_i of M (regarded as a differentiable manifold) by the equation

$$\tilde{p} = \sum_{i=0}^{\infty} (-1)^i p_i = \sum_{i=0}^{\infty} c_i \sum_{j=0}^{\infty} (-1)^j c_j.$$

***Proof*:**
[16, theorem 4.6.1]

5.3.1 Virtual Signature and Intersection Theory

Let M^n be a closed oriented smooth manifold, $j : V^{n-2} \to M^n$ an embedding of the closed oriented codimension-2 submanifold V^{n-2}. Let $N_{V/M}$ be the normal bundle of V with respect to M, and let j^* be the cohomology pullback.

Then by some basic Riemannian geometry we have that we can embed TV into TM and we have:

$$TM|_V = TV \oplus NV$$

where $NV := N_{V/M}$ is the normal bundle. Thus, if p is the total Pontryagin class:

$$j^* p(TM) = p(TV) \cdot p(NV) \text{ mod torsion}$$

Let $v \in H^2(M^n, \mathbb{Z})$ be the cohomology class Poincaré dual to V. By theorem 5.3.1 and theorem 5.3.2:

$$p(NV) = j^*(1 + v^2)$$

Thus:

$$p(V) = j^* \left((1 + v^2)^{-1} p(M)\right)$$

Now, taking the L-genus and taking advantage of the multiplicative and naturality property we get:

$$L(j^*(TM)) = j^*(L(TM)) = L(TV \oplus NV) = L(TV) \cdot L(NV)$$

Moving things around:

$$L(TV) = j^*(L(TM)) \cdot L(NV)^{-1}$$

Since we know the normal bundle has a single Pontryagin root $\gamma = v^2$, we get:

$$L(NV) = \frac{v}{\tanh(v)}$$

Finally, as $L(TM) = \sum_{i=0}^{\infty} L_i(TM) = \sum_{i=0}^{\infty} L_i(p_1(TM), ..., p_i(TM))$, we get:

$$L(TV) = \sum_{i=0}^{\infty} L_i(p_1(V), ..., p_i(V)) = j^* \left[\frac{\tanh(v)}{v} \sum_{i=0}^{\infty} L_i(p_1(M), ..., p_i(M)) \right]$$

Using this, we take advantage of Hirzebruch's Signature Theorem to get $\tau(V)$. First, by Poincaré duality, if $x \in H^{n-2}(M, \mathbb{Z}) \otimes \mathbb{Q}$, then:

$$j^*(x)[V] = vx[M]$$

Now, by Hirzebruch's signature theorem:

$$\tau(V) = \int_M \tanh(v) \sum_{i=0}^{\infty} L_i(p_1(M), ..., p_i(M))$$

Using the computational methods available to us we get:

$$\begin{aligned}
n = 2 \qquad & \tau(V^0) = \int_{M^2} v \\
n = 6 \qquad & \tau(V^4) = \int_{M^6} \frac{1}{3}(-v^3 + p_1 v) \\
n = 10 \qquad & \tau(V^8) = \int_{M^{10}} \frac{1}{45}(6v^5 - 5p_1 v^3 + (7p_2 - p_1^2)v)
\end{aligned}$$

Now here is where interpreting Chern classes naturally in intersection theory comes in. Say we have a collection $v_1, ..., v_r \in H^2(M, \mathbb{Z})$. v_1 is associated to V^{n-2}. Now say v_2 is restricted to V^{n-2} and is associated to $V^{n-4} \subseteq V^{n-2}$. Finally, v_r is restricted to $V^{n-2(r-1)}$ and is associated to V^{n-2r}. Then if $j : V^{n-2r} \to M^n$, then:

$$j^*(x)[V^{n-2r}] = v_1 v_2 \cdots v_r x[M^n]$$

Then applying the L-power series successively, we get the signature formula:

$$\tau(V^{n-2r}) = \int_{M^n} \tanh v_1 \tanh v_2 \cdots \tanh v_r \sum_{i=0}^{\infty} L_i(p_1(M^n), ..., p_i(M^n))$$

In [12], it was proven that every 2-dimensional integral cohomology class of a compact oriented differentiable manifold M^n can be represented by a submanifold $V^{n-2} \subseteq M^n$. By applying this result iteratively, we can justify the above equation. Thus, $\tau(V^{n-2r})$ depends only on the (unordered) choice of cohomology classes $v_1, v_2, ..., v_r$. Thus, we can write:

Definition 5.3.4: Virtual Signature

$$\tau(v_1, ..., v_r) = \int_{M^n} \tanh v_1 \tanh v_2 \cdots \tanh v_r \sum_{i=0}^{\infty} L_i(p_1(M^n), ..., p_i(M^n))$$

Every virtual signature occurs as the signature of a submanifold of M^n and is thus an integer.

Recall tanh satisfies:

$$\tanh(u+v) = \frac{\tanh(u) + \tanh(v)}{1 + \tanh(u)\tanh(v)}$$

or as Hirzebruch writes it:

$$\tanh(u+v) = \tanh(u) + \tanh(v) - \tanh(u)\tanh(v)\tanh(u+v)$$

This immediately gives:

Theorem 5.3.5: Functional Equation of the Virtual Signature

The virtual signature is a function which associates an integer $\tau(v_1, v_2, \ldots, v_r)$ to each (unordered) r-tuple $(v_1, v_2, \ldots, v_r)$ of 2-dimensional integral cohomology classes of a compact oriented differentiable manifold M^n. The function τ is zero if $n - 2r \not\equiv 0$ modulo 4, if $2r > n$, or if one of the classes v_i is zero. It satisfies the functional equation

$$\tau(v_1, \ldots, v_r, u+v) = \tau(v_1, \ldots, v_r, u) + \tau(v_1, \ldots, v_r, v) - \tau(v_1, \ldots, v_r, u, v, u+v)$$

If $n = 4k + 2$:

$$\tau(u+v) = \tau(u) + \tau(v) - \tau(u, v, u+v)$$

6

Riemann-Roch Theorems

In this final chapter, we shall prove the Hirzebruch-Riemann-Roch theorem. We start off with giving a proof of the classical Riemann-Roch theorem for curves and also a simple extension to the Riemann-Roch for vector bundles over curves as it provides insight on the algebra of increasing the dimension of the bundle beyond the dimension of the base space. We shall then start the proof of Hirzebruch-Riemann-Roch for line bundles by connecting the χ_y-characteristic to the T_y-characteristic by showing they are both equal to the signature τ when $y = 1$, and the $y = 0$ case (which is the theorem) then follows from the uniqueness theorem of chapter 3. It is then extended to Hirzebruch-Riemann-Roch for vector bundles by once again taking advantage of the splitting principle, giving the final complete proof.

The next section will be dedicated to very rapidly going over the Grothendieck-Riemann-Roch theorem. As a detailed proof would require build-up in intersection theory, the proof shall assume all the necessary background knowledge that can be found in [20].

The final section covers references to papers that further generalized the Riemann-Roch theorem in various different analytic, geometric, and algebraic directions.

6.1 Classical Riemann-Roch Proofs

Let us start with the most classical version of the Riemann-Roch Theorem from an algebraic perspective, namely it will be presented similarly to the proof in Hartshorne (see [35]) with some extra details added as compared to Hartshorne so that we don't reference some of the heavy machinery he built up (though Serre Duality is essentially unavoidable in this approach). By the correspondence between Riemann surfaces and algebraic curves, there is a purely geometric-analytic proof of this (this can be found in my notes in [33] or in Phillip A. Griffiths' excellent book *Introduction to Algebraic Curves* [29]).

Theorem 6.1.1: RR For Riemann Surface with Line Bundles

Let M be a compact Riemann surface and $\mathcal{L}$ a complex analytic line bundle on M. Then

1. There exists a Cartier divisor D such that $\mathcal{L} \cong \mathcal{O}_M(D)$ so that $\deg(\mathcal{L}) := \deg(D)$
2. The Riemann-Roch equation holds:

$$\dim_{\mathbb{C}} H^0(M, \mathcal{L}) - \dim_{\mathbb{C}} H^0(M, \omega_M \otimes \mathcal{L}^\vee) = \deg(\mathcal{L}) + 1 - g$$

where $g = \dim H^0(M, \omega_M) = \dim H^1(M, \mathcal{O}_M)$ is the genus of M.

***Proof*:**
Any compact Riemann surface M is an algebraic variety, in particular a curve, and by [3, section 6.1] there is an embedding $M \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$ for some N. By the GAGA principle ([34]), $\mathcal{L}$ is (the analytification of) an algebraic line bundle. As M is a variety, by [3, section 4.3] there exists a (Weil) divisor D such that $\mathcal{L} = \mathcal{O}_M(D)$. Let us show $\deg(\mathcal{L}) = \deg(D)$ is well-defined, i.e. independent of linear equivalence. Let $f \in \mathcal{M}(M)$ be nonconstant. Then as shown in [33, section 6.3] there exists a branched covering $f : M \to \mathbb{P}^1_{\mathbb{C}}$ where:

$$\#(f^{-1}(0)) = \#(f^{-1}(\infty)) = \text{deg. of map}$$

Hence:

$$\deg(\operatorname{div}(f)) = \#(f^{-1}(0)) - \#(f^{-1}(\infty)) = 0$$

Thus, $D \sim E$ implies $\deg(D) = \deg(E)$, hence $\deg(\mathcal{L})$ is well-defined.

For the 2nd point, if $D = 0$, then $\mathcal{O}_M(0) = \mathcal{O}_M$, and (writing $l(D) = \dim H^0(M, \mathcal{O}_M(D))$ and $i(D) = \dim H^1(M, \mathcal{O}_M(D))$) we have $H^0(M, \mathcal{O}_M) = \mathbb{C}$ so $l(0) = 1$ and $i(0) = g$ and

$$l(0) - i(0) = 1 - g = \deg(0) + 1 - g$$

by [33, section 6.3], hence assume $D \neq 0$ (though the beginning of this discussion doesn't need this assumption). First use Serre Duality to get:

$$H^0(M, \omega_M \otimes \mathcal{L}^\vee) \cong H^1(M, \mathcal{L})^\vee$$

Reducing the left hand side to the Euler characteristic $\chi(M, \mathcal{O}_M(D))$. To get the right hand side, we take advantage of additivity of χ. For any point $P \in M$, we get an exact sequence:

$$0 \to \mathcal{O}_M(-P) \to \mathcal{O}_M \to \kappa_P \to 0$$

where κ_P is the skyscraper sheaf:

$$(\kappa_P)_x = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

Note that as this sheaf is flasque (flabby), H^i disappears for $i > 0$ so $\chi(M, \kappa_P) = 1$. Furthermore, counting the points of D with multiplicity, $\chi(M, \kappa_D) = \deg(D)$.

Tensoring with $\mathcal{O}_M(D)$, and applying our theory from [3, section 4.3] we get:

$$0 \to \mathcal{O}_M(D - P) \to \mathcal{O}_M(D) \to \kappa_P \otimes \mathcal{O}_M(D) \to 0$$

Thus, by additivity of χ:

$$\chi(M, \mathcal{O}_M(D)) = \chi(M, \kappa_P \otimes \mathcal{O}_M(D)) + \chi(M, \mathcal{O}_M(D - P))$$

Now, if $D > 0$, we can pick a point P appearing in D. Then $\deg(D - P) = \deg(D) - 1$. We can thus do induction on $\deg(D)$. If $\deg(D) = 0$, then $D = 0$, which is exactly the case already handled above, giving the base case. Now suppose the result holds for $\deg(D) = n - 1$ and consider $\deg(D) = n$. Then $\deg(D - P) = n - 1$ from which we get by the inductive hypothesis

$$\deg(D - P) + 1 - g = \chi(M, \mathcal{O}_M(D - P))$$

adding back in P, we get:

$$\deg(D) + (-1 + 1) + 1 - g = \chi(M, \mathcal{O}_M(D))$$

giving the desired result.

For arbitrary D, we can split $D = D^+ - D^-$ where $D^+, D^- \geq 0$. Then we get a short exact sequence:

$$0 \to \mathcal{O}_M \to \mathcal{O}_M(D^-) \to \kappa_{D^-} \to 0$$

Then by additivity:

$$\chi(M, \mathcal{O}_M(D^-)) = \chi(M, \mathcal{O}_M) + \chi(M, \kappa_{D^-}) = 1 - g + \deg(D^-)$$

Now, tensor with $\mathcal{O}_M(D)$ to get the short exact sequence:

$$0 \to \mathcal{O}_M(D) \to \mathcal{O}_M(D + D^-) \to \kappa_{D^-} \to 0$$

where the last term is the same (as can be verified point-wise). Applying additivity of χ again we get:

$$\chi(M, \mathcal{O}_M(D + D^-)) = \chi(M, \mathcal{O}_M(D)) + \chi(M, \kappa_{D^-}) = \chi(M, \mathcal{O}_M(D)) + \deg(D^-)$$

As $D + D^- = D^+$, re-arranging we get:

$$\chi(M, \mathcal{O}_M(D)) = -\deg(D^-) + \chi(M, \mathcal{O}_M(D^+))$$

Applying our earlier result we get:

$$\chi(M, \mathcal{O}_M(D)) = -\deg(D^-) + \deg(D^+) + 1 - g = \deg(D) + 1 - g$$

as we sought to show.

As $\mathcal{L}$ is a line bundle, we can take the value of its first Chern class $c_1(\mathcal{L}) \in H^2(M; \mathbb{Z})$. By letting $[M] \in H_2(M; \mathbb{Z}) \cong \mathbb{Z}$ be the fundamental class, then:

$$\deg(\mathcal{L}) = c_1(\mathcal{L})[M] \in \mathbb{Z}$$

Then theorem 6.1.1 becomes:

$$\begin{aligned}\chi(M,\mathcal{L}) &= c_1(\mathcal{L})[M] + 1 - g \\ &= \left[c_1(\mathcal{L}) + \frac{1}{2}(2-2g)\right][M] \\ &= \left[c_1(\mathcal{L}) + \frac{1}{2}c_1(TM)\right][M]\end{aligned}$$

By Bass cancellation (see [12, section 5.1]) or by some result from the theory of characteristic classes (see [12, section 3.2]), vector bundles of rank exceeding the dimension of the base split off trivial summands; in particular as $\dim_{\mathbb{C}} M = 1$, any vector bundle of rank $r > 1$ splits as $\mathbb{I}^{r-1} \oplus L$ for some line bundle L, so only a line bundle's worth of new information remains cohomologically. Let us show this explicitly to further motivate HRR.

Theorem 6.1.2: Atiyah-Serre On Vector Bundles

Let M be either a compact complex manifold or an algebraic variety; in either case of dimension $d = \dim_{\mathbb{C}} M$. Let E be a rank $r > d$ vector bundle on M that is smooth or algebraic depending on whether M is a manifold or variety respectively [a]. Then there is a trivial bundle $\mathbb{I}^{r-d}$ of rank $r-d$ that fits into the short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

and $\mathrm{rk}(E'') = d$.

[a] If algebraic, assume $\Gamma_{\mathrm{alg}}(M, \mathcal{O}_M(E)) \to E_x$ is surjective for all x; i.e. it is generated by its global sections

***Proof*:**
If E is smooth, then by partitions of unity it is always generated by its global sections. As algebraic vector bundles need not be generated by their global sections, we must assume they are (for example $\mathcal{O}_{\mathbb{P}^1}(-1)$ does not satisfy this condition). Without loss of generality, we shall work over $\overline{k} = \mathbb{C}$.

Then, for each $x \in M$, $\Gamma(M, \mathcal{O}_M(E)) \twoheadrightarrow E_x$ surjective onto the fiber E_x. Then we can choose a finite dimensional subspace $W(x) \subseteq \Gamma(M, \mathcal{O}_M(E))$ such that $W(x) \to E_x$ is surjective. Then either in a $\mathbb{C}$-neighbourhood or a Zariski-neighbourhood of x, $W(x)$ surjects onto the fibers in the neighbourhood (namely, the determinant is smooth/regular, and hence we can follow a similar argument in both cases). As M is [quasi]-compact, there exists a finite open cover with this property.

Label these choices $W(i)$. Taking all their generators, we have a finite dimensional subspace $W \subseteq \Gamma(M, \mathcal{O}_M(E))$ such that $W \twoheadrightarrow E_x$ surjects for all $x \in M$ (namely $\sigma \mapsto \sigma(x)$ surjects). Let:

$$\ker(x) = \ker(W \to E_x)$$

Then as $\dim E_x = r$,

$$\dim \ker(x) = \dim W - r$$

As the dimension is constant, $\dim \ker(x)$ is independent of x, thus we can take their union over M. To be precise, let $\mathbb{P}(\ker(x)) \subseteq \mathbb{P}(W)$ be the projectivizations of $\ker(x) \subseteq W$ (i.e. collection of all lines). Then consider

$$Z = \cup_{x \in M} \mathbb{P}(\ker(x))$$

and take its Zariski closure if necessary. Then:

$$\dim Z = \dim M + \dim W - r - 1 = \dim W + d - r - 1$$

where the -1 comes from the fact that $\dim(\mathbb{P}(V)) = \dim V - 1$. Hence, we have:

$$\operatorname{codim}(Z \hookrightarrow \mathbb{P}(W)) = r - d$$

Then it is simple algebraic/affine geometry to see there exists a projective (linear) subspace $T \subseteq \mathbb{P}(W)$ with $\dim T = r - d - 1$ such that

$$T \cap Z = \emptyset$$

Let $T = \mathbb{P}(S)$ for some subspace $S \subseteq W$ (so $\dim S = r - d$). Then by construction:

$$M \times S = M \times \mathbb{C}^{r-d} = \mathbb{I}^{r-d}$$

From this, we can define a map $\mathbb{I}^{r-d} \hookrightarrow E$ sending $(x, s) \mapsto s(x) \in E$. As $T \cap Z = \emptyset$, $s(x)$ is never zero, and hence $\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$ has full rank at each point. Letting $E'' = E/\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$, we get a vector bundle of dimension d and a short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

as we sought to show.

Now, as shown in chapter 2 we have that $c_i(\mathbb{I}^r) = 0$ for all $i \geq 1$ (its characteristic classes vanish), so by additivity of characteristic classes we have that $c_1(E) = c_1(E'')$. As $c_1(E) = c_1\left(\bigwedge^r E\right)$, we get:

$$c_1(E) = c_1\left(\bigwedge^{\operatorname{rk}(E'')} E''\right)$$

Using this, we can prove the following generalization:

Theorem 6.1.3: RR For Riemann Surface with Vector Bundles

Let M be a compact Riemann surface and E a complex vector bundle over M of rank r. Then:

$$\chi(M, E) = c_1(E)[M] + r(1 - g)$$

where $c_1(E)$ is the first Chern class of E.

***Proof*:**
Start with embedding M in $\mathbb{P}^N$ via an algebraic equation and push forward E so that it is an algebraic vector bundle. Then E need not be generated by global holomorphic functions, however by twisting it enough

$$E(n) := E \otimes \mathcal{O}_M(n) \cong E \otimes \mathcal{O}_M(1)^{\otimes n}$$

we shall have a globally generated sheaf (by Serre's Theorem A), and hence $E(n)$ shall have enough global sections (see [3, section 4.4] for the details). From this, we can apply theorem 6.1.2:

$$0 \to \mathbb{I}^{r-1} \to E(n) \to E'' \to 0$$

where $\mathrm{rk}(E'') = 1$. Then twisting it by $\mathcal{O}_M(-n)$ we get:

$$0 \to \bigoplus^{r-1} \mathcal{O}_M(-n) \to E \to E''(-n) \to 0 \tag{6.1}$$

Then:

$$\chi(M, \mathcal{O}_M(E)) = \chi(M, E''(-n)) + \chi\left(M, \bigoplus^{r-1} \mathcal{O}_M(-n)\right)$$

As $\mathrm{rk}(E''(-n)) = 1$, by ordinary Riemann-Roch we have:

$$\chi(M, E''(-n)) = c_1(E''(-n))[M] + 1 - g$$

giving us:

$$\chi(M, \mathcal{O}_M(E)) = c_1(E''(-n))[M] + 1 - g + \chi\left(M, \bigoplus^{r-1} \mathcal{O}_M(-n)\right)$$

From this, we shall proceed with inducting on the rank r. The base case $r = 1$ was theorem 6.1.1. Assume Riemann-Roch holds for $r-1$, so that by the inductive hypothesis:

$$\chi\left(M, \bigoplus^{r-1} \mathcal{O}_M(-n)\right) = c_1\left(\bigoplus^{r-1} \mathcal{O}_M(-n)\right)[M] + (r-1)(1-g)$$

Then:

$$\chi(M, \mathcal{O}_M(E)) = c_1(E''(-n))[M] + 1 - g + c_1\left(\bigoplus^{r-1} \mathcal{O}_M(-n)\right)[M] + (r-1)(1-g)$$

Then we can combine the Chern classes by additivity on equation (6.1), and so:

$$\chi(M, \mathcal{O}_M(E)) = c_1(E)[M] + r(1-g)$$

as we sought to show.

We have thus shown that:

$$\chi(M, \mathcal{O}_M(E)) = \left[c_1(E) + \frac{\mathrm{rk}(E)}{2} c_1(TM)\right][M]$$

6.2 Hirzebruch-Riemann-Roch Theorem

Let us now proceed to prove the main result. The first subsection builds up the important connections between the three main functions: the χ_y-characteristic, the T_y-characteristic and the signature. These two shall be equal to the Euler characteristic when $y = -1$ and the signature when $y = 1$. We have shown that the T_y-characteristic is invariant under pullback, and so we shall have to do the same for the χ_y-characteristic. In the next subsection, we combine these results to show the result holds over split manifolds, then over general projective varieties with the trivial bundle, then over line bundles, and finally over vector bundles.

6.2.1 Index, T_y-characteristic, χ_y-characteristic

With $y = 1$, we get the following key result:

Theorem 6.2.1: Euler Characteristic and Signature

Let M be a compact Kähler manifold of dimension n. Then:

$$\chi_1(M) = \sum_{p=0}^{n} \chi^p(M) = \sum_{p,q} (-1)^q h^{p,q}(M)$$

is equal to the *signature* $\tau(M)$

Throughout this proof, let M_n represent an n-dimensional manifold, and M represent the same manifold when it is not necessary to remember the dimension.

***Proof*:**
This proof will be very rapid, a more detailed exposition can be found in [16, theorem 15.8.2].

If n is odd then by proposition 3.3.2:

$$\chi^p(M_n) = (-1)^n \chi^{n-p}(M_n) = -\chi^{n-p}(M_n)$$

Thus $\sum_{p=0}^{n} \chi^p(M_n) = 0$. On the other hand, $\tau(M_n) = 0$ for odd n. Thus for n odd the theorem is true for arbitrary compact complex manifolds.

Now suppose n is even and $z_j = x_{2j-1} + ix_{2j}$ are local complex coordinates. In this proof, we shall use the orientation given by the natural order $dx_1 \wedge dx_2 \wedge \cdots \wedge dx_{2n}$. Often in the literature on Kähler manifolds, the orientation for M_n is given by $dx_1 \wedge dx_3 \wedge \cdots \wedge dx_{2n-1} \wedge dx_2 \wedge dx_4 \wedge \cdots \wedge dx_{2n}$ (ex. see [17]). The two orientations differ by a sign $(-1)^{\frac{n(n-1)}{2}}$.

Let us now decompose $H^n(M, \mathbb{R})$ into the orthogonal subspaces on which it will be easy to compute τ. Let $B^{p,q}$ be the complex vector space of harmonic forms of type (p, q). On a compact Kähler manifold the fundamental form ω is a harmonic form of type $(1, 1)$ whose product with any other harmonic form is again harmonic (a consequence of the Kähler identities, see [16, sections 15.6 and 15.7]). Define a homomorphism

$$L : B^{p,q} \to B^{p+1,q+1}$$

by associating to each form $\alpha \in B^{p,q}$ the form $L(\alpha) = \omega\alpha \in B^{p+1,q+1}$. Then, since ω is real, $\overline{L\alpha} = L\bar{\alpha}$. Recall there is an anti-isomorphism:

$$\star : B^{p,q} \to B^{n-p,n-q}$$

where $\star(\alpha) = \overline{\star(\bar{\alpha})}$. From it, define the homomorphism:

$$\Lambda : B^{p,q} \to B^{p-1,q-1}$$

defined by $\Lambda = (-1)^{p+q} \star L \star$. Then $\overline{\Lambda\alpha} = \Lambda\bar{\alpha}$.

The kernel of Λ is denoted by $B_0^{p,q}$ and called the subspace of effective harmonic forms of type (p, q). Then:

$$\Lambda L^k : B_0^{p-k,q-k} \to B^{p-1,q-1}$$

$(p + q \leq n, k \geq 1)$ is (up to a non-zero scalar factor) equal to L^{k-1} and

$$L^k : B_0^{p-k,q-k} \to B^{p,q}$$

$(p + q \leq n)$ is injective.

For $p + q \leq n$, by taking advantage of the Hodge decomposition and the commutativity of $\star$, we get:

$$B^{p,q} = B_0^{p,q} \oplus LB_0^{p-1,q-1} \oplus \cdots \oplus L^r B_0^{p-r,q-r}$$

where $r = \min(p, q)$. Define $B_k^{p,q} = L^k B_0^{p-k,q-k}$. The elements of $B_k^{p,q}$ are called harmonic forms of type (p, q) and class k. Next, for $\varphi \in B_k^{p,q}$ where $p + q = n$:

$$\star\varphi = (-1)^{q+k}\bar{\varphi}$$

so $\bar{\varphi} \in B_k^{q,p}$. Now, by using theorem 3.1.7 with the above results, the cohomology group $H^n(M_n, \mathbb{C})$ can be decomposed into:

$$H^n(M_n, \mathbb{C}) = \sum_{\substack{p+q=n \\ k \leq \min(p,q)}} B_k^{p,q}.$$

The summands in the direct sum decomposition are mutually orthogonal with respect to the scalar product:

$$(\alpha, \beta) = \int_{M_n} \alpha \wedge \star\beta$$

Indeed, the scalar product can be non-zero only if $\alpha \wedge \star\beta$ is of type (n, n). Therefore $B_k^{p,q}$, $B_{k'}^{p',q'}$ are orthogonal for $(p, q) \neq (p', q')$. If $\alpha \in B_k^{p,q}$ and $\beta \in B_{k'}^{p,q}$ for $k > k'$ and $p + q = n$ then

$$(\alpha, \beta) = (L^k\alpha_0, L^{k'}\beta_0) \text{ with } \alpha_0, \beta_0 \text{ effective } (\Lambda\alpha_0 = \Lambda\beta_0 = 0).$$

Since L and Λ are adjoint operators, $(L\alpha, \varphi) = (\alpha, \Lambda\varphi)$, and therefore:

$$(\alpha, \beta) = (\alpha_0, \Lambda^k L^{k'}\beta_0) = 0$$

Now, the real cohomology groups $H^n(M_n, \mathbb{R})$ can be identified with the real vector space of real harmonic forms. In particular, there is the direct sum decomposition:

$$H^n(M_n, \mathbb{R}) = \sum_{\substack{p+q=n \\ k \leq p \leq q}} E_k^{p,q}$$

where $E_k^{p,q}$ is the real vector space of real harmonic forms α which can be written in the form $\alpha = \varphi + \bar{\varphi}$ with $\varphi \in B_k^{p,q}$ (and hence $\bar{\varphi} \in B_k^{q,p}$). By definition of $\tau(M_n)$, we have:

$$Q(\alpha, \beta) = \int_{M_n} \alpha \wedge \beta \quad (\alpha, \beta \in H^n(M_n, \mathbb{R})).$$

By the corresponding results for the complex decomposition, the real vector space summands in the real decomposition are mutually orthogonal with respect to this quadratic form. Now $\star\varphi = (-1)^{q+k}\bar{\varphi}$ implies that the quadratic form $(-1)^{q+k}Q(\alpha,\beta)$ is positive definite when restricted to $E_k^{p,q}$. Therefore

$$\tau(M_n) = \sum_{\substack{p+q=n \\ k\le p\le q}} (-1)^{q+k} \dim_{\mathbb{R}} E_k^{p,q}$$

Clearly $\dim_{\mathbb{R}} E_k^{p,q} = 2\dim_{\mathbb{C}} B_k^{p,q}$ for $p<q$. If $n=2m$ then $\dim_{\mathbb{R}} E_k^{m,m} = \dim_{\mathbb{C}} B_k^{m,m}$.
Therefore

$$\tau(M_n) = \sum_{\substack{p+q=n \\ k\le \min(p,q)}} (-1)^{q+k} \dim_{\mathbb{C}} B_k^{p,q}$$

Now, let $h^{p,q} = \dim_{\mathbb{C}} B^{p,q}$. Then:

$$h^{p-k,q-k} - h^{p-k-1,q-k-1} = \dim_{\mathbb{C}} B_k^{p,q} \text{ for } p+q\le n.$$

Since $h^{r,s} = h^{s,r} = h^{n-r,n-s}$:

$$h^{p-k-1,q-k-1} = h^{p+k+1,q+k+1} \text{ for } p+q=n.$$

Combining this all, we get:

$$\begin{aligned}
\tau(M_n) &= \sum_{\substack{k\ge 0 \\ p+q=n}} (-1)^{q-k}h^{p-k,q-k} + \sum_{\substack{k\ge 0 \\ p+q=n}} (-1)^{q+k+1}h^{p+k+1,q+k+1} \\
&= \sum_{p+q\le n} (-1)^q h^{p,q} + \sum_{p+q>n} (-1)^q h^{p,q} \\
&= \sum_{p,q} (-1)^q h^{p,q}
\end{aligned}$$

where in the second step the added terms with $p+q \not\equiv n \bmod 2$ cancel in pairs by the symmetry $h^{p,q} = h^{q,p}$, as we sought to show.

Theorem 6.2.2: T-Characteristic and Signature

Let M_n be a compact almost complex manifold of complex dimension n. Then:

$$T_1(M_n) = \sum_{p=0}^{n} T^p(M_n)$$

is equal to the *signature* $\tau(M)$

***Proof*:**
Let $Q(y;x)$ be the associated power series for T_y. Then:

$$Q(1;x) = \frac{x(2e^{2x} - (e^{2x} - 1))}{e^{2x} - 1} = \frac{x(e^{2x} + 1)}{e^{2x} - 1} = x\coth(x) = \frac{x}{\tanh(x)}$$

which by theorem 5.2.4 is the power series associated to the signature.

By combining theorem 6.2.2 with proposition 4.3.5 and applying theorem 5.3.5, for a split manifold M of complex dimension n with diagonal classes $a_1, \ldots, a_n$ we get:

$$2^n T(M) = \sum_{l=0}^{n} \sum_{1 \leq i_1 < \cdots < i_l \leq n} \tau(a_{i_1}, \ldots, a_{i_l})_M$$

From this we get:

Theorem 6.2.3: Splitting and Virtual Todd Genus

Let M be a compact almost complex split manifold of dimension n, and let $\ell_1, \ldots, \ell_r$ be elements of $H^2(M, \mathbb{Z})$. The virtual Todd genus $T(\ell_1, \ldots, \ell_r)_M$ multiplied by 2^{n-r} is an integer.

This also follows, without the split hypothesis, from theorem 4.4.6.

Next, recall theorem 3.4.9 which shows that χ_y is the unique function with the desired properties. In section 4.3 and section 4.2.1, the virtual T_y-characteristic was shown to satisfy every condition of theorem 3.4.9 except (1), the normalization $T_y(M,E) = \chi_y(M,E)$. With the results on the signature, we have:

Theorem 6.2.4: Virtual Characteristics Equivalence

Let M be a projective variety and $L_1, ..., L_r$ complex line bundles over M with cohomology classes $\ell_1, ..., \ell_r \in H^2(M, \mathbb{Z})$. Then:

$$\chi_1(\{L_1, ..., L_r\})_M = T_1(\{L_1, ..., L_r\})_M = \tau(\{\ell_1, ..., \ell_r\})_M$$

***Proof*:**
By the above theorems we have $\chi_1(M) = T_1(M) = \tau(M)$. By taking a submanifold $V \hookrightarrow M$ dual to the classes $\ell_1, ..., \ell_r$ and considering the characteristics and signature of V, it is easy to show the equality holds. Thus, the virtual signatures are equal.

Next, we present a theorem by Borel that shows that the Euler characteristic is invariant under pullback to a split manifold:

Theorem 6.2.5: Euler-characteristic Invariant Under Splitting Pullback

Let M be a Kähler manifold, E a complex vector bundle of rank q over M, M^Δ the split manifold with projection $\varphi : M^\Delta \to M$. Let $F = \mathbf{F}(q)$ be the fibre of φ. Then:

$$\chi_y(M^\Delta, \varphi^*(E)) = \chi_y(M, E)\chi_y(F)$$

In particular, as $\chi(\mathbf{F}(q)) = 1$:

$$\chi(M^\Delta, \varphi^*(E)) = \chi(M, E)$$

***Proof*:**
The second equality comes from $\chi(\mathbf{F}(q)) = 1$. For the first equation, the proof is long and would require knowledge of spectral sequences. For an introduction to spectral sequences, I recommend [8], and for the proof see [16, Appendix 2].

6.2.2 Hirzebruch-Riemann-Roch

Let M be a projective variety, $T(M)$ be the Todd genus, and $\chi(M)$ be the arithmetic genus. We shall show that $T(M) = \chi(M)$.

Theorem 6.2.6: Result over split manifold

Let M be a projective variety which is also a complex split manifold. Then:

$$\chi(M) = T(M)$$

***Proof*:**
Let $m = \dim M$, and $A_1, ..., A_m$ the complex diagonal line bundles defined over M. Then by proposition 4.3.5:

$$(1+y)^m T(M) = \sum_{l=0}^{m} y^l \sum_{1\leq i_1<\cdots<i_l\leq m} T_y(\{a_{i_1}, \ldots, a_{i_l}\})_M$$

where the T_y are polynomials, and by theorem 3.4.6

$$(1+y)^m \chi(M) = \sum_{l=0}^{m} y^l \sum_{i_1<\cdots<i_l} \chi_y(\{A_{i_1}, \ldots, A_{i_l}\})_M$$

As M is algebraic, by theorem 3.4.8 χ_y is a polynomial. Then if the two agree on some $y \neq -1$, we are done. By choosing $y = 1$, theorem 6.2.4 shows they are equal, completing the proof.

Now, as we know we can pull back any vector bundle E over M via the map $M^\Delta \to M$ where M^Δ is the flag bundle (split manifold) with fibre $\mathbf{F}(q)$, and by theorem 4.4.7 we have $T(M) = T(M^\Delta)$. We showed the same for the χ_y-characteristic; we now show the same for the arithmetic genus:

Theorem 6.2.7: Euler-characteristic Invariant Under Splitting

Let E be a complex $\mathrm{GL}_q(\mathbb{C})$-bundle over a projective variety M, M^Δ the split manifold with fibre the flag manifold $\mathbf{F}(q)$. Let φ be the map and $\varphi^*(E)$ be the pullback. Then

$$\chi(M) = \chi(M^\Delta)$$

***Proof*:**
First, theorem 2.2.3 implies that M^Δ is algebraic. Let φ be the projection from M^Δ on to M. The bundle $\varphi^* E$ over M^Δ splits: let $E_1, \ldots, E_q$ be the corresponding $\mathbb{C}^*$-bundles and let γ_i be the image of $c_1(E_i)$ under the natural homomorphism $H^2(M^\Delta, \mathbb{Z}) \to H^2(M^\Delta, \mathbb{C})$. By theorem 3.3.5 the cohomology classes γ_i are of type $(1,1)$. The cohomology homomorphism φ^* maps $H^*(M, \mathbb{C})$ injectively into $H^*(M^\Delta, \mathbb{C})$, and since φ is a complex analytic map, φ^* maps cohomology classes of type (p,q) to cohomology classes of type (p,q). As $H^*(M^\Delta, \mathbb{C})$ is generated by $\varphi^* H^*(M, \mathbb{C})$ and the γ_i (see [12]), since each γ_i is of type $(1,1)$, all cohomology classes of type $(0,p)$ in $H^*(M^\Delta, \mathbb{C})$ must lie in $\varphi^* H^*(M, \mathbb{C})$. Therefore $h^{0,p}(M^\Delta) = h^{0,p}(M)$ and hence $\chi(M) = \chi(M^\Delta)$, as we sought to show.

Theorem 6.2.8: Hirzebruch-Riemann-Roch for Line Bundles

Let M be a projective variety. Then:

$$\chi(M) = T(M)$$

If L is a line bundle, then:

$$\chi(M, L) = \int_M \mathrm{ch}(L) \cdot \mathrm{td}(M)$$

***Proof*:**
Let M^Δ be the split manifold associated with the tangent bundle TM. Then:

$$\chi(M) \overset{\text{th } 6.2.7}{=} \chi(M^\Delta) \overset{\text{th } 6.2.6}{=} T(M^\Delta) \overset{\text{th } 4.4.5}{=} T(M)$$

For adding the line bundle, consider $L_1, ..., L_r$ complex line bundles over M. The χ-version of the uniqueness theorem (the final clause of theorem 3.4.9), applied to the virtual T at $y = 0$ with the normalization $T(M) = \chi(M)$ just proven (its conditions (2) and (3) hold by the results of sections 4.3 and 4.2.1), gives:

$$\chi(\{L_1, ..., L_r\})_M = T(\{L_1, ..., L_r\})_M$$

For $r = 1$, $L_1 = L$, $\chi(L)_M = T(L)_M$. By proposition 3.4.2:

$$\chi(L) = \chi(M) - \chi(M, L^{-1})$$

and by proposition 4.3.2

$$T(L) = T(M) - T(M, L^{-1})$$

replacing L by L^{-1}. As $T(M) = \chi(M)$ and $T(L) = \chi(L)$, we manipulate and get:

$$\chi(M, L) = T(M, L)$$

as we sought to show.

Finally:

Theorem 6.2.9: Hirzebruch-Riemann-Roch

Let M be a projective variety and E a complex vector bundle (the result extends to coherent sheaves via finite locally free resolutions). Then:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \cdot \mathrm{td}(M)$$

***Proof*:**
Let E be a complex vector bundle over M, and take the usual $\varphi : M^\Delta \to M$ such that $\varphi^*(E)$ splits. Then by theorem 4.4.5

$$T(M, E) = T(M^\Delta, \varphi^*(E))$$

(the cited theorem is stated for line-bundle coefficients, but its fiber-integration proof applies verbatim to an arbitrary bundle E, since $\mathrm{ch}(\varphi^* E) = \varphi^* \mathrm{ch}(E)$). By theorem 6.2.5:

$$\chi(M^\Delta, \varphi^*(E)) = \chi(M, E)\chi(\mathbf{F}(q)) = \chi(M, E)$$

Thus, we can reduce to line bundles, in particular given $\varphi^*(E)$ splits into $A_1, ..., A_q$, by use of the properties of T shown in section 4.3:

$$T(M^\Delta, \varphi^*(E)) = \sum_{i=1}^{q} T(M^\Delta, A_i)$$

and by theorem 3.3.7:

$$\chi(M^\Delta, \varphi^*(E)) = \sum_{i=1}^{q} \chi(M^\Delta, A_i)$$

As M^Δ is projective by theorem 2.2.3, by theorem 6.2.8 for all $1 \leq i \leq q$:

$$\chi(M^\Delta, A_i) = T(M^\Delta, A_i)$$

But then combining everything we get:

$$\chi(M, E) = T(M, E)$$

as we sought to show.

6.3 *Grothendieck-Riemann-Roch

This section is simply for exposure and does not go into the necessary background, but it does provide an outline of the proof of the Grothendieck-Riemann-Roch theorem.

The Grothendieck-Riemann-Roch (GRR) theorem is a generalization of the Hirzebruch-Riemann-Roch theorem as a property of proper morphisms. Recall the commutative diagram presented at the end of section 2.4:

$$\begin{tikzcd} K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(Y) \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TY)"] \\ A_*(X;\mathbb{Q}) \arrow[r, "f_*"] & A_*(Y;\mathbb{Q}) \end{tikzcd}$$

The Hirzebruch-Riemann-Roch theorem can be thought of as:

$$\begin{tikzcd} K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(\mathrm{pt}) \arrow[d, "\mathrm{ch}(-)"] \\ A_*(X;\mathbb{Q}) \arrow[r, "f_*"] & \mathbb{Q} \end{tikzcd}$$

as we will demonstrate in this section. This section provides a brief overview of the theorem and its proof, following the exposition in Ravi Vakil's notes on intersection theory [19]. The main idea in Hirzebruch's proof of using the equivalence between the Euler characteristic and T-characteristic going through the signature τ is replaced by showing any proper morphism f can be decomposed into a closed immersion and a projection. The projection will in fact reduce further to only needing to prove the result on $\mathbb{P}^n \to \mathrm{pt}$ which exactly mirrors the Hirzebruch-Riemann-Roch proof, mimicking the strategy done in the introduction of chapter 4. The proof using closed immersion is where the virtual results are replaced. The codimension information is still present but generalized into intersection with *normal cones* (think of the construction of a vector bundle from a locally free sheaf, $\mathrm{Spec}_X(\mathrm{Sym}(\mathcal{F}))$, and replace $\mathrm{Sym}(\mathcal{F})$ with R, the associated graded algebra of the ideal sheaf of the immersion, which can hold "singular" information).

Build-up

The key objects in the Grothendieck-Riemann-Roch theorem are defined in a purely algebraic context. The *Grothendieck group of vector bundles*, denoted $K^0(X)$, is the abelian group generated by the isomorphism classes of vector bundles (locally free sheaves of finite rank) on a scheme X where the relation $[E] = [E'] + [E'']$ is imposed for every short exact sequence $0 \to E' \to E \to E'' \to 0$ of vector bundles. The tensor product of vector bundles gives $K^0(X)$ a commutative ring structure. Similarly, the *Grothendieck group of coherent sheaves*, denoted $K_0(X)$, is the abelian group generated by isomorphism classes of coherent sheaves on X, with the same relation for short exact sequences. If X is a smooth variety, the natural map $K^0(X) \to K_0(X)$ is an isomorphism, and we denote the group by $K(X)$.

For a proper morphism $f : X \to Y$, take the pushforward map $f_* : K_0(X) \to K_0(Y)$; this is the map denoted $f_!$ on K-theory in the diagrams above. Pushing forward a coherent sheaf $\mathcal{F}$ via f does not generally preserve exactness, namely the functor f_* is only left-exact (see [35], [3]). The failure of exactness is measured by the higher direct image sheaves, $R^i f_*\mathcal{F}$. For a short exact sequence of coherent sheaves $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ on X, applying the pushforward functor yields a long exact sequence of coherent sheaves on Y:

$$0 \to f_*\mathcal{F}' \to f_*\mathcal{F} \to f_*\mathcal{F}'' \to R^1 f_*\mathcal{F}' \to R^1 f_*\mathcal{F} \to \cdots$$

To ensure that the pushforward is a homomorphism on the Grothendieck group, it must preserve the defining additive relation. This is achieved by defining the pushforward via an alternating sum (see [35]):

$$f_*([\mathcal{F}]) = \sum_{i\geq 0}(-1)^i[R^i f_*\mathcal{F}]$$

With this definition, the long exact sequence ensures that $f_*([\mathcal{F}]) = f_*([\mathcal{F}']) + f_*([\mathcal{F}''])$ in $K_0(Y)$.

This pushforward generalizes the Euler-Poincaré characteristic: by taking $Y = \mathrm{Spec}k$ to be a point, then $K_0(Y) \cong \mathbb{Z}$ (a coherent sheaf on a point is a finite-dimensional vector space, and its class is its dimension). For a proper scheme X over k, the higher direct image sheaves $R^i f_*\mathcal{F}$ correspond to the cohomology groups $H^i(X, \mathcal{F})$. Specifically, $R^i f_*\mathcal{F}$ is a sheaf on a point, so its class in $K_0(\mathrm{Spec}k)$ is simply $\dim_k H^i(X, \mathcal{F})$. Therefore, the pushforward to a point is:

$$f_*([\mathcal{F}]) = \sum_{i\geq 0}(-1)^i \dim_k H^i(X, \mathcal{F}) = \chi(X, \mathcal{F})$$

Next, the intersection-theoretic Chern classes are defined as operators on the Chow groups. For a vector bundle E on a scheme X, the *i-th operational Chern class*, $c_i(E)$, is a homomorphism

$$c_i(E) \cap - : A_k(X) \to A_{k-i}(X)$$

for all k. These classes satisfy properties analogous to those of the topological Chern classes used in the Hirzebruch-Riemann-Roch theorem, such as the Whitney sum formula. For smooth varieties, these operational classes correspond to the classes defined via topology (see [20]).

The *Chern character* is a ring homomorphism $\mathrm{ch} : K(X) \to A_*(X) \otimes \mathbb{Q}$. It is defined to be additive over direct sums, which makes it compatible with the Grothendieck group structure. For a vector bundle E with Chern roots $\gamma_1, \cdots, \gamma_r$ (formal symbols such that $c_k(E)$ is the k-th elementary symmetric polynomial in the γ_i), the Chern character is

$$\mathrm{ch}(E) = \sum_{i=1}^{r} e^{\gamma_i} = \mathrm{rank}(E) + c_1(E) + \frac{1}{2}(c_1(E)^2 - 2c_2(E)) + \cdots$$

This definition extends to all elements of $K(X)$. Its key properties are:

- $\mathrm{ch}([E] + [F]) = \mathrm{ch}(E) + \mathrm{ch}(F)$
- $\mathrm{ch}([E] \otimes [F]) = \mathrm{ch}(E) \cdot \mathrm{ch}(F)$

The *Todd class* of a vector bundle E with Chern roots $\gamma_1, \cdots, \gamma_r$ is

$$\mathrm{td}(E) = \prod_{i=1}^{r} \frac{\gamma_i}{1 - e^{-\gamma_i}} \in A_*(X) \otimes \mathbb{Q}.$$

The Todd class is multiplicative on short exact sequences: if $0 \to E' \to E \to E'' \to 0$ is exact, then $\mathrm{td}(E) = \mathrm{td}(E') \cdot \mathrm{td}(E'')$. Thus, the Chern character and Todd class are defined verbatim to how they were defined in this paper up to change of mathematical domain.

Grothendieck-Riemann-Roch

The commutative diagram in the beginning of this section can be translated into the following:

Theorem 6.3.1: Grothendieck-Riemann-Roch

Let $f : X \to Y$ be a proper morphism of smooth quasi-projective varieties. Then for any $\alpha \in K(X)$, the following equality holds in $A_*(Y) \otimes \mathbb{Q}$:

$$\mathrm{ch}(f_![\alpha]) \cdot \mathrm{td}(TY) = f_*(\mathrm{ch}([\alpha]) \cdot \mathrm{td}(TX)).$$

Let us reduce this proof in the way stated at the beginning:

Proposition 6.3.2: Reduction to Projections and Immersions

To prove the Grothendieck-Riemann-Roch theorem for any proper morphism $f : X \to Y$ between smooth quasi-projective varieties, it suffices to prove GRR for the cases where f is a closed immersion and where f is a projection.

***Proof*:**
The GRR theorem can be stated as the commutativity of the diagram involving the transformation $T_Z(\alpha) = \mathrm{ch}(\alpha) \cdot \mathrm{td}(TZ)$. If we have two morphisms $f : X \to Z$ and $g : Z \to Y$, the pushforward is functorial, so $(g \circ f)_* = g_* \circ f_*$ for both K-theory and Chow groups. If the diagram commutes for f and g, then

$$T_Y \circ (g \circ f)_* = T_Y \circ g_* \circ f_* = g_* \circ T_Z \circ f_* = g_* \circ f_* \circ T_X = (g \circ f)_* \circ T_X.$$

So the diagram also commutes for the composition $g \circ f$. Any proper morphism $f : X \to Y$ between smooth varieties can be factored as a closed immersion $i : X \hookrightarrow \mathbb{P}^n \times Y$ followed by a projection $p : \mathbb{P}^n \times Y \to Y$. Thus, proving the theorem for closed immersions and projections suffices.

Proposition 6.3.3: GRR for Projections

The Grothendieck-Riemann-Roch theorem holds for the projection $\pi : \mathbb{P}^n \times Y \to Y$.

***Proof*:**
We need to show that for any $\alpha \in K(\mathbb{P}^n \times Y)$, $\pi_*(T_{\mathbb{P}^n \times Y}(\alpha)) = T_Y(\pi_*\alpha)$. By the projective bundle formula, it suffices to check for elements of the form $\alpha = \beta \otimes \gamma$, where $\beta \in K(\mathbb{P}^n)$ and $\gamma \in K(Y)$. The tangent bundle splits as $T(\mathbb{P}^n \times Y) \cong \pi_1^* T\mathbb{P}^n \oplus \pi_2^* TY$, so the Todd class is multiplicative: $\mathrm{td}(T(\mathbb{P}^n \times Y)) = \pi_1^*\mathrm{td}(T\mathbb{P}^n) \cdot \pi_2^*\mathrm{td}(TY)$. The Chern character is also multiplicative for exterior products. The right side of the GRR formula becomes:

$$\begin{aligned}
\pi_*(T_{\mathbb{P}^n \times Y}(\beta \otimes \gamma)) &= \pi_*(\mathrm{ch}(\beta \otimes \gamma) \cdot \mathrm{td}(T_{\mathbb{P}^n \times Y})) \\
&= \pi_*(\pi_1^*(T_{\mathbb{P}^n}(\beta)) \cdot \pi_2^*(T_Y(\gamma))) \\
&\overset{!}{=} \pi_*(\pi_1^*(T_{\mathbb{P}^n}(\beta))) \cdot T_Y(\gamma)
\end{aligned}$$

where $\overset{!}{=}$ comes from intersection theory (see ex. [19]). The term $\pi_*(\pi_1^*(T_{\mathbb{P}^n}(\beta)))$ is the push-forward of a class from $\mathbb{P}^n$ to a point, which is computed by the degree map on 0-cycles (the algebraic analogue of integration in de Rham theory). The left side of the GRR formula is:

$$\begin{aligned} T_Y(\pi_*(\beta \otimes \gamma)) &= T_Y(\pi_*(\beta) \cdot \gamma) \\ &= \mathrm{ch}(\pi_*(\beta)) \cdot T_Y(\gamma) \end{aligned}$$

Comparing the two sides, the GRR formula for the projection π reduces to showing that $\mathrm{ch}(\pi_*(\beta)) = \int_{\mathbb{P}^n} T_{\mathbb{P}^n}(\beta)$ for any $\beta \in K(\mathbb{P}^n)$. This is precisely the Hirzebruch-Riemann-Roch theorem for $\mathbb{P}^n$ (pushing forward to a point), which can be verified directly (recall the computations done in the beginning of chapter 4; they are exactly this!). To recall the result of that section and generalizing a bit, for a line bundle $\mathcal{O}(k)$, $\pi_*[\mathcal{O}(k)] = \chi(\mathbb{P}^n, \mathcal{O}(k)) = \binom{n+k}{n}$. Then:

$$\binom{n+k}{n} = \int_{\mathbb{P}^n} \mathrm{ch}(\mathcal{O}(k)) \cdot \mathrm{td}(T\mathbb{P}^n).$$

Since the classes of line bundles generate $K(\mathbb{P}^n)$, the theorem holds for projections.

Proposition 6.3.4: GRR for Closed Immersions

The Grothendieck-Riemann-Roch theorem holds for any closed immersion $i : X \hookrightarrow Y$ of smooth varieties.

This is where the virtual machinery developed in this paper is replaced with the machinery of intersection theory.

***Proof*:**
This is an overview of the proof in [19] so that the reader gets a sense of the development of the machinery of intersection theory.

First, Vakil proves a special case where X is embedded as the zero section in a projective bundle over itself. Then, the general case is reduced to this special case by using the deformation to the normal cone (one of the key ideas of intersection theory).

Step 1. The special case $i : X \hookrightarrow Y = \mathbb{P}(N \oplus \mathcal{O}_X)$
Let N be a vector bundle on X, and consider the immersion i of X as the zero section into the projective bundle $Y = \mathbb{P}(N \oplus \mathcal{O}_X)$. The normal bundle of this immersion is $N_{X/Y} \cong N$. For any vector bundle $\mathcal{E}$ on X, one can construct a Koszul resolution of $i_*\mathcal{E}$ on Y. This resolution allows for an explicit computation of $\mathrm{ch}(i_*\mathcal{E})$ in terms of the Chern classes of $\mathcal{E}$ and the bundles involved in the geometry of the projective bundle Y. An algebraic manipulation, relying on the properties of Chern characters and Todd classes under short exact sequences, shows that the GRR formula holds in this specific situation. The formula essentially reduces to an identity involving the Todd class of the normal bundle N.

Step 2. Deformation to the normal cone
For a general closed immersion $i : X \to Y$, the deformation to the normal cone is a flat family $M \to \mathbb{P}^1$ whose fiber over $0 \in \mathbb{P}^1$ is Y and whose fiber over $\infty \in \mathbb{P}^1$ is the special fiber $M_\infty = \mathrm{Bl}_X(Y) \cup_E \mathbb{P}(C_{X/Y} \oplus \mathcal{O}_X)$, where E is the exceptional divisor of the blow-up of Y along X. The key insight is that because the fibers are rationally equivalent, any intersection-theoretic formula that holds on one fiber must hold on the other (which matches with the intuition for flat families as presented in [35], along with the added rational-equivalence requirement).

Now, if the formula holds for the immersions into the special fiber M_∞, it must hold for the original immersion $i : X \to Y$. The special fiber consists of two pieces glued along the exceptional divisor E. The GRR formula is additive over unions, and the contribution from the blow-up part will cancel, leaving only the contribution from the piece $\mathbb{P}(C_{X/Y} \oplus \mathcal{O}_X)$.

This reduces the problem to proving GRR for the immersion of X into $\mathbb{P}(C_{X/Y} \oplus \mathcal{O}_X)$, in particular this reduces the general problem to the special case proved in Step 1, completing the proof of the Grothendieck-Riemann-Roch theorem.

6.4 ** Current Generalization Results

Since the publication of the Hirzebruch-Riemann-Roch theorem in 1954, many generalizations have taken place. In the purely algebraic perspective, Grothendieck proved the Grothendieck-Riemann-Roch theorem in 1957. In 1963, Atiyah and Singer proved the *Atiyah-Singer Index Theorem* (ASIT), a generalization in a more analytic context by taking advantage of the fact that the Dolbeault operator $\overline{\partial}$ is an *elliptic operator*. Their original 1963 proof used cobordism, much as Hirzebruch's did; later proofs instead used K-theory and *Bott periodicity* (a 2-periodic phenomenon) in place of cobordism.

These two results have since been pushed further. ASIT has been pushed as far as being well-posed on quasi-conformal manifolds (a generalization of a conformal structure to a more topological setting) by Connes, Sullivan, and Teleman in 1994 ([22]). There are also proofs of the index theorem via heat-equation methods ([2]).

For the generalizations of GRR, the result has been pushed as far as Noetherian schemes by Navarro, A. and Navarro, J. in 2017 ([5]). The result in intersection theory strongly suggests that these results can be generalized to derived schemes, and indeed Lowrey and Schürg proved exactly this in 2012 ([28]). From an arithmetic (Arakelov-theoretic) point of view, Riemann-Roch-type isometries were extended to non-compact orbifolds such as modular curves in 2016 ([1]).

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